Lecture Notes of MTH420 Measures and Integrals
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CHAPTER 1

The Riemann integral: scope and limitations

Let $f : [a, b] \to \mathbb{R}$ be a bounded real function, where $a, b \in \mathbb{R}$ satisfying $a < b$. A *partition* of $[a, b]$ is a finite set $P = \{a_0, a_1, a_2, \ldots, a_n\}$ with

$$a = a_0 < a_1 < a_2 < \cdots < a_n = b.$$

The partition P gives rise to the *upper* and *lower sums*

$$U(P, f) = \sum_{i=1}^{n} M_i(a_i - a_{i-1}), \quad L(P, f) = \sum_{i=1}^{n} m_i(a_i - a_{i-1})$$

where

$$M_i = \sup_{x \in [a_{i-1}, a_i]} f(x), \quad m_i = \inf_{x \in [a_{i-1}, a_i]} f(x)$$

for each $1 \leq i \leq n$. (Note that M_i, m_i are real numbers since f is bounded on each interval $[a_{i-1}, a_i]$.)

In order to define the Riemann integral of f, one first shows that for any given partition P, $L(P, f) \leq U(P, f)$, and next that for any *refinement*, i.e. a partition $P' \supseteq P$, we must have $L(P, f) \leq L(P', f)$ and $U(P', f) \leq U(P, f)$. Finally, since for any two partitions P_1 and P_2, their union $P_1 \cup P_2$ is a refinement of both, we see that $L(P, f) \leq U(Q, f)$ for *any* partitions P, Q. The set $\{L(P, f) : P \text{ is a partition of } [a, b]\}$ is thus bounded from above in $\mathbb{R}$, and we call its supremum the *lower integral* $\underline{\int_a^b} f dx$ of f on $[a, b]$. Similarly, the infimum of the set of upper sums is the *upper integral* $\overline{\int_a^b} f dx$. The function f is now said to be Riemann-integrable on $[a, b]$ if these two numbers coincide, and their common value is the *Riemann integral* of f, denoted by $\int_a^b f dx$. See Figure 1.1.

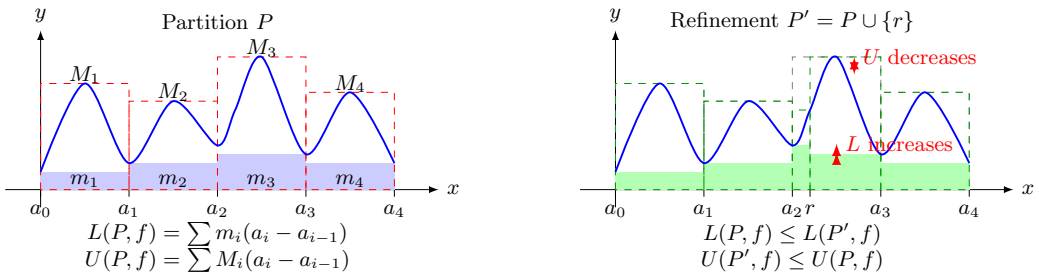


FIGURE 1.1. Riemann integration

This definition does not provide a convenient criterion for checking the integrability of particular functions; however, the following formulation provides a useful criterion for integrability, see Theorem 7.2.8 in [**1**] or Theorem 14.12 in [**11**] for a proof.

THEOREM 1.1 (Riemann's criterion). *$f : [a, b] \to \mathbb{R}$ is Riemann-integrable if and only if for every $\varepsilon > 0$ there exists a partition P_ε such that $U(P_\varepsilon, f) - L(P_\varepsilon, f) < \varepsilon$.*

Riemann's criterion still does not give us a precise picture of the *class* of Riemann-integrable functions. However, we can show (see [**1**] or [**11**]) that any bounded *monotone* or *continuous* functions on $[a, b]$ belong to this class.

We can relax the continuity requirement. A trivial step is to assume f is bounded and continuous on $[a, b]$ except at finitely many points. Then f is Riemann integrable. To see this, split the interval into pieces on which f is continuous. Then f is integrable on each and hence one can derive integrability of f on the whole interval. As an example consider a function f equal to zero for all $x \in [0, 1]$ except $a_1, \ldots, a_n$ where it equals 1. It is integrable with integral over $[0, 1]$ equal to 0.

Since the purpose of this module is to present *Lebesgue's theory of integration*, we should discuss *why* we need a new theory of integration at all: what, if anything, is wrong with the simple Riemann integral described above?

First, scope: it doesn't deal with all the kinds of functions that we hope to handle.

The results that are most easily proved rely on continuous functions on bounded intervals; in order to handle integrals over unbounded intervals, e.g.

$$\int_0^\infty e^{-x^2} dx$$

or the integral of an unbounded function:

$$\int_0^1 \frac{1}{\sqrt{x}} dx,$$

we have to resort to 'improper' Riemann integrals, defined by a limit process: e.g. considering the integrals

$$\int_0^n e^{-x^2} dx, \quad \int_\varepsilon^1 \frac{1}{\sqrt{x}} dx,$$

and letting $n \to \infty$ or $\varepsilon \to 0^+$ respectively. This isn't a serious flaw.

Second, dependence on intervals: we have no easy way of integrating over more general sets, or of integrating functions whose values are distributed 'awkwardly' over sets that differ greatly from intervals. For example, consider the upper and lower sums for the indicator function $\mathbf{1}_{\mathbb{Q}}$ of $\mathbb{Q}$ over $[0, 1]$; however we partition $[0, 1]$, each subinterval must contain both rational and irrational points; thus each upper sum is 1 and each lower sum is 0. Hence we cannot calculate the Riemann integral of f over the interval $[0, 1]$; it is simply 'too discontinuous'. (You may easily convince yourself that f is discontinuous at all points of $[0, 1]$.)

Third, lack of completeness: rather more importantly from the point of view of applications, the Riemann integral doesn't interact well with taking the limit of a sequence of functions. One may expect results of the following form: if a sequence f_n of Riemann-integral functions converges (in some appropriate sense) to f, then $\int_a^b f_n dx \to \int_a^b f dx$.

We give two counterexamples showing what difficulties can arise if the functions (f_n) converge to f pointwise, i.e. $f_n(x) \to f(x)$ for all $x \in [a, b]$.

1. The limit need not be Riemann integrable, and so the convergence question does not even make sence. Here we may take $f = \mathbf{1}_{\mathbb{Q}\cap[0,1]}$, $f_n = \mathbf{1}_{A_n}$ where $A_n = \{q_1, q_2, \ldots, q_n\}$, and the sequence $(q_n)_{n=1}^\infty$ is an enumeration of the rational points in $[0, 1]$, so that (f_n) is even monotone increasing.

2. The limit is Riemann integrable, but the convergence of Riemann integrals does not hold. Let $f = 0$, consider $[a, b] = [0, 1]$, and set

$$f_n(x) = \begin{cases} 4n^2x, & \text{if } 0 \le x < \frac{1}{2n}, \\ 4n - 4n^2x, & \text{if } \frac{1}{2n} \le x < \frac{1}{n}, \\ 0, & \text{if } \frac{1}{n} \le x \le 1. \end{cases}$$

f_n is a continuous function with integral 1 for all $n \in \mathbb{N}$. On the other hand, the sequence $f_n(x)$ converges pointwise to $f = 0$ since for all $x > 0$, $f_n(x) = 0$ for n sufficiently large (such that $\frac{1}{n} < x$). See Figure 1.2.

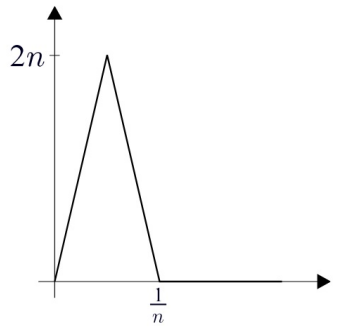


FIGURE 1.2. Graph of f_n.

To avoid problems of this kind, we can introduce the idea of *uniform* convergence: a sequence (f_n) in $C([0, 1])$ converges uniformly to f if the sequence $a_n = \sup\{|f_n(x) - f(x)| : 0 \le x \le 1\}$ converges to 0, where $C([0, 1])$ denotes the space containing all the continuous function on $[0, 1]$. In this case one can easily prove the convergence of the Riemann integrals:

$$\int_0^1 f_n(x)dx \to \int_0^1 f(x)dx.$$

However, the 'distance' $\sup\{|f_n(x) - f(x)| : 0 \le x \le 1\}$ has nothing to do with integration and the uniform convergence is too restrictive for many applications. A more natural concept of 'distance', given by $\int_0^1 |f(x) - g(x)|dx$, leads to another problem. Defining

$$g_n(x) = \begin{cases} 0, & \text{if } 0 \le x \le \frac{1}{2}, \\ n\left(x - \frac{1}{2}\right), & \text{if } \frac{1}{2} < x < \frac{1}{2} + \frac{1}{n}, \\ 1, & \text{if } \frac{1}{2} + \frac{1}{n} \le x \le 1, \end{cases}$$

it can be shown that $\int_0^1 |g_n(x) - g_m(x)|dx \to 0$ as $m, n \to \infty$; in Figure 1.3 the shaded area vanishes. (We say that (f_n) is a *Cauchy sequence* in this distance.) Yet there is no continuous function f to which this sequence converges since the pointwise limit is $f(x) = 1$ for $x > \frac{1}{2}$ and 0 otherwise, so that $f = \mathbf{1}_{(\frac{1}{2},1]}$. So the space $C([0, 1])$ is too small from this point of view.

This is rather similar to the situation which leads one to work with $\mathbb{R}$ rather than just with the set of rational numbers $\mathbb{Q}$ (there are Cauchy sequences without limits in $\mathbb{Q}$, for example a sequence of rational approximations of $\sqrt{2}$). Recalling the crucial importance completeness in the case of $\mathbb{R}$, we naturally look for a theory of integration which does not

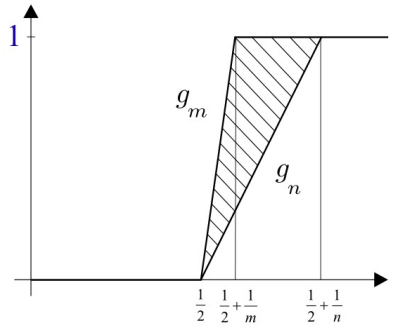


FIGURE 1.3. Graph of g_n, g_m.

have this shortcoming. In the process, we shall find that our new theory, which will include the Riemann integral as a special case, also solves the other problems listed.

REMARK (Finishing remark). This chapter is prepared based on Chapter 1.2 in [**5**], and you may have a look at Chapter 1.3 in [**5**] for its probabilistic connections.

CHAPTER 2

Measure

2.1. Null sets

The idea of a 'negligible' set relates to one of the limitations of the Riemann integral, as we saw in the previous chapter. Since the function $f = \mathbf{1}_{\mathbb{Q}}$ takes a non-zero value only on $\mathbb{Q}$, and equals 1 there, the 'area under its graph' (if such makes sense) must be very closely linked to the 'length' of the set $\mathbb{Q}$. This is why it turns out that we cannot integrate f in the Riemann sense: the sets $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are so different from intervals that it is not clear how we should measure their 'lengths' and it is clear that the 'integral' of f over $[0, 1]$ should equal the 'length' of the set of rational numbers in $[0, 1]$. So how *should* we define this concept for more general sets?

The obvious way of defining the 'length' of a set is to start with intervals. Suppose that I is a bounded interval of any kind, i.e. $I = [a, b]$, $I = [a, b)$, $I = (a, b]$ or $I = (a, b)$. We simply define the length of I as $\ell(I) = b - a$ in each case.

As a particular case, we have $\ell(\{a\}) = \ell([a, a]) = 0$. It is then natural to say that a one-element set is 'null'. Before we extend this idea to more general sets, first consider the length of a finite set. A finite set is not an interval but since a single point has length 0, and adding finitely many such lengths together should still gives 0. The underlying concept here is that if we decompose a set into a finite number of disjoint intervals, we compute the length of this set by adding the lengths of the pieces.

As we have seen, in general it may not be always possible to decompose a set into intervals. Therefore, we consider systems of intervals that cover a given set. We shall generalize the above idea by allowing a countable number of covering intervals. Thus, we arrive at the following more general definition of sets of 'zero length':

DEFINITION 2.1. *A null set $A \subseteq \mathbb{R}$ is a set that may be covered by a sequence of intervals of arbitrarily small total length, i.e. given any $\varepsilon > 0$, we can find a countable collection of intervals, denoted $\{I_n : n \geq 1\}$, such that*

$$A \subseteq \bigcup_{n=1}^{\infty} I_n \quad \textit{and} \quad \sum_{n=1}^{\infty} \ell(I_n) < \varepsilon.$$

(We also say simply that 'A is null'.)

Note that the intervals do not need to be disjoint. It follows at once from the definition that the empty set is null.

Next, any one-element set $\{x\}$ is a null set. We can take $I_1 = [x, x]$, $I_n = [0, 0]$ for $n \geq 2$. (We could equally have taken $I_n = (0, 0) = \emptyset$ for $n \geq 2$.)

More generally, any countable set $A = \{x_1, x_2, \dots\}$ is null, and we can take $I_n = [x_n, x_n]$ for all $n \in \mathbb{N}$. In fact, we have the following general result:

THEOREM 2.2. *If $(N_n)_{n\geq 1}$ is a sequence of null sets, then their union*

$$N = \bigcup_{n=1}^{\infty} N_n$$

is also null.

PROOF. We fix any $\varepsilon > 0$, and our goal is to cover the set N by countably many intervals with total length less than ε. The proof goes in three steps.

Step 1. We carefully cover each N_n by intervals.

Since N_1 is null, there exist intervals I_k^1, $k \geq 1$, such that

$$\sum_{k=1}^{\infty} \ell(I_k^1) < \frac{\varepsilon}{2}, \quad N_1 \subseteq \bigcup_{k=1}^{\infty} I_k^1.$$

For N_2, we find a system of intervals I_k^2, $k \geq 1$, with

$$\sum_{k=1}^{\infty} \ell(I_k^2) < \frac{\varepsilon}{2^2}, \quad N_2 \subseteq \bigcup_{k=1}^{\infty} I_k^2.$$

You can see a cunning plan of making the total lengths smaller at each step at a geometric rate. In general, we cover N_n with intervals I_k^n, $k \geq 1$, whose total length is less than $\frac{\varepsilon}{2^n}$:

$$\sum_{k=1}^{\infty} \ell(I_k^n) < \frac{\varepsilon}{2^n}, \quad N_n \subseteq \bigcup_{k=1}^{\infty} I_k^n.$$

Step 2. The intervals I_k^n form a sequence.

We arrange the countable family of intervals $\{I_k^n\}_{k\geq 1, n\geq 1}$ into a sequence J_j, $j \geq 1$. For instance, we put $J_1 = I_1^1$, $J_2 = I_2^1$, $J_3 = I_1^2$, $J_4 = I_1^3$ etc., so that none of the I_k^n are skipped. The union of the new system of intervals is the same as the union of the old one and so

$$N = \bigcup_{n=1}^{\infty} N_n \subseteq \bigcup_{n=1}^{\infty} \bigcup_{k=1}^{\infty} I_k^n = \bigcup_{j=1}^{\infty} J_j.$$

Step 3. Compute the total length of J_j.

This is tricky because we have a series of numbers with two indices:

$$\sum_{j=1}^{\infty} \ell(J_j) = \sum_{n=1,k=1}^{\infty} \ell(I_k^n).$$

Now we wish to write this as a series of numbers each being the sum of a series. We can rearrange the double sum because the components are non-negative (see Theorem 0.0.2 in [**16**]), and so

$$\sum_{n=1,k=1}^{\infty} \ell(I_k^n) = \sum_{n=1}^{\infty} \left(\sum_{k=1}^{\infty} \ell(I_k^n) \right) < \sum_{n=1}^{\infty} \frac{\varepsilon}{2^n} = \varepsilon,$$

which completes the proof.

□

It seems that null sets are closely related to countable sets, and this is no surprise as any proper interval is uncountable. So any countable subset is quite 'sparse' when compared with an interval, hence makes no real contribution to its 'length'.

However, uncountable sets *can* be null, provided their points are sufficiently 'sparsely distributed', as the following famous example, due to Cantor, shows:

1. Start with the interval $[0,1]$, remove the 'middle third', that is the interval $(\frac{1}{3}, \frac{2}{3})$, obtaining the set C_1, which consists of the two intervals $[0, \frac{1}{3}]$ and $[\frac{2}{3}, 1]$.
2. Next remove the middle third of each of these two intervals, leaving C_2, consisting of four intervals, each of length $\frac{1}{3^2}$, etc. (See Figure 2.1.)
3. At the nth stage, we have a set C_n, consisting of 2^n disjoint closed intervals, each of length $\frac{1}{3^n}$. Thus, the total length of C_n is $(\frac{2}{3})^n$.

FIGURE 2.1. Cantor set construction (C_3).

We call

$$C = \bigcap_{n=1}^{\infty} C_n$$

the *Cantor set*.

Now we show that C is null as promised. Given any $\varepsilon > 0$, choose n so large that $(\frac{2}{3})^n < \varepsilon$. Since $C \subseteq C_n$, and C_n consists of a finite sequence of intervals of total length less than ε, we see that C is a null set.

All that remains is to check that C is an uncountable set. This is similar to the proof of the uncountability of $\mathbb{R}$ since every x in C can be expressed in ternary form:

$$x = \sum_{k=1}^{\infty} \frac{a_k}{3^k} = 0.a_1 a_2 \dots$$

where a_k equal to 0 or 2 for all $k \in \mathbb{N}$.

REMARK. $\star$ *Why* is the Cantor set null, even though it is uncountable? Clearly it is the distribution of its points, the fact that it is 'spread out' all over $[0,1]$, which causes the trouble. This makes it the source of many examples which show that intuitively 'obvious' things are not always true! For example, we can use the Cantor set to define a function, called Cantor-Lebesgue function, with very odd properties:

If $x \in [0,1]$ has ternary expansion (a_n), i.e. $x = 0.a_1 a_2 \dots$ with $a_n = 0, 1$ or 2, define N as the first index n for which $a_n = 1$, and set $N = \infty$ if none of the a_n are 1 (i.e. when $x \in C$). Now set $b_n = \frac{a_n}{2}$ for $n < N$ and $b_N = 1$, and let $F(x) = \sum_{n=1}^{N} \frac{b_n}{2^n}$ for each $x \in [0,1]$. Clearly, this function is monotone increasing and has $F(0) = 0$, $F(1) = 1$. It is constant on the middle thirds (i.e. the complement of C), so all its increase occurs on the Cantor set. Since we have shown that C is a null set, F 'grows' from 0 to 1 entirely on a 'negligible' set. We can also show that F is continuous, see Proposition 2.1 in [**7**]. See Figure 2.2 for the graph.

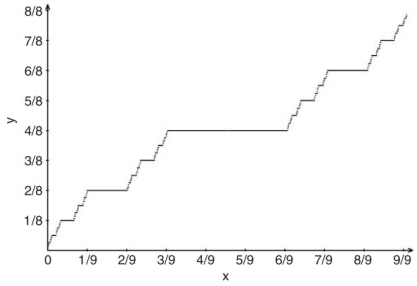


FIGURE 2.2. The graph of the Cantor-Lebesgue function F [**7**].

2.2. Outer measure

The simple concept of null sets provides the key to our idea of length, since it tells us what we can 'ignore'. A quite general notion of 'length' is now provided by:

DEFINITION 2.3. *The (Lebesgue) outer measure of any set $A \subseteq \mathbb{R}$ is given by*

$$m^*(A) = \inf Z_A$$

where

$$Z_A = \left\{ \sum_{n=1}^{\infty} \ell(I_n) : I_n \text{ are intervals, } A \subseteq \bigcup_{n=1}^{\infty} I_n \right\}.$$

We say the $(I_n)_{n\geq 1}$ *cover* the set A. So the outer measure is the infimum of lengths of all possible covers of A. (Note again that some of the I_n may be empty.)

Clearly, $m^*(A) \geq 0$ for any $A \subseteq \mathbb{R}$. For some sets A, the series $\sum_{n=1}^{\infty} \ell(I_n)$ may diverge for any covering of A. So $m^*(A)$ may equal to ∞. Since we wish to be able to add the outer measures of various sets, we have to adopt a convention to deal with infinity. An obvious choice is $a + \infty = \infty$, $\infty + \infty = \infty$ and a less obvious but quite practical assumption is $0 \times \infty = 0$.

The set Z_A is bounded from below by 0, so the infimum always exists. If $r \in Z_A$, then $[r, \infty] \subseteq Z_A$ (clearly, we may expand the first interval of any cover to increase the total length by any number). This shows that Z_A is either $\{\infty\}$ or the interval $(x, \infty]$ or $[x, \infty]$ for some real number x. For the later case, the infimum of Z_A is x.

First we show that the concept of null set is consistent with that of outer measure:

THEOREM 2.4. *$A \subseteq \mathbb{R}$ is a null set if and only if $m^*(A) = 0$.*

PROOF. Suppose that A is a null set. We wish to show that $\inf Z_A = 0$. To this end, we show that for any $\varepsilon > 0$ we can find an element $z \in Z_A$ such that $z < \varepsilon$.

By the definition of null set we can find a sequence (I_n) of intervals covering A with $\sum_{n=1}^{\infty} \ell(I_n) < \varepsilon$ and so $\sum_{n=1}^{\infty} \ell(I_n)$ is the required element z of Z_A.

Conversely, if $A \subseteq \mathbb{R}$ has $m^*(A) = 0$, then by the definition of $\inf$, given any $\varepsilon > 0$, there is $z \in Z_A$, $z < \varepsilon$. But a member of Z_A is the total length of some covering of A. That is, there is a covering (I_n) of A with total length less than ε, so A is null. □

This combines our general outer measure with the special case of 'zero measure'. Note that $m^*(\emptyset) = 0$, $m^*(\{x\}) = 0$ for any $x \in \mathbb{R}$, and $m^*(\mathbb{Q}) = 0$ (and in fact, for any countable X, $m^*(X) = 0$).

Next we observe that m^* is monotone: the bigger the set, the greater its outer measure.

PROPOSITION 2.5. *If $A \subseteq B$, then $m^*(A) \leq m^*(B)$.*

PROOF. If the intervals I_n cover B, then they also cover A: $A \subseteq B \subseteq \cup_{n=1}^{\infty} I_n$, hence $Z_B \subseteq Z_A$. The infimum of a larger set cannot be greater than the infimum of a smaller set, hence the result. □

The second step is to relate outer measure to the length of an interval. This innocent result contains the crux of the theory, since it shows that the formal definition of m^*, which is applicable to *all* subsets of $\mathbb{R}$, coincides with the intuitive idea for intervals. In the proof, we will use the famous Heine-Borel theorem, see Theorem A.5.

THEOREM 2.6. *The outer measure of an interval equals its length.*

PROOF. If I is unbounded, then we can use the corresponding results for bounded intervals to show it. Indeed, suppose $I = (a, \infty)$ for example, then

$$n = \ell((a, a+n)) = m^*((a, a+n)) \leq m^*((a, \infty)) = m^*(I),$$

where we have used Proposition 2.5 in the inequality. Letting n go to infinity, we have $m^*(I) = \infty = \ell(I)$ as desired. Now, we restrict ourselves to bounded intervals.

Step 1. $m^*(I) \leq \ell(I)$.

We claim that $\ell(I) \in Z_I$. Take the following sequence of intervals: $I_1 = I$, $I_n = [0, 0]$ for $n \geq 2$. This sequence covers the set I, and the total length is equal to the length of I hence $\ell(I) \in Z_I$. This is sufficient since the infimum of Z_I cannot exceed any of its elements.

Step 2. $\ell(I) \leq m^*(I)$.

(1) $I = [a, b]$. We shall show that for any $\varepsilon > 0$

$$\ell([a, b]) \leq m^*([a, b]) + \varepsilon. \tag{2.1}$$

This is sufficient since we may obtain the required inequality passing to the limit, $\varepsilon \to 0^+$.

So, we take an arbitrary $\varepsilon > 0$. By the definition of outer measure, we can find a sequence of intervals I_n covering $[a, b]$ such that

$$\sum_{n=1}^{\infty} \ell(I_n) \leq m^*([a, b]) + \frac{\varepsilon}{2}. \tag{2.2}$$

We shall slightly increase each of the intervals to an open one. Let the endpoints of I_n be a_n, b_n, and we take

$$J_n = \left(a_n - \frac{\varepsilon}{2^{n+2}}, b_n + \frac{\varepsilon}{2^{n+2}}\right).$$

It is clear that

$$\ell(I_n) = \ell(J_n) - \frac{\varepsilon}{2^{n+1}}$$

so that

$$\sum_{n=1}^{\infty} \ell(I_n) = \sum_{n=1}^{\infty} \ell(J_n) - \frac{\varepsilon}{2}.$$

We insert this in (2.2) and we have

$$\sum_{n=1}^{\infty} \ell(J_n) \leq m^*([a, b]) + \varepsilon. \tag{2.3}$$

The new sequence of intervals covers $[a,b]$. So, by the Heine-Borel theorem (see Theorem A.5), we can choose a finite number of J_n to cover $[a,b]$ (the set $[a,b]$ is compact in $\mathbb{R}$), i.e. there exists $m \in \mathbb{N}$ such that

$$[a,b] \subseteq \bigcup_{k=1}^{m} J_{n_k}.$$

It follows that

$$\ell([a,b]) \leq \sum_{k=1}^{m} \ell(J_{n_k}). \tag{2.4}$$

Now it is sufficient to put (2.3) and (2.4) together in order to deduce (2.1).

(2) $I = (a,b)$. As before, it is sufficient to show (2.1) with $[a,b]$ replaced by (a,b). Let us fix any sufficiently small $\varepsilon > 0$, and we deduce

$$\begin{aligned} \ell((a,b)) &= \ell\left(\left[a+\frac{\varepsilon}{2}, b-\frac{\varepsilon}{2}\right]\right) + \varepsilon \\ &\leq m^*\left(\left[a+\frac{\varepsilon}{2}, b-\frac{\varepsilon}{2}\right]\right) + \varepsilon \quad \text{(by (1))} \\ &\leq m^*((a,b)) + \varepsilon \qquad\qquad \text{(by Proposition 2.5)}. \end{aligned}$$

(3) $I = [a,b)$ or $I = (a,b]$. We have

$$\begin{aligned} \ell(I) = \ell((a,b)) &\leq m^*((a,b)) \quad \text{(by (2))} \\ &\leq m^*(I) \qquad \text{(by Proposition 2.5)} \end{aligned}$$

which completes the proof.

□

The next theorem gives us an important technical tool which will be used in many proofs.

THEOREM 2.7. *Outer measure is countably subadditive, i.e. for any countable collection of sets* $\{E_n\}$

$$m^*\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} m^*(E_n).$$

(Note that both sides may be infinite here.)

PROOF. If the right-hand side is infinite, then the inequality is of course true. So, suppose that $\sum_{n=1}^{\infty} m^*(E_n) < \infty$. For each given $\varepsilon > 0$ and $n \geq 1$, we can find a covering sequence $(I_k^n)_{k\geq 1}$ of E_n with

$$\sum_{k=1}^{\infty} \ell(I_k^n) \leq m^*(E_n) + \frac{\varepsilon}{2^n}.$$

The iterated series converges:

$$\sum_{n=1}^{\infty}\left(\sum_{k=1}^{\infty} \ell(I_k^n)\right) \leq \sum_{n=1}^{\infty} m^*(E_n) + \varepsilon < \infty$$

and since all its terms are non-negative,

$$\sum_{n=1}^{\infty}\left(\sum_{k=1}^{\infty} \ell(I_k^n)\right) = \sum_{n,k=1}^{\infty} \ell(I_k^n),$$

see Theorem 0.0.2 in [**16**]. The system of intervals $(I_k^n)_{k,n\geq 1}$ covers $\cup_{n=1}^{\infty} E_n$, hence

$$m^*\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n,k=1}^{\infty} \ell(I_k^n) \leq \sum_{n=1}^{\infty} m^*(E_n) + \varepsilon.$$

To complete the proof, we let ε decrease to zero. □

A similar result is of course true for a finite family $(E_n)_{n=1}^j$:

$$m^*\left(\bigcup_{n=1}^{j} E_n\right) \leq \sum_{n=1}^{j} m^*(E_n).$$

It is a corollary to Theorem 2.7 with $E_k = \emptyset$ for $k > j$.

We conclude this section with a simple and intuitive property of outer measure. Note that the length of an interval does not change if we shift it along the real line: $\ell([a,b]) = \ell([a+t, b+t]) = b - a$ for example. Since the outer measure is defined in terms of the lengths of intervals, it is natural to expect it to share this property. For $A \subseteq \mathbb{R}$ and $t \in \mathbb{R}$, we set $A + t = \{a + t : a \in A\}$.

PROPOSITION 2.8. *Outer measure is translation invariant, i.e.*

$$m^*(A) = m^*(A + t)$$

for each $A \subseteq \mathbb{R}$ and $t \in \mathbb{R}$.

PROOF. If a system I_n of intervals covers A, then the intervals $I_n + t$ cover $A + t$. Conversely, if J_n cover $A + t$ then $J_n - t$ cover A. Moreover, the total length of a family of intervals does not change when we shift each by a number. So, we have a one-to-one correspondence between the interval coverings of A and $A + t$, and this correspondence preserves the total length of the covering. This implies that the sets Z_A and Z_{A+t} are the same so their infima are equal. □

2.3. Lebesgue measurable sets and Lebesgue measure

With outer measure, countable subadditivity (see Theorem 2.7) is as far as we can get. We wish, however, to ensure that if sets (E_n) are pairwise disjoint (i.e. $E_i \cap E_j = \emptyset$ if $i \neq j$), then the inequality in Theorem 2.7 becomes an equality. We call this *countable additivity*. It turns out that this will not in general be true for outer measure, although examples where it fails are quite difficult to construct (see Section 2.6 below or Appendix of [**5**]). In order to define the 'good' sets which have this property, we introduce the following definition.

DEFINITION 2.9. *A set $E \subseteq \mathbb{R}$ is (Lebesgue) measurable if for every set $A \subseteq \mathbb{R}$, we have*

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c) \tag{2.5}$$

where $E^c = \mathbb{R} \setminus E$, and we write $E \in \mathcal{M}$.

We obviously have $A = (A \cap E) \cup (A \cap E^c)$. Hence, by Theorem 2.7, we have

$$m^*(A) \leq m^*(A \cap E) + m^*(A \cap E^c)$$

for any A and E. So our future task of verifying (2.5) has simplified: $E \in \mathcal{M}$ if and only if the following inequality holds

$$m^*(A) \geq m^*(A \cap E) + m^*(A \cap E^c) \text{ for all } A \subseteq \mathbb{R}. \tag{2.6}$$

Now we give some examples of measurable sets.

THEOREM 2.10. *(i) Any null set is measurable.*
(ii) Any interval is measurable.

PROOF. (i) If N is a null set, then (see Theorem 2.4) $m^*(N) = 0$. So for any $A \subseteq \mathbb{R}$, we have

$$\begin{aligned} m^*(A \cap N) &\leq m^*(N) = 0 \quad \text{since } A \cap N \subseteq N \\ m^*(A \cap N^c) &\leq m^*(A) \quad \text{since } A \cap N^c \subseteq A. \end{aligned}$$

Adding together gives (2.6).

(ii) Let $E = I$ be an interval. Suppose, for example, that $I = [a, b]$. Take any $A \subseteq \mathbb{R}$ and $\varepsilon > 0$. Find a covering of A with

$$m^*(A) \leq \sum_{n=1}^{\infty} \ell(I_n) \leq m^*(A) + \varepsilon.$$

Clearly the intervals $I'_n = I_n \cap [a, b]$ cover $A \cap [a, b]$, hence

$$m^*(A \cap [a, b]) \leq \sum_{n=1}^{\infty} \ell(I'_n).$$

The intervals $I''_n = I_n \cap (-\infty, a)$, $I'''_n = I_n \cap (b, \infty)$ cover $A \cap [a, b]^c$ so

$$m^*(A \cap [a, b]^c) \leq \sum_{n=1}^{\infty} \ell(I''_n) + \sum_{n=1}^{\infty} \ell(I'''_n).$$

Putting the above three inequalities together, we deduce

$$\begin{aligned} m^*(A \cap [a, b]) + m^*(A \cap [a, b]^c) &\leq \sum_{n=1}^{\infty} [\ell(I'_n) + \ell(I''_n) + \ell(I'''_n)] \\ &= \sum_{n=1}^{\infty} \ell(I_n) \leq m^*(A) + \varepsilon. \end{aligned}$$

Letting ε decrease to zero, we obtain (2.6).

If I is unbounded, eg. $I = [a, \infty)$, then the proof is even simpler since it is sufficient to consider $I'_n = I_n \cap [a, \infty)$ and $I''_n = I_n \cap (-\infty, a)$.

□

The fundamental properties of the class $\mathcal{M}$ of all Lebesgue-measurable subsets of $\mathbb{R}$ can now be proved. They fall into two categories: first we show that certain set operations on sets in $\mathcal{M}$ again produce sets in $\mathcal{M}$ (these are what we call 'closure properties') and second we prove that for sets in $\mathcal{M}$ the outer measure m^* has the property of countable additivity announced above.

THEOREM 2.11. *(i) $\mathbb{R} \in \mathcal{M}$;*
(ii) If $E \in \mathcal{M}$, then $E^c \in \mathcal{M}$;
(iii) If $E_n \in \mathcal{M}$ for all $n \in \mathbb{N}$, then $\cup_{n=1}^{\infty} E_n \in \mathcal{M}$.
Moreover, if $E_n \in \mathcal{M}$, $n \in \mathbb{N}$, and $E_j \cap E_k = \emptyset$ for $j \neq k$, then

$$m^* \left(\bigcup_{n=1}^{\infty} E_n \right) = \sum_{n=1}^{\infty} m^*(E_n). \tag{2.7}$$

REMARK. This result is the most important theorem in this chapter and provides the basis for all that follows. Conditions (i)-(iii) mean that $\mathcal{M}$ is a $\sigma-$algebra[1]. In other words, we say that a family of sets is a $\sigma-$algebra if it contains the base set and is closed under complements and countable unions. A $[0,\infty]-$valued function defined on a $\sigma-$algebra is called a *measure* if it satisfies $m^*(\emptyset) = 0$ and (2.7) for pairwise disjoint sets, i.e. it is *countably additive*.

An alternative approach to measure theory is to begin with the above properties as *axioms*, i.e. to call a triple $(\Omega, \mathcal{F}, \mu)$ a *measure space* if Ω is an abstractly given set, $\mathcal{F}$ is a $\sigma-$algebra of subsets of Ω, and $\mu : \mathcal{F} \to [0,\infty]$ is a function satisfying $\mu(\emptyset) = 0$ and (2.7) (with μ instead of m^*). The task of defining Lebesgue measure on $\mathbb{R}$ then becomes that of verifying, with $\mathcal{M}$ and $m = m^*$ on $\mathcal{M}$ defined as above, that the triple $(\mathbb{R}, \mathcal{M}, m)$ satisfies these axioms, i.e. becomes a measure space.

PROOF OF THEOREM 2.11. (i) Let $A \subseteq \mathbb{R}$. Note that $A \cap \mathbb{R} = A$, and $A \cap \mathbb{R}^c = A \cap \emptyset = \emptyset$. Now (2.5) reads $m^*(A) = m^*(A) + m^*(\emptyset)$ and is of course true since $m^*(\emptyset) = 0$.

(ii) Suppose $E \in \mathcal{M}$ and take any $A \subseteq \mathbb{R}$. We have to show (2.5) for E^c, i.e.

$$m^*(A) = m^*(A \cap E^c) + m^*(A \cap (E^c)^c).$$

Since $(E^c)^c = E$, this reduces to the condition for E which holds by hypothesis.

We split the proof of (iii) and the last assertion into several steps.

Step 1. We will show if pairwise disjoint E_k, $k \in \mathbb{N}$, are in $\mathcal{M}$, then their union is in $\mathcal{M}$ and (2.7) holds.

Let $A \subseteq \mathbb{R}$. As $E_1 \in \mathcal{M}$, we have

$$m^*(A) = m^*(A \cap E_1) + m^*(A \cap E_1^c). \tag{2.8}$$

Now, apply (2.5) for E_2 with $A \cap E_1^c$ in place of A:

$$\begin{aligned} m^*(A \cap E_1^c) &= m^*((A \cap E_1^c) \cap E_2) + m^*((A \cap E_1^c) \cap E_2^c) \\ &= m^*(A \cap (E_1^c \cap E_2)) + m^*(A \cap (E_1^c \cap E_2^c)) \\ &= m^*(A \cap E_2) + m^*(A \cap (E_1 \cup E_2)^c) \end{aligned}$$

where we have used the fact E_1 and E_2 are disjoint, and de Morgan's law $E_1^c \cap E_2^c = (E_1 \cup E_2)^c$ in the last equality. Substituting this into (2.8), we get

$$m^*(A) = m^*(A \cap E_1) + m^*(A \cap E_2) + m^*(A \cap (E_1 \cup E_2)^c).$$

Continue this process, and after n steps we expect

$$m^*(A) = \sum_{k=1}^{n} m^*(A \cap E_k) + m^*\left(A \cap \left(\bigcup_{k=1}^{n} E_k\right)^c\right). \tag{2.9}$$

Let us demonstrate this by induction. The case $n = 1$ is shown in (2.8). Suppose that

$$m^*(A) = \sum_{k=1}^{n-1} m^*(A \cap E_k) + m^*\left(A \cap \left(\bigcup_{k=1}^{n-1} E_k\right)^c\right). \tag{2.10}$$

[1] In some other books, it is called $\sigma-$field.

Since $E_n \in \mathcal{M}$, we may apply (2.5) for E_n with $A \cap (\cup_{k=1}^{n-1} E_k)^c$ in place of A:

$$m^*\left(A \cap \left(\bigcup_{k=1}^{n-1} E_k\right)^c\right) = m^*\left(A \cap \left(\bigcup_{k=1}^{n-1} E_k\right)^c \cap E_n\right) + m^*\left(A \cap \left(\bigcup_{k=1}^{n-1} E_k\right)^c \cap E_n^c\right). \tag{2.11}$$

Now, we make the following observations:

$$\left(\bigcup_{k=1}^{n-1} E_k\right)^c \cap E_n = E_n \quad (E_i \text{ are pairwise disjoint}),$$

$$\left(\bigcup_{k=1}^{n-1} E_k\right)^c \cap E_n^c = \left(\bigcup_{k=1}^{n} E_k\right)^c \quad (\text{by de Morgan's law}).$$

Inserting these into (2.11), we find

$$m^*\left(A \cap \left(\bigcup_{k=1}^{n-1} E_k\right)^c\right) = m^*(A \cap E_n) + m^*\left(A \cap \left(\bigcup_{k=1}^{n} E_k\right)^c\right).$$

Inserting this into the induction hypothesis (2.10) yields

$$m^*(A) = \sum_{k=1}^{n-1} m^*(A \cap E_k) + m^*(A \cap E_n) + m^*\left(A \cap \left(\bigcup_{k=1}^{n} E_k\right)^c\right)$$

as required to complete the induction step. Thus (2.9) holds for all $n \in \mathbb{N}$ by induction.

Since

$$\left(\bigcup_{k=1}^{n} E_k\right)^c \supseteq \left(\bigcup_{k=1}^{\infty} E_k\right)^c,$$

from (2.9) by monotonicity (Proposition 2.5) we get

$$m^*(A) \geq \sum_{k=1}^{n} m^*(A \cap E_k) + m^*\left(A \cap \left(\bigcup_{k=1}^{\infty} E_k\right)^c\right).$$

The inequality remains true after we pass to the limit $n \to \infty$:

$$m^*(A) \geq \sum_{k=1}^{\infty} m^*(A \cap E_k) + m^*\left(A \cap \left(\bigcup_{k=1}^{\infty} E_k\right)^c\right). \tag{2.12}$$

By countable subadditivity (Theorem 2.7), we have

$$\sum_{k=1}^{\infty} m^*(A \cap E_k) \geq m^*\left(A \cap \left(\bigcup_{k=1}^{\infty} E_k\right)\right)$$

and so

$$m^*(A) \geq m^*\left(A \cap \left(\bigcup_{k=1}^{\infty} E_k\right)\right) + m^*\left(A \cap \left(\bigcup_{k=1}^{\infty} E_k\right)^c\right) \tag{2.13}$$

as required. This implies $\cup_{k=1}^{\infty} E_k \in \mathcal{M}$ and the two sides of (2.13) are equal. The right hand side of (2.12) is squeezed between the left and right of (2.13) which yields

$$m^*(A) = \sum_{k=1}^{\infty} m^*(A \cap E_k) + m^*\left(A \cap \left(\bigcup_{k=1}^{\infty} E_k\right)^c\right).$$

The equality here is a consequence of the assumption that E_k are pairwise disjoint. It holds for any set A so we may insert $A = \cup_{j=1}^{\infty} E_j$. The last term on the right is zero because we have $m^*(\emptyset) = 0$. As $(\cup_{j=1}^{\infty} E_j) \cap E_n = E_n$, (2.7) follows.

Step 2. We will show if $E_1, E_2 \in \mathcal{M}$, then $E_1 \cup E_2 \in \mathcal{M}$ (not necessarily disjoint). The proof is very similar to the beginning part of Step 1. We will use (2.8) and let us apply (2.5) for E_2 with $A \cap E_1^c$ in place of A:

$$\begin{aligned} m^*(A \cap E_1^c) &= m^*(A \cap E_1^c \cap E_2) + m^*(A \cap E_1^c \cap E_2^c) \\ &= m^*(A \cap E_1^c \cap E_2) + m^*(A \cap (E_1 \cup E_2)^c), \end{aligned} \tag{2.14}$$

where we have used de Morgan's law in the last equality. By subadditivity of m^*, we have

$$m^*(A \cap E_1) + m^*(A \cap E_1^c \cap E_2) \geq m^*(A \cap (E_1 \cup E_2)).$$

Inserting this and (2.14) into (2.8) yields

$$m^*(A) \geq m^*(A \cap (E_1 \cup E_2)) + m^*(A \cap (E_1 \cup E_2)^c)$$

as required.

Step 3. We claim if $E_k \in \mathcal{M}$, $k \in \mathbb{N}$, then $\cup_{k=1}^{n} E_k \in \mathcal{M}$ (not necessarily disjoint). We argue by induction. There is nothing to prove for $n = 1$. Suppose the claim is true for $n - 1$. Then

$$E_1 \cup \cdots \cup E_n = (E_1 \cup \cdots \cup E_{n-1}) \cup E_n$$

so that the results follows from Step 2.

Step 4. We claim if $E_1, E_2 \in \mathcal{M}$, then $E_1 \cap E_2 \in \mathcal{M}$. Observe that $E_1^c, E_2^c \in \mathcal{M}$ by (ii), $E_1^c \cup E_2^c$ by Step 2, $(E_1^c \cup E_2^c)^c \in \mathcal{M}$ by (ii) again, but by de Morgan's law the last set is equal to $E_1 \cap E_2$.

Step 5. We claim that if $E_1, E_2, \ldots$ are in $\mathcal{M}$, then so is $\cup_{k=1}^{\infty} E_k$. Let $E_k \in \mathcal{M}$, $k \in \mathbb{N}$. We define an auxiliary sequence of pairwise disjoint sets F_k with the same union as E_k:

$$\begin{aligned} F_1 &= E_1, \\ F_2 &= E_2 \setminus E_1 = E_2 \cap E_1^c, \\ F_3 &= E_3 \setminus (E_1 \cup E_2) = E_3 \cap (E_1 \cup E_2)^c, \\ &\cdots \\ F_k &= E_k \setminus (E_1 \cup \cdots \cup E_{k-1}) = E_k \cap (E_1 \cup \cdots \cup E_{k-1})^c. \end{aligned}$$

By Steps 3 and 4, we know that all F_k are in $\mathcal{M}$. By the construction, they are pairwise disjoint so by Step 1 their union is in $\mathcal{M}$. We shall show that

$$\bigcup_{k=1}^{\infty} F_k = \bigcup_{k=1}^{\infty} E_k.$$

This will complete the proof since the latter is now in $\mathcal{M}$. The inclusion

$$\bigcup_{k=1}^{\infty} F_k \subseteq \bigcup_{k=1}^{\infty} E_k$$

is obvious since for each k, $F_k \subseteq E_k$ by definition. For the inverse, let $a \in \cup_{k=1}^{\infty} E_k$. Put $S = \{n \in \mathbb{N} : a \in E_n\}$ which is non-empty since a belongs to the union. Let $n_0 = \min S \in S$. If $n_0 = 1$, then $a \in E_1 = F_1$. Suppose $n_0 > 1$. So

$a \in E_{n_0}$ and, by the definition of n_0, $a \notin E_1, \ldots, a \notin E_{n_0-1}$. By the definition of F_{n_0}, this means that $a \in F_{n_0}$ so a is in $\cup_{k=1}^{\infty} F_k$.

□

Using de Morgan's law, you should easily verify an additional property of $\mathcal{M}$.

COROLLARY 2.12. *If $E_k \in \mathcal{M}$, $k \in \mathbb{N}$, then*

$$E = \bigcap_{k=1}^{\infty} E_k \in \mathcal{M}.$$

We can therefore summarize the properties of the family $\mathcal{M}$ of Lebesgue measurable sets as follows:

$\mathcal{M}$ is closed under countable unions, countable intersections, and complements. It contains intervals and all null sets.

DEFINITION 2.13. *We shall write $m(E)$ instead of $m^*(E)$ for any E in $\mathcal{M}$ and call $m(E)$ the Lebesgue measure of the set E.*

Finally, we summarise what we have shown as follows:

Lebesgue measure $m : \mathcal{M} \to [0, \infty]$ is a countably additive set function defined on the $\sigma-$algebra $\mathcal{M}$ of measurable sets. Lebesgue measure of an interval is equal to its length. Lebesgue measure of a null set is zero.

2.4. Basic properties of Lebesgue measure

Since Lebesgue measure is nothing else than the outer measure restricted to a special class of sets, some properties of the outer measure are automatically inherited by Lebesgue measure. Let us start with the finite additivity of Lebesgue measure.

As $\emptyset \in \mathcal{M}$, we can take $E_i = \emptyset$ for all $i > n$ in (2.7) to conclude that Lebesgue measure is additive: if $E_i \in \mathcal{M}$ are pairwise disjoint, then

$$m\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} m(E_i).$$

Some basic properties of Lebesgue measure can be found in the following:

PROPOSITION 2.14. *Suppose that $A, B \in \mathcal{M}$.*

(i) If $A \subseteq B$, then $m(A) \leq m(B)$.
(ii) If $A \subseteq B$ and $m(A)$ is finite, then $m(B \setminus A) = m(B) - m(A)$.
(iii) m is translation invariant. More precisely, $A+t \in \mathcal{M}$ for all $t \in \mathbb{R}$, and $m(A+t) = m(A)$.

PROOF. (i) Proposition 2.5 tells us the outer measure is monotone. Since m is just the restriction of m^* to $\mathcal{M}$, then the same is true for m.

(ii) We write B as a disjoint union $B = A \cup (B \setminus A)$. Then, by additivity of m, we have $m(B) = m(A) + m(B \setminus A)$. Subtracting $m(A)$ (here it is important that $m(A)$ is finite), we get the result.

(iii) Let $t \in \mathbb{R}$, and $E \subseteq \mathbb{R}$. Since $A \in \mathcal{M}$, we have

$$\begin{aligned} m^*(E) = m^*(E-t) &= m^*((E-t) \cap A) + m^*((E-t) \cap A^c) \\ &= m^*(E \cap (A+t)) + m^*(E \cap (A+t)^c) \end{aligned}$$

where we have used the translation invariance of m^* (see Proposition 2.8) in the first and third equalities. This implies $A + t \in \mathcal{M}$, and $m(A + t) = m(A)$ follows from the translation invariance of m^* again.

□

Recalling that the *symmetric difference* $A\triangle B$ of two sets is defined by $A\triangle B = (A \setminus B) \cup (B \setminus A)$, the following result is easy to check:

PROPOSITION 2.15. *If $A \in \mathcal{M}$ and $m^*(A\triangle B) = 0$, then $B \in \mathcal{M}$ and $m(A) = m(B)$.*

PROOF. The set $A\triangle B$ is null, hence its subsets $A \setminus B$ and $B \setminus A$ are also null, which implies they are measurable. It follows that $A \cap B = A \setminus (A \setminus B)$ is measurable, and also $B = (A \cap B) \cup (B \setminus A) \in \mathcal{M}$. Now, $m(B) = m(A \cap B) + m(B \setminus A)$ as the sets on the right are disjoint. Since $m(B \setminus A) = 0 = m(A \setminus B)$, we deduce $m(B) = m(A \cap B) = m(A \cap B) + m(A \setminus B) = m(A)$. □

As we noted in Section A.3, every open set in $\mathbb{R}$ can be expressed as the union of a countable number of open intervals. This ensures that open sets in $\mathbb{R}$ are Lebesgue measurable, since $\mathcal{M}$ contains intervals and is closed under countable unions. Moreover, we can approximate the Lebesgue measure of any $A \in \mathcal{M}$ from above by the measures of a sequence of open sets containing A, as shown in the following result:

THEOREM 2.16 (Outer regularity of Lebesgue measure). *(i) For any $\varepsilon > 0$ and $A \subseteq \mathbb{R}$, we can find an open set O such that*

$$A \subseteq O \quad \textit{and} \quad m(O) \le m^*(A) + \varepsilon.$$

Consequently, for any $E \in \mathcal{M}$, we can find an open set O containing E such that $m(O \setminus E) < \varepsilon$.

(ii) For any $A \subseteq \mathbb{R}$, we can find a sequence of open sets O_n such that

$$A \subseteq \bigcap_{n=1}^{\infty} O_n \quad \textit{and} \quad m\left(\bigcap_{n=1}^{\infty} O_n\right) = m^*(A).$$

Moreover, if $A \in \mathcal{M}$, then there exists a sequence of open sets O_n such that

$$A \subseteq B, \quad m\,(B) = m(A) \quad \textit{and} \quad m(B \setminus A) = 0.$$

where $B = \cap_{n=1}^{\infty} O_n$.

PROOF. (i) By definition of $m^*(A)$, we can find a sequence (I_n) of intervals with $A \subseteq \cup_{n=1}^{\infty} I_n$ and $\sum_{n=1}^{\infty} \ell(I_n) \le m^*(A) + \frac{\varepsilon}{2}$. Each I_n is contained in an open interval whose length is very close to that of I_n; if the left and right endpoints of I_n are a_n and b_n respectively, let $J_n = (a_n - \frac{\varepsilon}{2^{n+2}}, b_n + \frac{\varepsilon}{2^{n+2}})$. Set $O = \cup_{n=1}^{\infty} J_n$, which is open. Then, $A \subseteq O$ and

$$m(O) \le \sum_{n=1}^{\infty} \ell(J_n) \le \sum_{n=1}^{\infty} \ell(I_n) + \frac{\varepsilon}{2} \le m^*(A) + \varepsilon.$$

When $m(E) < \infty$, the final assertion follows from Proposition 2.14 (ii), since then $m(O \setminus E) = m(O) - m(E) \le \varepsilon$. When $m(E) = \infty$, we first write $\mathbb{R}$ as a countable union of finite intervals: $\mathbb{R} = \cup_{n=1}^{\infty}(-n, n)$. Now $E_n = E \cap (-n, n)$ has finite

measure, so we can find an open $O_n \supseteq E_n$ with $m(O_n \setminus E_n) \leq \frac{\varepsilon}{2^n}$. The set $O = \cup_{n=1}^{\infty} O_n$ is open and contains E. Now

$$O \setminus E = \left(\bigcup_{n=1}^{\infty} O_n \right) \setminus \left(\bigcup_{n=1}^{\infty} E_n \right) \subseteq \bigcup_{n=1}^{\infty} (O_n \setminus E_n)$$

so that $m(O \setminus E) \leq \sum_{n=1}^{\infty} m(O_n \setminus E_n) \leq \varepsilon$ as required.

(ii) Let O_n be the open set obtained in (i) for $\varepsilon = \frac{1}{n}$, $n \in \mathbb{N}$. Setting $E = \cap_{n=1}^{\infty} O_n$, we find $A \subseteq E$ and $m(E) \leq m(O_n) \leq m^*(A) + \frac{1}{n}$ for each $n \in \mathbb{N}$. Letting $n \to \infty$, we have $m^*(A) \leq m(E) \leq m^*(A)$ as desired. For the last assertion, we can use the last assertion of (i). Indeed, for each $n \in \mathbb{N}$, we can find open $O_n \supseteq A$ such that $m(O_n \setminus A) < \frac{1}{n}$. Setting $B = \cap_{n=1}^{\infty} O_n$, we have $A \subseteq B$ and

$$0 \leq m(B \setminus A) = m\left(\left(\bigcap_{n=1}^{\infty} O_n \right) \setminus A \right) \leq m(O_n \setminus A) < \frac{1}{n}.$$

Letting $n \to \infty$, we find $m(B \setminus A) = 0$ and

$$m(B) = m(B \setminus A) + m(A) = m(A).$$

□

Finally, we show that monotone sequences of measurable sets behave as one would expect with respect to m.

THEOREM 2.17. *Suppose that $A_n \in \mathcal{M}$ for all $n \in \mathbb{N}$. Then, we have:*

(i) if $A_n \subseteq A_{n+1}$ for all $n \in \mathbb{N}$, then

$$m\left(\bigcup_{n=1}^{\infty} A_n \right) = \lim_{n \to \infty} m(A_n);$$

(ii) if $A_n \supseteq A_{n+1}$ for all $n \in \mathbb{N}$ and $m(A_1) < \infty$, then

$$m\left(\bigcap_{n=1}^{\infty} A_n \right) = \lim_{n \to \infty} m(A_n).$$

PROOF. (i) Let $B_1 = A_1$, and $B_i = A_i \setminus A_{i-1}$ for $i \geq 2$. Then $\cup_{i=1}^{\infty} B_i = \cup_{i=1}^{\infty} A_i$ and the $B_i \in \mathcal{M}$ are pairwise disjoint, so that

$$\begin{aligned} m\left(\bigcup_{i=1}^{\infty} A_i \right) &= m\left(\bigcup_{i=1}^{\infty} B_i \right) \\ &= \sum_{i=1}^{\infty} m(B_i) && \text{(by countable additivity)} \\ &= \lim_{n \to \infty} \sum_{i=1}^{n} m(B_i) \\ &= \lim_{n \to \infty} m\left(\bigcup_{i=1}^{n} B_i \right) && \text{(by additivity)} \\ &= \lim_{n \to \infty} m(A_n), \end{aligned}$$

since $A_n = \cup_{i=1}^{n} B_i$ by construction.

(ii) Observing that $A_1 \setminus A_1 = \emptyset \subseteq A_1 \setminus A_2 \subseteq \cdots \subseteq A_1 \setminus A_n \subseteq \ldots$, we use (i) to deduce

$$m\left(\bigcup_{n=1}^{\infty}(A_1 \setminus A_n)\right) = \lim_{n\to\infty} m(A_1 \setminus A_n).$$

Since $m(A_1)$ is finite (so $m(A_n) < \infty$), $m(A_1 \setminus A_n) = m(A_1) - m(A_n)$. On the other hand, $\cup_{n=1}^{\infty}(A_1 \setminus A_n) = A_1 \setminus (\cap_{n=1}^{\infty} A_n)$, so that

$$m(A_1) - m\left(\bigcap_{n=1}^{\infty} A_n\right) = m\left(\bigcup_{n=1}^{\infty}(A_1 \setminus A_n)\right) = m(A_1) - \lim_{n\to\infty} m(A_n).$$

The result follows.

□

REMARK. The condition $m(A_1) < \infty$ in Theorem 2.17 (ii) is very important, as the result is violated by the example $A_n = (n, \infty)$ for all $n \in \mathbb{N}$.

2.5. Borel sets

The definition of $\mathcal{M}$ does not easily lend itself to verification that a particular set belongs to $\mathcal{M}$. It is therefore useful to introduce another construction which shows more directly how open sets and the structure of $\sigma-$algebras lie at the heart of many of the concepts we have developed.

We begin with an auxiliary construction enabling us to produce new $\sigma-$algebras.

THEOREM 2.18. *The intersection of a family of $\sigma-$algebras is a $\sigma-$algebra.*

PROOF. Let F_α be $\sigma-$algebras for $\alpha \in \Lambda$ (the index set Λ can be arbitrary). Put

$$\mathcal{F} = \bigcap_{\alpha\in\Lambda} \mathcal{F}_\alpha.$$

We verify the conditions of the definition.

(i) $\mathbb{R} \in \mathcal{F}_\alpha$ for all $\alpha \in \Lambda$ so $\mathbb{R} \in \mathcal{F}$.
(ii) If $E \in \mathcal{F}$, then $E \in \mathcal{F}_\alpha$ for all $\alpha \in \Lambda$. Since the $\mathcal{F}_\alpha$ are $\sigma-$algebras, $E^c \in \mathcal{F}_\alpha$ for all $\alpha \in \Lambda$ and so $E^c \in \mathcal{F}$.
(iii) If $E_k \in \mathcal{F}$ for $k \in \mathbb{N}$, then $E_k \in \mathcal{F}_\alpha$ for all α, k, hence $\cup_{k=1}^{\infty} E_k \in \mathcal{F}_\alpha$ for all $\alpha \in \Lambda$, and so $\cup_{k=1}^{\infty} E_k \in \mathcal{F}$.

□

DEFINITION 2.19. *Set*

$$\mathcal{B} = \bigcap \{\mathcal{F} : \mathcal{F} \text{ is a } \sigma-\text{algebra containing all intervals}\}.$$

We say that $\mathcal{B}$ is the $\sigma-$algebra generated by all intervals and we call the elements of $\mathcal{B}$ Borel sets. It is obviously the smallest $\sigma-$algebra containing all intervals. In general, we say that $\mathcal{G}$ is the $\sigma-$algebra generated by a family of sets A if $\mathcal{G} = \cap\{\mathcal{F} :$ $\mathcal{F}$ is a $\sigma-$algebra such that $\mathcal{F} \supseteq A\}$.

EXAMPLE 2.20 (Borel sets). *The following examples illustrate how the closure properties of the $\sigma-$algebra $\mathcal{B}$ may be used to verify that most familiar sets in $\mathbb{R}$ belong to $\mathcal{B}$.*

(i) By construction, all intervals belong to $\mathcal{B}$. Since $\mathcal{B}$ is a $\sigma-$algebra, all open sets must belong to $\mathcal{B}$, as any open set in $\mathbb{R}$ is a countable union of open intervals.

(ii) Countable sets are Borel sets, since each is a countable union of closed intervals of the form $[a, a]$; in particular, $\mathbb{N}$ and $\mathbb{Q}$ are Borel sets. Hence, as the complement of a Borel set, the set of irrational numbers is also Borel. Similarly, finite and cofinite sets (the complement of finite sets) are Borel sets.

The definition of $\mathcal{B}$ is also very flexible as long as we start with *all* intervals of a particular type and these collections generate the same Borel $\sigma-$algebra:

THEOREM 2.21. *If instead of the family of all intervals we take all open intervals, all closed intervals, all intervals of the form (a, ∞) (or of the form $[a, \infty), (-\infty, b)$, or $(-\infty, b]$), all open sets, or all closed sets, then the $\sigma-$algebra generated by them is the same as $\mathcal{B}$.*

PROOF. Consider for example the $\sigma-$algebra generated by the family of open intervals $\mathcal{O}$ and denote it by $\mathcal{C}$:

$$\mathcal{C} = \bigcap \{\mathcal{F} : \mathcal{F} \supseteq \mathcal{O},\ \mathcal{F} \text{ is a } \sigma-\text{algebra}\}.$$

We have to show that $\mathcal{B} = \mathcal{C}$. Since open intervals are intervals, $\mathcal{O} \subseteq \mathcal{I}$ (the family of all intervals). It follows that all the $\sigma-$algebras containing $\mathcal{I}$ must also contain $\mathcal{O}$. Therefore, $\mathcal{C} \subseteq \mathcal{B}$.

We shall show that $\mathcal{C}$ contains all intervals. This will be sufficient, since $\mathcal{B}$ is the intersection of such $\sigma-$algebras which implies $\mathcal{B} \subseteq \mathcal{C}$. To this end, consider intervals $[a, b), [a, b], (a, b]$ (the intervals of the form (a, b) are in $\mathcal{C}$ by definition):

$$[a, b) = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b\right),$$

$$[a, b] = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b + \frac{1}{n}\right),$$

$$(a, b] = \bigcap_{n=1}^{\infty} \left(a, b + \frac{1}{n}\right).$$

$\mathcal{C}$ as a $\sigma-$algebra is closed with respect to countable intersection, so it contains the sets on the right. The argument for unbounded intervals is similar. The proof is complete for open intervals, and other cases can be treated similarly. □

REMARK. Since $\mathcal{M}$ is a $\sigma-$algebra containing all intervals, and $\mathcal{B}$ is the smallest such $\sigma-$algebra, we have the inclusion $\mathcal{B} \subseteq \mathcal{M}$, i.e. every Borel set in $\mathbb{R}$ is Lebesgue measurable. The question therefore arises whether these $\sigma-$algebras might be the same. In fact, the inclusion is proper, and it is difficult to construct a set in $\mathcal{M} \setminus \mathcal{B}$ (see Appendix in [**5**]). However, by Theorem 2.16 (ii), given any $E \in \mathcal{M}$ we can find a Borel set $B \supseteq E$ of the form $B = \cap_{n=1}^{\infty} O_n$, where the (O_n) are open sets, and such that $m(E) = m(B)$. In particular,

$$m(B \triangle E) = m(B \setminus E) = 0,$$

and $B \triangle E \in \mathcal{M}$. However, we cannot conclude that every null set will be a Borel set (if $\mathcal{B}$ did contain all null sets then by Theorem 2.16 (ii), we would have $\mathcal{B} = \mathcal{M}$), and this points to an 'incompleteness' in $\mathcal{B}$.

DEFINITION 2.22. *A measure space $(X, \mathcal{F}, \mu)$ is complete if for all $F \in \mathcal{F}$ with $\mu(F) = 0$ and $N \subseteq F$, we have $N \in \mathcal{F}$ (and so $\mu(N) = 0$). The completion of a $\sigma-$algebra $\mathcal{G}$, relative to a given measure μ, is defined as the smallest $\sigma-$algebra $\mathcal{F}$ containing $\mathcal{G}$ such that if $N \subseteq E \in \mathcal{G}$ and $\mu(E) = 0$, then $N \in \mathcal{F}$.*

We will show that $\mathcal{M}$ is the completion of $\mathcal{B}$ relative to m and $(\mathbb{R}, \mathcal{M}, m)$ is complete (whereas the measure space $(\mathbb{R}, \mathcal{B}, m)$ is not complete).

PROPOSITION 2.23. $\star$ *The completion of the $\sigma-$algebra $\mathcal{G}$ on the space X relative to the measure μ is of the form $\{E \cup N : E \in \mathcal{G}, N \subseteq F \in \mathcal{G}$ with $\mu(F) = 0\}$.*

PROOF. $\star$ Let us first show the family $\mathcal{F} := \{E \cup N : E \in \mathcal{G}, N \subseteq F \in \mathcal{G}$ with $\mu(F) = 0\}$ is a $\sigma-$algebra. $\mathcal{F}$ contains the base set X since $X \in \mathcal{G}$. If $E_i \cup N_i \in \mathcal{F}$, $N_i \subseteq F_i \in \mathcal{G}$ and $\mu(F_i) = 0$ for all $i \in I$, then $\cup_{i\in I}(E_i \cup N_i) = (\cup_{i\in I} E_i) \cup (\cup_{i\in I} N_i)$ is in $\mathcal{G}$ since the first set on the right is in $\mathcal{G}$ and the second is a subset of a null set $\cup_{i\in I} F_i \in \mathcal{G}$. If $E \cup N \in \mathcal{F}$, $N \subseteq F \in \mathcal{G}$ and $\mu(F) = 0$, then

$$\begin{aligned}(E \cup N)^c = (E \cup (F \setminus (F \setminus N)))^c &= (E \cup (F \cap (F \setminus N)^c))^c \\ &= ((E \cup F) \cap (E \cup (F \setminus N)^c)))^c = (E \cup F)^c \cup (E^c \cap (F \setminus N))\end{aligned}$$

which is also in $\mathcal{F}$ since $(E \cup F)^c \in \mathcal{G}$ and $E^c \cap (F \setminus N)$ is a subset of a null set $F \in \mathcal{G}$. Thus, $\mathcal{F}$ is a $\sigma-$algebra. Consider any other $\sigma-$algebra $\mathcal{H}$ containing $\mathcal{F}$ and all subsets of null sets. Since $\mathcal{H}$ is closed with respect to the unions, it contains $\mathcal{F}$ and so $\mathcal{F}$ is the smallest $\sigma-$algebra with this property. □

REMARK. $\star$ This allows us to extend the measure μ uniquely to a measure $\bar{\mu}$ on $\mathcal{F}$ by setting $\bar{\mu}(E \cup N) = \mu(E)$ for $E \in \mathcal{G}$. Why unique? Suppose $E_1 \cup N_1 = E_2 \cup N_2$ satisfying $E_1, E_2 \in \mathcal{G}$, $N_1 \subseteq F_1 \in \mathcal{G}$, $N_2 \subseteq F_2 \in \mathcal{G}$ and $\mu(F_1) = \mu(F_2) = 0$. We claim $\mu(E_1) = \mu(E_2)$. Since $N_1 \setminus N_2 = E_2 \setminus E_1 \in \mathcal{G}$ and $N_1 \setminus N_2 \subseteq F_1$, we deduce $\mu(E_2 \setminus E_1) = \mu(N_1 \setminus N_2) = 0$. Similarly, we have $\mu(E_1 \setminus E_2) = 0$. Therefore,

$$\mu(E_1) = \mu(E_1 \cap E_2) + \mu(E_1 \setminus E_2) = \mu(E_1 \cap E_2) = \mu(E_1 \cap E_2) + \mu(E_2 \setminus E_1) = \mu(E_2)$$

as desired.

THEOREM 2.24. $\mathcal{M}$ *is the completion of $\mathcal{B}$ on $\mathbb{R}$ relative to m.*

PROOF. By Theorem 2.4 and Theorem 2.10 (i), $\mathcal{M}$ is a complete $\sigma-$algebra containing $\mathcal{B}$, and this means that $\mathcal{M}$ also contains the completion $\mathcal{C}$ of $\mathcal{B}$.

Next, we show that $\mathcal{M}$ is the minimal such $\sigma-$algebra, i.e. $\mathcal{M} \subseteq \mathcal{C}$. Consider $E \in \mathcal{M}$, and choose $B = \cap_{n=1}^{\infty} O_n \in \mathcal{B}$ as described in Theorem 2.16 (ii) such that $B \supseteq E$, $m(B) = m(E)$, and $m(B \setminus E) = 0$. Set $N = B \setminus E$, and we have $N \in \mathcal{M}$. Using Theorem 2.16 (ii) again, we can find $L \supseteq N$, $L \in \mathcal{B}$ and $m(L) = 0$. In other words, N is a subset of a null set in $\mathcal{B}$, and therefore $E = B \setminus N$ belongs to the completion $\mathcal{C}$ of $\mathcal{B}$. Thus, $\mathcal{M} \subseteq \mathcal{C}$ and so they are equal. □

Despite these technical differences, measurable sets are never far from 'nice' sets. In addition to approximations from above by open sets, as observed in Theorem 2.16, we can approximate the measure of any $E \in \mathcal{M}$ from below by those of closed subsets.

THEOREM 2.25 (Inner regularity of Lebesgue measure). *If $E \in \mathcal{M}$, then for given $\varepsilon > 0$ there exists a closed set $F \subseteq E$ such that $m(E \setminus F) < \varepsilon$. Moreover, there exists $B \subseteq E$ in the form $B = \cup_{n=1}^{\infty} F_n$, where all the F_n are closed sets, and $m(E \setminus B) = 0$.*

PROOF. The complement E^c is measurable and by Theorem 2.16 we can find an open set O containing E^c such that $m(O \setminus E^c) < \varepsilon$. But $O \setminus E^c = O \cap E = E \setminus O^c$, and $F = O^c$ is closed and contained in E. Hence this F is what we need. The proof of the last assertion is similar to that of Theorem 2.16 (ii). □

2.6. ⋆ A non-measurable set in $\mathbb{R}$

The construction of a non-measurable set needs the following axiom from set theory:

AXIOM 2.26 (The axiom of choice). *Suppose that $\mathcal{A} = \{A_\alpha : \alpha \in \Lambda\}$ is a non-empty collection of non-empty sets, indexed by the set Λ. Then there exists a choice function $f : \Lambda \to \cup_{\alpha\in\Lambda} A_\alpha$, i.e. $f(\alpha) \in A_\alpha$ for all $\alpha \in \Lambda$.*

PROPOSITION 2.27. *There exists a subset $E \subseteq [0, 1]$ which is not Lebesgue measurable.*

PROOF. First, we define the following equivalence relation on $[0, 1]$: $x \sim y$ if $y - x$ is a rational number (which will be in $[-1, 1]$). This relation is easily seen to be reflexive, symmetric and transitive. Hence it partitions $[0, 1]$ into disjoint equivalence classes $[A_\alpha]$, where for each α, any two elements x, y of $[A_\alpha]$ differ by a rational, while elements of different classes will always differ by an irrational. Thus, each A_α is countable, since $\mathbb{Q}$ is, but there are uncountably many different classes, as $[0, 1]$ is uncountable.

Now, applying the axiom of choice, we can get a set $E \subseteq [0, 1]$ which contains exactly one member a_α from each of the $[A_\alpha]$. Next, let x be any element of $[0, 1]$. Then it must lie in some equivalence class $[A_\alpha]$ of $[0, 1]$, and thus differs from a_α by some rational number in $[-1, 1]$. In other words, we have

$$[0, 1] \subseteq \bigcup_{q\in\mathbb{Q}\cap[-1,1]} (E + q). \tag{2.15}$$

On the other hand, we clearly have

$$\bigcup_{q\in\mathbb{Q}\cap[-1,1]} (E + q) \subseteq [-1, 2]. \tag{2.16}$$

Note that the different translates $E + q$ are disjoint, because E contains only one element from each equivalent class $[A_\alpha]$.

We claim that E is not Lebesgue measurable. To see this, suppose for contradiction that E was Lebesgue measurable. Then the translates $E + q$ would also be Lebesgue measurable (see Proposition 2.14 (iii)). By countable additivity, we thus have

$$m\left(\bigcup_{q\in\mathbb{Q}\cap[-1,1]} (E + q)\right) = \sum_{q\in\mathbb{Q}\cap[-1,1]} m(E + q).$$

By translation invariance of m, (2.15) and (2.16), we infer

$$1 \leq \sum_{q\in\mathbb{Q}\cap[-1,1]} m(E) \leq 3.$$

On the other hand, the sum $\sum_{q\in\mathbb{Q}\cap[-1,1]} m(E)$ is either zero (if $m(E) = 0$) or infinite (if $m(E) > 0$) leading to the desired contradiction. □

2.7. Finishing remarks

This chapter is prepared mainly based on Chapter 2 of [**5**]. For the last section, I have consulted the Appendix of [**5**] and Section 1.2.3 in [**16**].

CHAPTER 3

Measurable functions

3.1. Lebesgue-measurable functions

From now on, we will allow functions to take infinite values, i.e. their range to be (part of) the 'extended real line' $\overline{\mathbb{R}} = [-\infty, \infty]$. Arithmetic in this set obeys the following rules: $a + \infty = \infty$ for all $a \in \mathbb{R}$, $a \times \infty = \infty$ for all $a > 0$, $a \times \infty = -\infty$ for $a < 0$, $\infty \times \infty = \infty$, and $0 \times \infty = 0$, with similar definitions for $-\infty$. However, we avoid forming 'sums' of the form $\infty + (-\infty)$.

The domain of the functions we shall be considering is usually $\mathbb{R}$. Now we have the freedom of defining f only 'up to null sets': once we have shown two functions f and g to be equal on $\mathbb{R} \setminus E$ where E is some null set, then $f = g$ for all practical purposes. To formalize this, we say that f has a property (P) *almost everywhere* (a.e.) if f has this property at all points of its domain, except possibly on some null set.

For example, the function

$$f(x) = \begin{cases} 1, & \text{if } x \neq 0, \\ 0, & \text{if } x = 0 \end{cases}$$

is almost everywhere continuous, since it is continuous on $\mathbb{R} \setminus \{0\}$, and the exceptional set $\{0\}$ is null. (Note that probabilists tend to say 'almost surely' (a.s.) instead of 'almost everywhere' (a.e.).)

The next definition will introduce the class of Lebesgue-measurable functions. The condition imposed on $f : \mathbb{R} \to \mathbb{R}$ will be necessary to give meaning to the (Lebesgue) integral $\int f dm$. Let us first give some motivation.

Integration is always concerned with the process of approximation. In the Riemann integral, we split the interval $I = [a, b]$ over which integrate into small intervals I_n. The simplest method of doing this is to divide the interval into N equal parts. Then, we construct approximating sums by multiplying the lengths of the small intervals by certain numbers c_n like $c_n = \inf_{I_n} f$, $c_n = \sup_{I_n} f$, or $c_n = f(x)$ for some $x \in I_n$:

$$\sum_{n=1}^{N} c_n \ell(I_n).$$

For large N, this sum is close to the Riemann integral $\int_a^b f(x)dx$ (given some regularity of f).

The approach to the Lebesgue integral is similar but there is a crucial difference. Instead of splitting the integration domain into small parts, we decompose the range of the function. Again, a simple way is to introduce short intervals J_n of equal length. To build the approximating sums, we first take the pre-images of J_n by f, i.e. $f^{-1}(J_n)$. These may be complicated sets, not necessarily intervals. Here the theory of measure developed previously comes into its own. We are able to measure sets provided they are measurable,

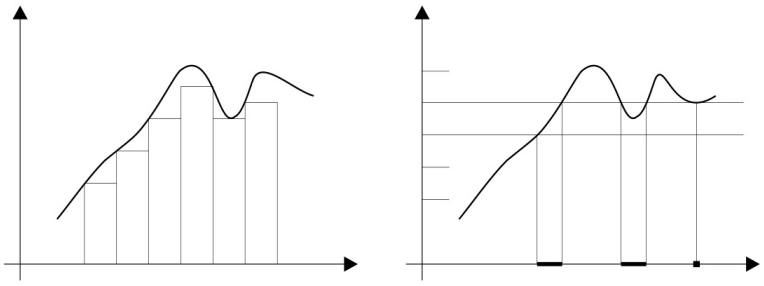


FIGURE 3.1. Riemann vs. Lebesgue.

i.e. they are in $\mathcal{M}$. Given that, we compute

$$\sum_{n=1}^{N} c_n m(f^{-1}(J_n))$$

where $c_n \in J_n$ or $c_n = \inf J_n$, for example.

The following definition guarantees that this procedure makes sense.

DEFINITION 3.1. *Suppose that E is a measurable set. We say that a function $f : E \to \mathbb{R}$ is (Lebesgue-)measurable if for any interval $I \subseteq \mathbb{R}$*

$$f^{-1}(I) = \{x \in E : f(x) \in I\} \in \mathcal{M}.$$

In what follows, the term measurable (without qualification) will refer to Lebesgue-measurable functions. If all the sets $f^{-1}(I) \in \mathcal{B}$, i.e. they are Borel sets, we call f Borel measurable, or simply a Borel function.

The underlying philosophy is one which is common for various mathematical notions: the pre-image of a *nice* set is *nice*. Remember continuous functions, for example, where the pre-image of any open set is required to be open. The actual meaning of the word *nice* depends on the particular branch of mathematics. In the above definitions, note that since $\mathcal{B} \subseteq \mathcal{M}$, every Borel function is (Lebesgue-)measurable.

We give some equivalent formulations:

THEOREM 3.2. *Let $E \in \mathcal{M}$, and $f : E \to \mathbb{R}$ be a function. The following conditions are equivalent:*

(a) f is measurable.
(b) for all $a \in \mathbb{R}$, $f^{-1}((a, \infty))$ is measurable.
(c) for all $a \in \mathbb{R}$, $f^{-1}([a, \infty))$ is measurable.
(d) for all $a \in \mathbb{R}$, $f^{-1}((-\infty, a))$ is measurable.
(e) for all $a \in \mathbb{R}$, $f^{-1}((-\infty, a])$ is measurable.

PROOF. Clearly, (a) implies any of the other conditions. We show that (b) implies (a). The proofs of the other implications are similar, and are left as exercises (which you should attempt).

We have to show that for any interval I, $f^{-1}(I) \in \mathcal{M}$. By (b), the particular case $I = (a, \infty)$ is valid. Suppose $I = (-\infty, a]$. Then

$$f^{-1}((-\infty, a]) = f^{-1}(\mathbb{R} \setminus (a, \infty)) = E \setminus f^{-1}((a, \infty)) \in \mathcal{M} \tag{3.1}$$

since both E and $f^{-1}((a,\infty))$ are in $\mathcal{M}$ (we use the closure properties of $\mathcal{M}$ established before). Next

$$f^{-1}((-\infty,b)) = f^{-1}\left(\bigcup_{n=1}^{\infty}\left(-\infty, b-\frac{1}{n}\right]\right) = \bigcup_{n=1}^{\infty} f^{-1}\left(\left(-\infty, b-\frac{1}{n}\right]\right).$$

By (3.1), $f^{-1}((-\infty, b-\frac{1}{n}]) \in \mathcal{M}$ and the same is true for the countable union. From this we can easily deduce that $f^{-1}([b,\infty)) \in \mathcal{M}$. Now, let $I = (a,b)$, and we infer

$$f^{-1}((a,b)) = f^{-1}((-\infty,b)\cap(a,\infty)) = f^{-1}((-\infty,b))\cap f^{-1}((a,\infty))$$

is in $\mathcal{M}$ as the intersection of two elements of $\mathcal{M}$. By the same reasoning, $\mathcal{M}$ contains

$$f^{-1}([a,b]) = f^{-1}((-\infty,b]\cap[a,\infty)) = f^{-1}((-\infty,b])\cap f^{-1}([a,\infty))$$

and half-open intervals are handled similarly. □

The following examples show that most of the functions encountered 'in practice' are measurable.

EXAMPLE 3.3. *(i) Constant functions are measurable. Let $f(x) \equiv c$. Then*

$$f^{-1}((a,\infty)) = \begin{cases} \mathbb{R}, & \text{if } a < c, \\ \emptyset, & \text{if } a \geq c, \end{cases}$$

and in both cases we have measurable sets.

(ii) Continuous functions are measurable. Since (a,∞) is an open set, $f^{-1}((a,\infty))$ is open by continuity of f. As we know, all open sets are measurable.

(iii) Recall the indicator function of a set A is defined by

$$\mathbf{1}_A(x) = \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{if } x \notin A. \end{cases}$$

Then

$$A \in \mathcal{M} \quad \Leftrightarrow \quad \mathbf{1}_A \text{ is measurable}$$

since

$$\mathbf{1}_A^{-1}((a,\infty)) = \begin{cases} \mathbb{R}, & \text{if } a < 0, \\ A, & \text{if } 0 \leq a < 1, \\ \emptyset, & \text{if } a \geq 1. \end{cases}$$

In general, we can extend the definition of measurable functions to the maps $T : X \to X'$ between two measurable spaces $(X,\mathcal{A})$ and $(X',\mathcal{A}')$ which respect the measurable structures, that is $\sigma-$algebras, on X and X'. More precisely, we have

DEFINITION 3.4. *Let $(X,\mathcal{A})$, $(X',\mathcal{A}')$ be measurable spaces. A map $T : X \to X'$ is called $\mathcal{A}/\mathcal{A}'-$measurable (or measurable unless this is too ambiguous) if the pre-image of every measurable set is a measurable set:*

$$T^{-1}(A') \in \mathcal{A} \quad \forall\, A' \in \mathcal{A}'.$$

From this definition, we can show a measurable function $f : E \to \mathbb{R}$ is actually an $\mathcal{M}/\mathcal{B}$ measurable mapping. Indeed, this is a direct consequence of the following useful lemma.

LEMMA 3.5. *Let $(X, \mathcal{A})$, $(X', \mathcal{A}')$ be measurable spaces and suppose the $\sigma-$algebra $\mathcal{A}'$ is generated by $\mathcal{G}' \subseteq P(X')$ (or $\mathcal{A}' = \sigma(\mathcal{G}')$ in short). Then, $T : X \to X'$ is $\mathcal{A}/\mathcal{A}'-$measurable if, and only if, $T^{-1}(\mathcal{G}') \subseteq \mathcal{A}$, i.e. if*

$$T^{-1}(G') \in \mathcal{A} \quad \forall\, G' \in \mathcal{G}'. \tag{3.2}$$

PROOF. If T is $\mathcal{A}/\mathcal{A}'-$measurable, we have $T^{-1}(\mathcal{G}') \subseteq T^{-1}(\mathcal{A}') \subseteq \mathcal{A}$, and (3.2) is obviously satisfied.

Conversely, consider the system $\Sigma' := \{A' \subseteq X' : T^{-1}(A') \in \mathcal{A}\}$. By (3.2), $\mathcal{G}' \subseteq \Sigma'$ and it is not difficult to see that Σ' is itself a $\sigma-$algebra since T^{-1} commutes with all set operations. Therefore,

$$\mathcal{A}' = \sigma(\mathcal{G}') \subseteq \Sigma' \implies T^{-1}(A') \in \mathcal{A} \quad \forall\, A' \in \mathcal{A}'.$$

□

More interesting properties of measurable mappings can be found in Chapter 7 of [**15**]. Henceforth, for a measurable function $f : (X, \mathcal{A}) \to \mathbb{R}$, we mean it is $\mathcal{A}/\mathcal{B}-$measurable. With this notion, most of the results related to the measurable function $f : E \to \mathbb{R}$ in this chapter can be generalized to a measurable function over a general measurable space $(X, \mathcal{A})$.

3.2. Properties of measurable functions

The class of measurable functions is very rich, as the following results show.

THEOREM 3.6. *The set of real-valued measurable functions defined on $E \in \mathcal{M}$ is a vector space and closed under multiplication, i.e. if f and g are measurable functions then $f + g$ and fg are also measurable (in particular, if g is a constant function $g \equiv c$, cf is measurable for all real c.)*

PROOF. Fix measurable functions $f, g : E \to \mathbb{R}$. First consider $f + g$. Our goal is to show that for each $a \in \mathbb{R}$,

$$B = (f+g)^{-1}((-\infty, a)) = \{t \in E : f(t) + g(t) < a\} \in \mathcal{M}.$$

Suppose that all the rational numbers are arranged in a sequence (q_n). Now

$$B = \bigcup_{n=1}^{\infty} \{t \in E : f(t) < q_n,\, g(t) < a - q_n\}.$$

In other words, we decompose the half-plane below the line $x + y = a$ into a countable union of unbounded 'boxes': $\left\{(x, y) \in \mathbb{R}^2 : x < q_n,\, y < a - q_n\right\}$, see Figure 3.2. Clearly,

$$\{t \in E : f(t) < q_n,\, g(t) < a - q_n\} = \{t \in E : f(t) < q_n\} \cap \{t \in E : g(t) < a - q_n\}$$

is measurable as an intersection of measurable sets. Hence $B \in \mathcal{M}$ as a countable union of elements of $\mathcal{M}$.

To deal with fg, we adopt a slightly indirect approach in order to remain 'one-dimensional': first note that if g is measurable, then so is $-g$. Hence $f - g = f + (-g)$ is measurable. Since $fg = \frac{1}{4}\left[(f+g)^2 - (f-g)^2\right]$, we need to prove that the square of a measurable function is measurable. So take a measurable $h : E \to \mathbb{R}$ and consider $\{x \in E : h^2(x) > a\}$. For $a < 0$ this set is $E \in \mathcal{M}$, and for $a \geq 0$

$$\left\{x \in E : h^2(x) > a\right\} = \left\{x \in E : h(x) > \sqrt{a}\right\} \cup \left\{x \in E : h(x) < -\sqrt{a}\right\}.$$

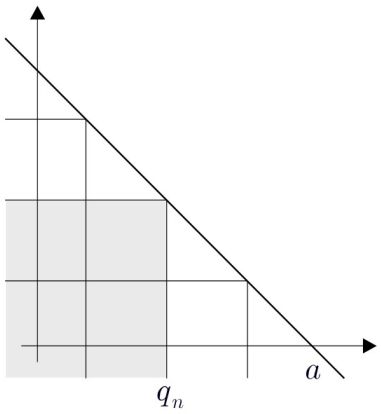


FIGURE 3.2. Use boxes to approximate a half-plane.

Both sets on the right are measurable, hence we have shown that h^2 is measurable. By similar methods, we can show cf is measurable for constant $c \in \mathbb{R}$. Applying all these, we can show fg is measurable. Moreover, the class of real-valued measurable functions forms a vector space. □

REMARK. An elegant proof of the theorem is based on the following lemma, which will also be useful later. Its proof makes use the topological fact that every open set in $\mathbb{R}^2$ can be expressed as a countable union of almost disjoint boxes, in analogy with open sets in $\mathbb{R}$ and intervals, see Lemma 1.2.11 in [**16**] for the proof. Here, we say two boxes are *almost disjoint* if their interiors are disjoint, thus for instance $[0,1]\times[0,1]$ and $[0,1]\times[1,2]$ are almost disjoint.

LEMMA 3.7. *Suppose that $F : \mathbb{R}^2 \to \mathbb{R}$ is a continuous function. If f and g are measurable, then $h(x) = F(f(x), g(x))$ is also measurable.*

It now suffices to take $F(u,v) = u + v$, $F(u,v) = uv$ to obtain a second proof of Theorem 3.6.

PROOF OF LEMMA 3.7. For any real a, we have

$$\{x : h(x) > a\} = \{x : (f(x), g(x)) \in G_a\}$$

where $G_a = \big\{(u,v) \in \mathbb{R}^2 : F(u,v) > a\big\} = F^{-1}((a,\infty))$. Suppose for the moment that G_a is a box: $G_a = I_1 \times I_2$, where I_1, I_2 are bounded intervals in $\mathbb{R}$.

It is clear from Figure 3.3 that

$$\begin{aligned}\{x : h(x) > a\} &= \{x : f(x) \in I_1 \text{ and } g(x) \in I_2\} \\ &= \{x : f(x) \in I_1\} \cap \{x : g(x) \in I_2\}\end{aligned}$$

which belongs to $\mathcal{M}$ as f, g are measurable. In general, we have to express the set G_a as a countable union of boxes. The set G_a is an open subset of $\mathbb{R}^2$ since F is continuous. Hence it can be written as

$$G_a = \bigcup_{n=1}^{\infty} R_n$$

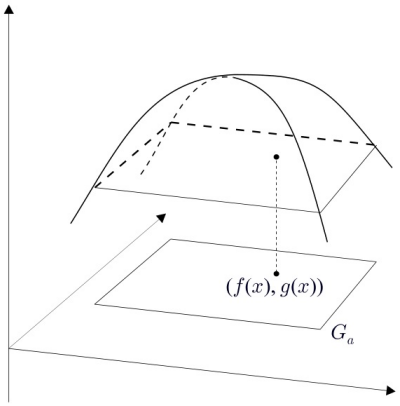


FIGURE 3.3. The set G_a.

where R_n is a collection of almost disjoint boxes $R_n = I_n \times J_n$ with I_n, J_n to be bounded intervals in $\mathbb{R}$. So

$$\{x : h(x) > a\} = \bigcup_{n=1}^{\infty} (\{x : f(x) \in I_n\} \cap \{x : g(x) \in J_n\})$$

is measurable due to the closure properties of $\mathcal{M}$. □

A simple application of Theorem 3.6 is to consider the product $f \cdot \mathbf{1}_A$. If f is a measurable function and A is a measurable set, then $f \cdot \mathbf{1}_A$ is measurable. This function is simply f on A and 0 outside A. Applying this to the set $A = \{x \in E : f(x) > 0\}$, we see that the positive part f^+ of a measurable function is measurable, as

$$f^+(x) = f \cdot \mathbf{1}_{\{f(x)>0\}}(x) = \begin{cases} f(x), & \text{if } f(x) > 0 \\ 0, & \text{if } f(x) \leq 0. \end{cases}$$

Similarly, the negative part f^- of f is measurable, since

$$f^-(x) = \begin{cases} 0, & \text{if } f(x) \geq 0 \\ -f(x), & \text{if } f(x) < 0. \end{cases}$$

PROPOSITION 3.8. *Let E be a measurable subset of $\mathbb{R}$.*

(i) $f : E \to \mathbb{R}$ is measurable if and only if both f^+ and f^- are measurable.
(ii) If f is measurable, then $|f|$ is measurable. However, the converse is false.

PROOF. (i) We have already shown that if f is measurable then f^+, f^- are measurable. Conversely, note that $f(x) = f^+(x) - f^-(x)$, so Theorem 3.6 gives the result.

(ii) The function $u \mapsto |u|$ is continuous, so Lemma 3.7 with $F(u, v) = |u|$ gives measurability of $|f|$ (an alternative is to use $|f| = f^+ + f^-$). To see that the converse is not true, take a non-measurable set A (which indeed exists and see Section 2.6 for details) and let $f = \mathbf{1}_A - \mathbf{1}_{A^c}$. It is nonmeasurable since $\{x : f(x) > 0\} = A$ is nonmeasurable. But $|f| = 1$ is clearly measurable.

□

Passage to the limit does not destroy measurability as shown in the following result:

THEOREM 3.9. *If (f_n) is a sequence of measurable functions defined on the measurable set E in $\mathbb{R}$, then the following are also measurable functions:*

$$\max_{n\le k} f_n, \quad \min_{n\le k} f_n, \quad \sup_{n\in\mathbb{N}} f_n, \quad \inf_{n\in\mathbb{N}} f_n, \quad \limsup_{n\to\infty} f_n, \quad \liminf_{n\to\infty} f_n.$$

PROOF. It is sufficient to note that the following are measurable sets:

$$\left\{x \in E : \left(\max_{n\le k} f_n\right)(x) > a\right\} = \bigcup_{n=1}^{k} \{x \in E : f_n(x) > a\},$$

$$\left\{x \in E : \left(\min_{n\le k} f_n\right)(x) > a\right\} = \bigcap_{n=1}^{k} \{x \in E : f_n(x) > a\},$$

$$\left\{x \in E : \left(\sup_{n\ge k} f_n\right)(x) > a\right\} = \bigcup_{n=k}^{\infty} \{x \in E : f_n(x) > a\},$$

$$\left\{x \in E : \left(\inf_{n\ge k} f_n\right)(x) \ge a\right\} = \bigcap_{n=k}^{\infty} \{x \in E : f_n(x) \ge a\}.$$

Moreover, we have $h_n = \sup_{m\ge n} f_m$ is measurable, and hence $\inf_{n\ge 1} h_n$ is measurable. From the definition of the upper limit as shown below

$$\limsup_{n\to\infty} f_n(x) = \inf_{n\ge 1}\left(\sup_{m\ge n} f_m(x)\right),$$

the upper limit $\limsup_{n\to\infty} f_n$ is measurable. The lower limit is done similarly. □

COROLLARY 3.10. *If a sequence f_n of measurable functions converges (pointwise) then the limit is a measurable function.*

PROOF. This is immediate since $\lim_{n\to\infty} f_n = \limsup_{n\to\infty} f_n$ which is measurable. □

REMARK. ⋆ Note that Theorems 3.6 and 3.9 have counterparts for Borel functions, i.e. they remain valid upon replacing 'measurable' by 'Borel' throughout.

Things are slightly more complicated when we consider the role of null sets. On the one hand, changing a function on a null set cannot destroy its measurability, i.e. any measurable function which is altered on a null set remains measurable. However, as not all null sets are Borel sets, we cannot conclude similarly for Borel sets, and thus the following results have no natural 'Borel' counterparts.

THEOREM 3.11. *If $E \in \mathcal{M}$, $f : E \to \mathbb{R}$ is measurable, and $g : E \to \mathbb{R}$ is a function satisfying the set $\{x \in E : f(x) \ne g(x)\}$ is null, then g is measurable.*

PROOF. Consider the difference $d(x) = g(x) - f(x)$. It is zero except on a null set so

$$\{x \in E : d(x) > a\} = \begin{cases} \text{a null set}, & \text{if } a \ge 0, \\ \text{a full set}, & \text{if } a < 0, \end{cases}$$

where a full set is the complement of a null set. Both null and full sets are measurable hence d is a measurable function. Thus $g = f + d$ is measurable. □

COROLLARY 3.12. *If (f_n) is a sequence of measurable functions and $f_n(x) \to f(x)$ almost everywhere in E, then f is measurable.*

PROOF. Let A be the null set such that $f_n(x)$ converges for all $x \in E \setminus A$. Then $\mathbf{1}_{A^c} f_n$ converge everywhere to $g = \mathbf{1}_{A^c} f$ which is therefore measurable (see Corollary 3.10). But $f = g$ almost everywhere, so f is measurable. □

Since we are able to adjust a function f on a null set without altering its measurability, the following definition is a useful means of concentrating on the values of f that 'really matter' for integration theory, by identifying its bounds 'outside null sets':

DEFINITION 3.13. *Suppose $f : E \to \mathbb{R}$ is measurable. The essential supremum ess* $\sup f$ *is defined as* $\inf\{b \in \mathbb{R} : f \leq b$ *a.e. on* $E\}$ *and the essential infimum ess* $\inf f$ *is* $\sup\{b \in \mathbb{R} : f \geq b$ *a.e. on* $E\}$.

Note that $\operatorname{ess\,sup} f$ can be $+\infty$. For $f : E \to \overline{\mathbb{R}}$, if $\operatorname{ess\,sup} f = -\infty$, then $f = -\infty$ a.e. since by definition of ess sup, $f \leq -n$ a.e. for all $n \geq 1$. Now if $\operatorname{ess\,sup} f$ is finite, set $A = \{x : \operatorname{ess\,sup} f < f(x)\}$ and A_n for $n \geq 1$ by

$$A_n = \left\{ x : \operatorname{ess\,sup} f < f(x) - \frac{1}{n} \right\}.$$

From the definition of ess sup, we know all the A_n are null sets, and so $A = \cup_{n=1}^{\infty} A_n$ is also a null set. Thus, we have verified:

$$f \leq \operatorname{ess\,sup} f \quad \text{a.e.}$$

The following is now straightforward to prove.

PROPOSITION 3.14. *If f, g are measurable functions, then*

$$\textit{ess}\sup(f+g) \leq \textit{ess}\sup f + \textit{ess}\sup g.$$

PROOF. Since $f \leq \operatorname{ess\,sup} f$ and $g \leq \operatorname{ess\,sup} g$ a.e., by adding we have $f + g \leq \operatorname{ess\,sup} f + \operatorname{ess\,sup} g$ a.e. It follows that the number $\operatorname{ess\,sup} f + \operatorname{ess\,sup} g$ belongs to the set $\{b : f + g \leq b \text{ a.e.}\}$, so the infimum of this set, $\operatorname{ess\,sup}(f+g)$, is smaller than this number. □

3.3. Finishing remarks

This chapter is prepared mainly based on Chapter 3 of [**5**].

CHAPTER 4

Integral

The theory developed below deals with Lebesgue measure for the sake of simplicity. However, all we need (except for the section where we discuss the Riemann integration) is the property of m being a measure, i.e. a countably additive (extended-)real valued function μ defined on a $\sigma-$algebra of subsets of a fixed set Ω. Therefore, the theory developed for the measure space $(\mathbb{R}, \mathcal{M}, m)$ in the following sections can be extended virtually without change to an abstractly given measure space $(\Omega, \mathcal{F}, \mu)$.

4.1. Definition of the integral

We are now able to resolve one of the problems we identified earlier: how to integrate functions like $\mathbf{1}_{\mathbb{Q}}$, which take only finitely many values, but where the sets on which these values are taken are not at all 'like intervals'.

DEFINITION 4.1. *A nonnegative function $\varphi : \mathbb{R} \to \mathbb{R}$ which takes only finitely many values, i.e. the range of φ is a finite set of distinct nonnegative reals $\{a_1, a_2, \ldots, a_n\}$, is a simple function if all the sets*

$$A_i = \varphi^{-1}(\{a_i\}) = \{x \in \mathbb{R} : \varphi(x) = a_i\}, \quad i = 1, 2, \ldots, n,$$

are measurable sets. Note that the sets $A_i \in \mathcal{M}$ are pairwise disjoint and their union is $\mathbb{R}$.

Clearly, we can write

$$\varphi(x) = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}(x)$$

so that (by Theorem 3.6) each simple function is measurable.

DEFINITION 4.2. *The (Lebesgue) integral over $E \in \mathcal{M}$ of the simple function φ is given by:*

$$\int_E \varphi dm = \sum_{i=1}^{n} a_i m(A_i \cap E).$$

(Note: Since we shall allow $m(A_i) = +\infty$, we use the convention $0 \times \infty = 0$ here.)

EXAMPLE 4.3. *Consider the simple function $\mathbf{1}_{\mathbb{Q}}$ which takes the value 1 on $\mathbb{Q}$ and 0 on $\mathbb{R} \setminus \mathbb{Q}$. By the above definition, we have*

$$\int_{\mathbb{R}} \mathbf{1}_{\mathbb{Q}} dm = 1 \times m(\mathbb{Q}) + 0 \times m(\mathbb{R} \setminus \mathbb{Q}) = 0$$

since $\mathbb{Q}$ is a null set. Recall that this function is not Riemann-integrable. Similarly, $\mathbf{1}_C$ has integral 0, where C is the Cantor set.

Now, we can introduce the Lebesgue integral of nonnegative functions as follows:

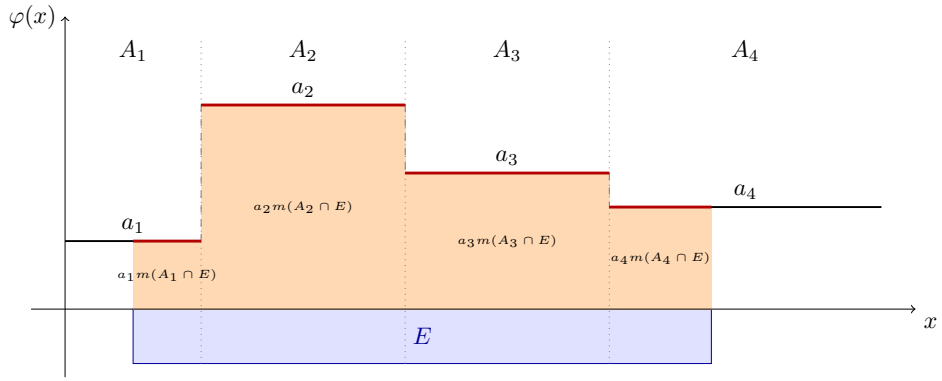


FIGURE 4.1. Integral of a simple function.

DEFINITION 4.4. *For any nonnegative measurable function f and $E \in \mathcal{M}$, the integral $\int_E f dm$ is defined as*

$$\int_E f dm = \sup Y(E, f)$$

where

$$Y(E, f) = \left\{ \int_E \varphi dm : 0 \le \varphi \le f, \, \varphi \text{ is simple} \right\}.$$

Note that the integral can be $+\infty$, and is always nonnegative. Clearly, the set $Y(E, f)$ is always of the form $[0, x]$ or $[0, x)$, where the value $x = +\infty$ is allowed.

If $E = [a, b]$, we write the integral as

$$\int_a^b f dm, \quad \int_a^b f(x) dm(x),$$

or even as $\int_a^b f(x) dx$ when no confusion is possible (and we set $\int_a^b f dm = -\int_b^a f dm$ if $a > b$). The notation $\int f dm$ means $\int_{\mathbb{R}} f dm$.

Clearly, if for some $A \in \mathcal{M}$ and a nonnegative measurable function g we have $g = 0$ on A^c, then any nonnegative simple function that lies below g must be zero on A^c. Applying this to $g = f \cdot \mathbf{1}_A$, we obtain the important identity

$$\int_A f dm = \int f \cdot \mathbf{1}_A dm.$$

If f is a simple function, we now have two definitions of the integral; thus for consistency you should check carefully that the above definitions coincide.

PROPOSITION 4.5. *For simple functions, Definitions 4.2 and 4.4 are equivalent.*

PROOF. Let $f = \sum_{i=1}^n c_i \mathbf{1}_{A_i}$. We have to show that

$$\sum_{i=1}^n c_i m(A_i \cap E) = \sup Y(E, f).$$

First, we may take $\varphi = f$ in the definition of $Y(E, f)$ so the number on the left belongs to $Y(E, f)$ and so

$$\sum_{i=1}^{n} c_i m(A_i \cap E) \leq \sup Y(E, f).$$

For the converse, take any $\alpha \in Y(E, f)$. So

$$\alpha = \int_E \psi dm = \sum_{j=1}^{k} d_j m(E \cap B_j)$$

for some simple function $\psi \leq f$. Now

$$\alpha = \sum_{j=1}^{k} \sum_{i=1}^{n} d_j m(E \cap B_j \cap A_i)$$

by the properties of measure (A_i form a partition of $\mathbb{R}$). If $B_j \cap A_i \neq \emptyset$, then for $x \in B_j \cap A_i$ we have $f(x) = c_i$, $\psi(x) = d_j$, and so $d_j \leq c_i$. It follows that

$$\alpha \leq \sum_{i=1}^{n} \sum_{j=1}^{k} c_i m(E \cap B_j \cap A_i) = \sum_{i=1}^{n} c_i m(E \cap A_i)$$

since B_j partition $\mathbb{R}$. Therefore,

$$\sum_{i=1}^{n} c_i m(A_i \cap E) \geq \sup Y(E, f)$$

and the equality follows. □

Furthermore, we can prove the following basic properties of integrals of simple functions:

THEOREM 4.6. *Let φ, ψ be simple functions. Then,*

(i) if $\varphi \leq \psi$ and $E \in \mathcal{M}$, then $\int_E \varphi dm \leq \int_E \psi dm$;
(ii) if A, B are disjoints sets in $\mathcal{M}$, then

$$\int_{A \cup B} \varphi dm = \int_A \varphi dm + \int_B \varphi dm;$$

(iii) for all constants $a > 0$,

$$\int_E a\varphi dm = a \int_E \varphi dm.$$

PROOF. (i) Notice that $Y(E, \varphi) \subseteq Y(E, \psi)$ (we use Definition 4.4), and the supremum of a bigger set is larger.

(ii) Let $\varphi = \sum_{i=1}^n c_i \mathbf{1}_{D_i}$. Employing the properties of m, we have

$$\begin{aligned}\int_{A\cup B} \varphi dm &= \sum_{i=1}^n c_i m(D_i \cap (A \cup B)) \\ &= \sum_{i=1}^n c_i [m(D_i \cap A) + m(D_i \cap B)] \\ &= \left[\sum_{i=1}^n c_i m(D_i \cap A)\right] + \sum_{i=1}^n c_i m(D_i \cap B) \\ &= \int_A \varphi dm + \int_B \varphi dm.\end{aligned}$$

(iii) If $\varphi = \sum_{i=1}^n c_i \mathbf{1}_{A_i}$, then $a\varphi = \sum_{i=1}^n ac_i \mathbf{1}_{A_i}$ and

$$\int_E a\varphi dm = \sum_{i=1}^n ac_i m(E \cap A_i) = a\sum_{i=1}^n c_i m(E \cap A_i) = a\int_E \varphi dm$$

as required.

□

Next, we show that the properties of the integrals of simple functions extend to the integrals of nonnegative measurable functions:

THEOREM 4.7. *Suppose f and g are nonnegative measurable functions.*

(i) If $A \in \mathcal{M}$ and $f \leq g$ on A, then

$$\int_A f dm \leq \int_A g dm.$$

(ii) If $A, B \in \mathcal{M}$ satisfying $B \subseteq A$, then

$$\int_B f dm \leq \int_A f dm.$$

(iii) For $a \geq 0$,

$$\int_A af dm = a\int_A f dm.$$

(iv) If A is null, then

$$\int_A f dm = 0.$$

(v) If $A, B \in \mathcal{M}$ and $A \cap B = \emptyset$, then

$$\int_{A\cup B} f dm = \int_A f dm + \int_B f dm.$$

PROOF. (i) Notice that $Y(A, f) \subseteq Y(A, g)$, and the supremum of a bigger set is larger.

(ii) If φ is a simple function lying below f on B, then extending it by zero outside B we obtain a simple function which is below f on A. The integrals of these simple functions are the same, so $Y(B, f) \subseteq Y(A, f)$ and we conclude as in (i).

(iii) The elements of the set $Y(A, af)$ are of the form $a \times x$ where $x \in Y(A, f)$ so the same relation holds between their suprema.

(iv) For any simple function φ, $\int_A \varphi dm = 0$. To see this, take $\varphi = \sum_{i=1}^n c_i \mathbf{1}_{E_i}$, we have $m(A \cap E_i) = 0$ for each $1 \leq i \leq n$, so $Y(a, f) = \{0\}$ and $\int_A f dm = 0$.

(v) The elements of $Y(A\cup B, f)$ are of the form $\int_{A\cup B}\varphi dm$, so by Theorem 4.6 (ii) they are of the form $\int_A \varphi dm + \int_B \varphi dm$. It follows that $Y(A\cup B, f) = Y(A, f)+Y(B, f)$. We can conclude by using the following result from mathematical analysis: if $E, F \subseteq \mathbb{R}$, then $\sup(E + F) = \sup E + \sup F$.

□

We now confirm that null sets are precisely the 'negligible sets' for integration theory:

THEOREM 4.8. *Suppose f is a nonnegative measurable function. Then $f = 0$ a.e. if and only if $\int_{\mathbb{R}} f dm = 0$.*

PROOF. First, note that if $f = 0$ a.e., then there exists some $A \subseteq \mathbb{R}$ satisfying $f = 0$ pointwise in A and $m(A^c) = 0$. So, by using Theorem 4.7 (v), we have

$$\int_{\mathbb{R}} f dm = \int_A f dm + \int_{A^c} f dm = \int_A f dm = 0$$

where we have also used Theorem 4.7 (iv) in the second equality.

Conversely, given $\int_{\mathbb{R}} f dm = 0$, let $E = \{x : f(x) > 0\}$. Our goal is to show that $m(E) = 0$. Set

$$E_n = f^{-1}\left(\left[\frac{1}{n}, \infty\right)\right) \quad \text{for} \quad n \in \mathbb{N}.$$

Clearly, (E_n) increase to E with

$$E = \bigcup_{n=1}^{\infty} E_n.$$

To show that $m(E) = 0$, it is sufficient to prove that $m(E_n) = 0$ for all $n \in \mathbb{N}$, see Theorem 2.17. The function $\varphi = \frac{1}{n}\mathbf{1}_{E_n}$ is simple and $\varphi \le f$ by the definition of E_n. So

$$0 \le \int_{\mathbb{R}} \varphi dm = \frac{1}{n} m(E_n) \le \int_{\mathbb{R}} f dm = 0$$

hence $m(E_n) = 0$ for all $n \in \mathbb{N}$. This implies $m(E) = 0$, and so $f = 0$ a.e. as desired. □

REMARK. By applying a similar proof as for Theorem 4.8 above, we can show the following famous Markov inequality:

$$\text{for all } E \in \mathcal{M},\ \alpha > 0: \quad m\left(f^{-1}([\alpha, \infty)) \cap E\right) \le \frac{1}{\alpha}\int_E f dm,$$

where f is any non-negative measurable function over $\mathbb{R}$. (Exercise!)

Using the results proved so far, the following 'a.e.' version of the monotonicity of the integral is not difficult to prove:

PROPOSITION 4.9. *If f and g are nonnegative measurable functions, then $f \le g$ a.e. implies $\int_{\mathbb{R}} f dm \le \int_{\mathbb{R}} g dm$.*

PROOF. Let $A = \{x \in \mathbb{R} : f(x) \le g(x)\}$, then A^c is null and $f\mathbf{1}_A \le g\mathbf{1}_A$ (pointwise). So $\int_{\mathbb{R}} f\mathbf{1}_A dm \le \int_{\mathbb{R}} g\mathbf{1}_A dm$ by Theorem 4.7 (i). But since A^c is null, $\int_{\mathbb{R}} f\mathbf{1}_{A^c} dm = 0 = \int_{\mathbb{R}} g\mathbf{1}_{A^c} dm$. So, by (v) of the same Theorem,

$$\int_{\mathbb{R}} f dm = \int_A f dm + \int_{A^c} f dm = \int_A f dm$$
$$\le \int_A g dm = \int_A g dm + \int_{A^c} g dm = \int_{\mathbb{R}} g dm.$$

□

4.2. Monotone convergence theorem

The crux of Lebesgue integration is its convergence theory. Before we start, we introduce the following notation:

NOTATION 4.10. *For a sequence (a_n) in $\mathbb{R}$, the notation $a_n \nearrow a$ means (a_n) is nondecreasing and $\lim_{n\to\infty} a_n = a$. Similarly, for a sequence (A_n) of sets in $\mathbb{R}$, the notation $A_n \nearrow A$ means $A_n \subseteq A_{n+1}$ for all $n \in \mathbb{N}$ and $\cup_{n=1}^{\infty} A_n = A$.*

Let us start with the following famous result:

THEOREM 4.11 (Fatou's lemma). *Let $E \in \mathcal{M}$. If (f_n) is a sequence of nonnegative measurable functions, then*

$$\liminf_{n\to\infty} \int_E f_n dm \geq \int_E \left(\liminf_{n\to\infty} f_n\right) dm.$$

PROOF. Set $f = \liminf_{n\to\infty} f_n$ and recall that $f = \lim_{n\to\infty} g_n$, where $g_n = \inf_{k\geq n} f_k$ (the sequence $(g_n(x))$ is nondecreasing for each $x \in \mathbb{R}$). Let φ be a simple function satisfying $\varphi \leq f$ pointwise in $\mathbb{R}$. To show

$$\int_E f dm \leq \liminf_{n\to\infty} \int_E f_n dm,$$

it is sufficient to prove

$$\int_E \varphi dm \leq \liminf_{n\to\infty} \int_E f_n dm$$

for any such φ.

The set where $f = 0$ is irrelevant since it does not contribute to $\int_E f dm$ so we can assume, without loss of generality, that $f > 0$ on E. Put

$$\overline{\varphi}(x) = \begin{cases} \varphi(x) - \varepsilon > 0, & \text{if } \varphi(x) > 0, \\ 0, & \text{if } \varphi(x) = 0 \text{ or } x \notin E, \end{cases}$$

where $\varepsilon > 0$ is sufficiently small to ensure $\overline{\varphi} \geq 0$. Observe that $\overline{\varphi}$ is also a simple function, and $\overline{\varphi} < f$ pointwise in E. Moreover, by setting

$$A_k = \{x \in \mathbb{R} : g_k(x) \geq \overline{\varphi}(x)\},$$

we have $A_k \nearrow \mathbb{R}$ since $g_n \nearrow f$ pointwise in $\mathbb{R}$. Next, for $k \geq n$, we deduce

$$\begin{aligned} \int_{A_n\cap E} \overline{\varphi} dm &\leq \int_{A_n\cap E} g_n dm \quad (\text{as } g_n \text{ dominates } \overline{\varphi} \text{ on } A_n) \\ &\leq \int_{A_n\cap E} f_k dm \quad (\text{by the definition of } g_n) \\ &\leq \int_E f_k dm \qquad (\text{as } E \text{ is the larger set}). \end{aligned}$$

It follows that

$$\int_{A_n\cap E} \overline{\varphi} dm \leq \liminf_{k\to\infty} \int_E f_k dm. \tag{4.1}$$

Writing $\overline{\varphi} = \sum_{i=1}^{l} c_i \mathbf{1}_{B_i}$, we let $n \to \infty$:

$$\int_{A_n\cap E} \overline{\varphi} dm = \sum_{i=1}^{l} c_i m(A_n \cap E \cap B_i) \to \sum_{i=1}^{l} c_i m(E \cap B_i) = \int_E \overline{\varphi} dm$$

and the inequality (4.1) remains true in the limit:

$$\int_E \overline{\varphi} dm \leq \liminf_{k\to\infty} \int_E f_k dm.$$

We will conclude by letting $\varepsilon \to 0^+$.

Suppose that $m(\{x \in E : \varphi(x) > 0\}) < \infty$. Then

$$\int_E \overline{\varphi} dm = \int_E \varphi dm - \varepsilon m(\{x \in E : \varphi(x) > 0\})$$

and we get the result by letting $\varepsilon \to 0^+$.

The case $m(\{x \in E : \varphi(x) > 0\}) = \infty$ has to be treated separately. Here $\int_E \varphi dm = \infty$, so $\int_E f dm = \infty$. We have to show that

$$\liminf_{k\to\infty} \int_E f_k dm = \infty.$$

Let c_i be the values of φ on $D = \{x \in E : \varphi(x) > 0\}$ and let $a = \frac{1}{2}\min\{c_i\}$ ($\{c_i\}$ is a finite set!). Similarly to above setting

$$D_n = \{x \in D : g_n(x) > a\},$$

we find

$$\int_{D_n} g_n dm \geq a m(D_n) \to \infty$$

since $D_n \nearrow D$ and $m(D) = \infty$. As before

$$\int_{D_n} g_n dm \leq \int_{D_n} f_k dm \leq \int_E f_k dm$$

for $k \geq n$, the limit $\liminf_{n\to\infty} \int_E f_n dm$ has to be infinite. □

EXAMPLE 4.12. *Let $f_n = \mathbf{1}_{[n,n+1]}$. Clearly, $\int_{\mathbb{R}} f_n dm = 1$ for all n, $\liminf_{n\to\infty} f_n(x) = 0$ $(= \lim_{n\to\infty} f_n(x))$ for all $x \in \mathbb{R}$. So the inequality in Fatou's lemma could be strict as the following shows:*

$$\lim_{n\to\infty} \int_{\mathbb{R}} f_n dm = 1 > 0 = \int_{\mathbb{R}} \left(\lim_{n\to\infty} f_n\right) dm.$$

THEOREM 4.13 (Monotone convergence theorem). *Let $E \in \mathcal{M}$. If (f_n) is a sequence of nonnegative measurable functions, and $f_n(x)$ increases monotonically to $f(x)$ as $n \to \infty$ for each x, i.e. $f_n \nearrow f$ pointwise, then*

$$\lim_{n\to\infty} \int_E f_n dm = \int_E f dm.$$

PROOF. Since $f_n \leq f$ pointwise, $\int_E f_n dm \leq \int_E f dm$ and so

$$\limsup_{n\to\infty} \int_E f_n dm \leq \int_E f dm.$$

Fatou's lemma gives

$$\int_E f dm \leq \liminf_{n\to\infty} \int_E f_n dm.$$

These together with the basic relation

$$\liminf_{n\to\infty} \int_E f_n dm \leq \limsup_{n\to\infty} \int_E f_n dm$$

gives

$$\int_E f dm = \liminf_{n\to\infty} \int_E f_n dm = \limsup_{n\to\infty} \int_E f_n dm.$$

Hence, the sequence $\int_E f_n dm$ tends to $\int_E f dm$. □

REMARK. Note that the limit functions in the above two theorems, i.e. $\liminf_{n\to\infty} f_n$ in Theorem 4.11 and f in Theorem 4.13, can have value $+\infty$. However, a tricky problem is to understand the measurability of functions $f : E \to \overline{\mathbb{R}}$, where $E \in \mathcal{M}$. A possible approach is that we could define an extended $\sigma-$algebra over $\overline{\mathbb{R}}$ by setting:

$$F \in \overline{\mathcal{B}} \quad \text{if and only if} \quad F = F_1 \uplus F_2 \text{ with } F_1 \in \mathcal{B} \text{ and } F_2 \subseteq \{-\infty, +\infty\}.$$

Then, we say $f : E \to \overline{\mathbb{R}}$ is measurable if and only if the $f^{-1}(F) \in \mathcal{M}$ for every $F \in \overline{\mathcal{B}}$ in the spirit of Definition 3.4. Following a similar proof, we can show the following equivalent definition for measurability in this situation:

$$\text{for all } a \in \mathbb{R},\ f^{-1}([a, \infty]) \text{ is measurable.}$$

Its integration can be defined in the same way as in Definition 4.4.

COROLLARY 4.14. *Suppose (f_n) and f are nonnegative and measurable. If (f_n) increases to f almost everywhere, then we have $\int_E f_n dm \nearrow \int_E f dm$ for all measurable $E \in \mathcal{M}$.*

PROOF. Suppose that $f_n \nearrow f$ a.e. and A is the set where the (nondecreasing) convergence holds, so that A^c is null. We can define

$$g_n = \begin{cases} f_n, & \text{on } A, \\ 0, & \text{on } A^c, \end{cases} \qquad g = \begin{cases} f, & \text{on } A, \\ 0, & \text{on } A^c. \end{cases}$$

Then, using $E = (E \cap A^c) \cup (E \cap A)$, we get

$$\int_E g_n dm = \int_{E\cap A} f_n dm + \int_{E\cap A^c} 0 dm = \int_{E\cap A} f_n dm + \int_{E\cap A^c} f_n dm = \int_E f_n dm$$

(since $E \cap A^c$ is null) and similarly $\int_E g dm = \int_E f dm$. The (nondecreasing) convergence $g_n \to g$ holds everywhere so by monotone convergence theorem, $\int_E g_n dm \to \int_E g dm$. It follows that $\int_E f_n dm \to \int_E f dm$. This, in combined with Proposition 4.9, implies $\int_E f_n dm \nearrow \int_E f dm$. □

To apply the monotone convergence theorem, it is convenient to approximate nonnegative measurable functions by increasing sequences of simple functions.

PROPOSITION 4.15. *For any nonnegative measurable f, there is a sequence s_n of nonnegative simple functions such that $s_n \nearrow f$ pointwise in $\mathbb{R}$.*

PROOF. $\star$ For $n \in \mathbb{N}$ and $0 \le k \le 2^{2n} - 1$, define

$$E_{n,k} := \left\{ x \in \mathbb{R} : \frac{k}{2^n} \le f(x) < \frac{k+1}{2^n} \right\}, \quad E_n := \{x \in \mathbb{R} : f(x) \ge 2^n\}$$

and let

$$s_n := \left(\sum_{k=0}^{2^{2n}-1} \frac{k}{2^n} \cdot \mathbf{1}_{E_{n,k}} \right) + 2^n \cdot \mathbf{1}_{E_n}.$$

If $f(x) = 0$, then $x \in E_{n,0}$ and so $s_n(x) = 0$ for all $n \in \mathbb{N}$. If $0 < f(x) < \infty$, then $x \in E_{n,\lfloor 2^n f(x)\rfloor}$ for all $n \in \mathbb{N}$ sufficiently large, where $\lfloor\cdot\rfloor$ is the integer part. Hence

$$0 \le s_n(x) = \frac{\lfloor 2^n f(x)\rfloor}{2^n} \le \frac{\lfloor 2^{n+1} f(x)\rfloor}{2^{n+1}} = s_{n+1}(x) \to f(x)$$

as $n \to \infty$, where we have used the facts: for all nonnegative number α, we have $2\lfloor\alpha\rfloor \le \lfloor 2\alpha\rfloor$ and $\lim_{n\to\infty} \frac{\lfloor 2^n\alpha\rfloor}{2^n} = \alpha$. (To prove the second fact, you may first write α as a binary number, then the multiplication by 2^n operation can be treated as shifting all the digits to the left by n units.) □

4.3. Integrable functions

Now, we can extend the integral to general real functions, using the positive part $f^+ = \max(f, 0)$, and the negative part $f^- = \max(-f, 0)$, of any measurable function $f : \mathbb{R} \to \mathbb{R}$.

DEFINITION 4.16. *If $E \in \mathcal{M}$ and the measurable function f has both $\int_E f^+ dm$ and $\int_E f^- dm$ finite, then we say that f is integrable on E, and define*

$$\int_E f dm = \int_E f^+ dm - \int_E f^- dm.$$

The set of all functions that are integrable over E is denoted by $\mathcal{L}^1(E)$. In what follows, E will be a fixed measurable set in $\mathbb{R}$ and we often simply write $\mathcal{L}^1$ for $\mathcal{L}^1(E)$.

REMARK. If only one of $\int_E f^+ dm$ and $\int_E f^- dm$ is finite, we can also define

$$\int_E f dm = \int_E f^+ dm - \int_E f^- dm.$$

For example, for all $c < 0$ and $A \in \mathcal{M}$ satisfying $m(A) = \infty$, we have

$$\int_{\mathbb{R}} c\mathbf{1}_A dm = -\int_{\mathbb{R}} (-c\mathbf{1}_A)\, dm = cm(A) = -\infty$$

and $c\mathbf{1}_A \notin \mathcal{L}^1(\mathbb{R})$. Moreover, for a measurable function $f : E \to \overline{\mathbb{R}}$, we say it is integrable if both $\int_E f^+ dm$ and $\int_E f^- dm$ are finite, where the last two integrals are defined in the same way as in Definition 4.4. Observe that if $E \in \mathcal{M}$ with $m(E) = 0$, then for all measurable $f : \mathbb{R} \to \overline{\mathbb{R}}$ we have $\int_E f dm = 0$ since $\int_E f^+ dm = \int_E f^- dm = 0$ by Theorem 4.7 (iv).

Note that for all measurable functions $f : \mathbb{R} \to \mathbb{R}$, f is integrable if and only if $|f|$ is integrable. (You may need to check Proposition 3.8 (ii) to understand why we need f to be measurable here.) Moreover, we have

$$\int_E |f| dm = \int_E f^+ dm + \int_E f^- dm.$$

(To prove these, we may use the result in Step 2 of the proof of Theorem 4.18 below.)

The properties of the integral of nonnegative functions extend to any, not necessarily nonnegative, integrable functions.

PROPOSITION 4.17. *If f and g are integrable, and $f \le g$ a.e., then $\int_{\mathbb{R}} f dm \le \int_{\mathbb{R}} g dm$.*

PROOF. If $f \le g$ a.e., then $f^+ \le g^+$ a.e. but $f^- \ge g^-$ a.e. By using Proposition 4.9, we imply $\int_{\mathbb{R}} f^+ dm \le \int_{\mathbb{R}} g^+ dm$ and $\int_{\mathbb{R}} g^- dm \le \int_{\mathbb{R}} f^- dm$. Adding and rearranging give the result. □

REMARK. A direct consequence of Proposition 4.17 is the following: if f and g are integrable, and $f = g$ a.e., then $\int_{\mathbb{R}} f dm = \int_{\mathbb{R}} g dm$.

REMARK. We observe that many proofs of results concerning integrable functions follow a standard pattern, utilising linearity and monotone convergence properties. To prove that a result holds for all functions in a space such as $\mathcal{L}^1(E)$ we proceed in four steps:

(i) verify that the required property holds for indicator functions,
(ii) use linearity to extend the property to nonnegative simple functions,
(iii) then use the monotone convergence theorem to show that the property is shared by all nonnegative measurable functions,
(iv) finally, extend to the whole class of functions by writing $f = f^+ - f^-$ and using linearity again.

The next result gives a good illustration of the technique. We wish to show that the mapping $f \mapsto \int_A f dm$ is linear, which will allow us to show that $\mathcal{L}^1$ is a vector space.

THEOREM 4.18. *For any integrable functions f, g on E, their sum $f + g$ is also integrable on E and*

$$\int_E (f+g) dm = \int_E f dm + \int_E g dm.$$

PROOF. We apply the technique described in the remark above.

Step 1. Suppose first that f, g are nonnegative simple functions. The result is a matter of routine calculation: let $f = \sum a_i \mathbf{1}_{A_i}$ and $g = \sum b_j \mathbf{1}_{B_j}$, where i, j ranges from some finite sets, $\uplus_i A_i = \uplus_j B_j = \mathbb{R}$ and $a_i, b_j \geq 0$ for all i, j. The sum $f + g$ is also a simple function which can be written in the form

$$f + g = \sum_{i,j} (a_i + b_j) \mathbf{1}_{A_i \cap B_j}.$$

Therefore,

$$\begin{aligned}
\int_E (f+g) dm &= \sum_{i,j} (a_i + b_j) m(A_i \cap B_j \cap E) \\
&= \sum_i \sum_j a_i m(A_i \cap B_j \cap E) + \sum_j \sum_i b_j m(A_i \cap B_j \cap E) \\
&= \sum_i a_i \sum_j m(A_i \cap B_j \cap E) + \sum_j b_j \sum_i m(A_i \cap B_j \cap E) \\
&= \sum_i a_i m\left(\bigcup_j (A_i \cap B_j \cap E) \right) + \sum_j b_j m\left(\bigcup_i (A_i \cap B_j \cap E) \right) \\
&= \sum_i a_i m\left(A_i \cap \left(\bigcup_j B_j \right) \cap E \right) + \sum_j b_j m\left(\left(\bigcup_i A_i \right) \cap B_j \cap E \right) \\
&= \sum_i a_i m(A_i \cap E) + \sum_j b_j m(B_j \cap E) = \int_E f dm + \int_E g dm
\end{aligned}$$

where we have used the additivity of m in the fourth equality and the facts that A_i, B_j cover $\mathbb{R}$ separately in the sixth equality.

Step 2. Now suppose that f, g are nonnegative measurable (not necessarily simple) functions. By Proposition 4.15, we can find sequences s_n, t_n of simple functions such that $s_n \nearrow f$ and $t_n \nearrow g$ pointwise. Clearly $(s_n + t_n) \nearrow (f + g)$. Hence, using the monotone convergence theorem and the additivity property for simple functions obtained in Step 1, we obtain

$$\begin{aligned}\int_E (f+g)dm &= \lim_{n\to\infty} \int_E (s_n + t_n)dm \\ &= \lim_{n\to\infty} \int_E s_n dm + \lim_{n\to\infty} \int_E t_n dm \\ &= \int_E f dm + \int_E g dm.\end{aligned}$$

This, in particular, implies that the integral of $f + g$ is finite if the integrals of f and g are finite.

Step 3. Finally, let f, g be arbitrary integrable functions. Since

$$\int_E |f+g|dm \le \int_E (|f| + |g|)dm$$

which follows from Proposition 4.9, we can use Step 2 to deduce that the left-hand side is finite. So the assertion is proved except the last formula.

We have

$$\begin{aligned}f + g &= (f+g)^+ - (f+g)^- \\ f + g &= (f^+ - f^-) + (g^+ - g^-)\end{aligned}$$

so

$$(f+g)^+ - (f+g)^- = f^+ - f^- + g^+ - g^-.$$

We rearrange the equality to have only additions on both sides

$$(f+g)^+ + f^- + g^- = f^+ + g^+ + (f+g)^-.$$

We have nonnegative functions on both sides, so by Step 2:

$$\int_E (f+g)^+ dm + \int_E f^- dm + \int_E g^- dm = \int_E f^+ dm + \int_E g^+ dm + \int_E (f+g)^- dm.$$

It follows that

$$\int_E (f+g)^+ dm - \int_E (f+g)^- dm = \int_E f^+ dm - \int_E f^- dm + \int_E g^+ dm - \int_E g^- dm.$$

By definition of the integral, the last relation implies the formula in the theorem.

□

The following result is a routine application of monotone convergence:

PROPOSITION 4.19. *If f is integrable and $c \in \mathbb{R}$, then cf is integrable and*

$$\int_E (cf)dm = c \int_E f dm.$$

PROOF. For $c \ge 0$, the claim is obvious for simple functions $f = \sum a_i \mathbf{1}_{A_i}$, as it is just elementary algebra. For nonnegative measurable f, and positive c, take $s_n \nearrow f$ by

Proposition 4.15 and note that $cs_n \nearrow cf$. Consequently, it follows from the monotone convergence theorem that

$$\int_E cf dm = \lim_{n\to\infty} \int_E cs_n dm = \lim_{n\to\infty} c \int_E s_n dm = c \lim_{n\to\infty} \int_E s_n dm = c \int_E f dm.$$

Finally for any f and c, we can employ the usual trick introducing the positive and negative parts to conclude. In this final step, we can show the integrability of cf by observing that:

$$\int_E |cf| dm = \int_E |c||f| dm = |c| \int_E |f| dm < \infty$$

where we have used Step 2 in the second equality. □

We complete the proof that $\mathcal{L}^1(E)$ is a vector space.

THEOREM 4.20. *For any measurable $E \subseteq \mathbb{R}$, $\mathcal{L}^1(E)$ is a vector space.*

PROOF. This is a direct consequence of Theorem 4.18 and Proposition 4.19. □

We can now answer an important question on the extent to which the integral determines the integrand.

THEOREM 4.21. *Let $f, g \in \mathcal{L}^1(\mathbb{R})$. If $\int_A f dm \leq \int_A g dm$ for all $A \in \mathcal{M}$, then $f \leq g$ almost everywhere. In particular, if $\int_A f dm = \int_A g dm$ for all $A \in \mathcal{M}$, then $f = g$ almost everywhere.*

PROOF. By Theorem 4.18 and Proposition 4.19, it is sufficient to show that $\int_A h dm \geq 0$ for all $A \in \mathcal{M}$ implies $h \geq 0$ (and then take $h = g - f$). Write $A = \{x \in \mathbb{R} : h(x) < 0\}$, and we have $A = \cup_{n=1}^{\infty} A_n$ where $A_n = \{x \in (-n, n) : h(x) \leq -\frac{1}{n}\}$. By monotonicity of the integral (see Proposition 4.17)

$$\int_{A_n} h dm \leq \int_{A_n} \left(-\frac{1}{n}\right) dm = -\frac{1}{n} m(A_n)$$

which is nonnegative but this can only happen if $m(A_n) = 0$. The sequence of sets A_n increases with n, hence $m(A) = 0$. Therefore, $h \geq 0$ almost everywhere.

A similar argument shows that if $\int_A h dm \leq 0$ for all $A \in \mathcal{M}$, then $h \leq 0$ a.e. This implies the second assertion of the theorem: put $h = g - f$ and $\int_A h dm$ is both nonnegative and nonpositive for all $A \in \mathcal{M}$, hence $h \geq 0$ and $h \leq 0$ a.e. which implies $h = 0$ a.e. □

The next Proposition lists further important properties of integrable functions.

PROPOSITION 4.22. *Let $E \in \mathcal{M}$ and $f \in \mathcal{L}^1(E)$. Then*

(i) $m(E) ess\inf_E f \leq \int_E f dm \leq m(E) esssup_E f$,
(ii) $|\int_E f dm| \leq \int_E |f| dm$,
(iii) Let $g : E \to \overline{\mathbb{R}}$ be measurable. If g is integrable, then g is finite a.e.

PROOF. (i) If $m(E) = 0$ or $\operatorname{ess\,sup}_E f = \infty$ or $\operatorname{ess\,inf}_E f = -\infty$, the inequalities trivially hold. So, we assume $\operatorname{ess\,sup}_E f, \operatorname{ess\,inf}_E f \in \mathbb{R}$. Suppose $m(E) < \infty$. Observing that the functions $(\operatorname{ess\,sup}_E f)\mathbf{1}_E$, $(\operatorname{ess\,inf}_E f)\mathbf{1}_E \in \mathcal{L}^1(\mathbb{R})$, it follows from the facts

$$(\operatorname{ess\,inf}_E f)\, \mathbf{1}_E \leq f\mathbf{1}_E \leq (\operatorname{ess\,sup}_E f)\, \mathbf{1}_E \quad \text{a.e.}$$

and Proposition 4.17 that

$$m(E) \operatorname{ess\,inf}_E f \leq \int_E f dm \leq m(E) \operatorname{ess\,sup}_E f.$$

When $m(E) = \infty$, we claim that

$$\operatorname{ess\,sup}_E f \geq 0 \quad \text{and} \quad \operatorname{ess\,inf}_E f \leq 0.$$

If the above claim holds, then the results easily follow. Let us prove the claim $\operatorname{ess\,sup}_E f \geq 0$, and the other one will follow by a similar argument. To derive a contradiction, we suppose $\operatorname{ess\,sup}_E f < 0$. From a result after Definition 3.13, we have

$$f \leq \operatorname{ess\,sup}_E f \text{ a.e.} \quad \Longleftrightarrow \quad -f = f^- \geq -\operatorname{ess\,sup}_E f > 0 \text{ a.e.}$$

By Proposition 4.9, we have

$$\int_E f^- dm \geq \int_E (-\operatorname{ess\,sup}_E f) dm = -\operatorname{ess\,sup}_E f \cdot m(E) = \infty$$

which contradicts the integrability of f. So, the claim follows and then the results hold.

(ii) Observe that $-|f| \leq f \leq |f|$ pointwise. Since $|f| \in \mathcal{L}^1(E)$, we can use Proposition 4.17 to get

$$-\int_E |f| dm \leq \int_E f dm \leq \int_E |f| dm,$$

and the inequalities follow.

(iii) In order to derive a contradiction, we suppose g is not finite in some set $A \subseteq E$ with $m(A) > 0$. In other words, we have $|g(x)| = \infty$ for all $x \in A$. Then the simple functions $s_n = n\mathbf{1}_A$ satisfy $s_n \leq |g|$ pointwise in E for all $n \in \mathbb{N}$, but $\int_E s_n dm = n \cdot m(A)$ and the supremum here is ∞. Thus $\int_E |g| dm = \infty$ which contradicts the integrability of $|g|$.

□

4.4. Dominated convergence theorem

Many questions in analysis centre on conditions under which the order of two limit processes, applied to certain functions, can be interchanged. Since integration is a limit process applied to measurable functions, it is natural to ask under what conditions on a pointwise (or pointwise a.e.) convergent sequence (f_n), the limit of the integrals is the integral of the pointwise limit function f, i.e. when can we state that $\lim \int f_n dm = \int (\lim f_n) dm$? The monotone convergence theorem (Theorem 4.13) provided the answer that this conclusion is valid for monotone increasing sequence of nonnegative measurable functions. The following example shows that for general sequences of integrable functions the conclusion will not hold without some further conditions:

EXAMPLE 4.23. *Let $f_n(x) = n\mathbf{1}_{[0,\frac{1}{n}]}(x)$ for all $x \in \mathbb{R}$ and $n \in \mathbb{N}$. Clearly, $f_n \to 0$ a.e. but $\int_{\mathbb{R}} f_n dm = 1$.*

The limit theorem which turns out to be the most useful in practice states that convergence holds for an a.e. convergent sequence which is *dominated* by an integrable function.

THEOREM 4.24 (Dominated convergence theorem). *Suppose $E \in \mathcal{M}$. Let (f_n) be a sequence of measurable functions such that $|f_n| \leq g$ a.e. on E for all $n \geq 1$, where g is integrable over E. If $f = \lim_{n\to\infty} f_n$ a.e. then f is integrable over E and*

$$\lim_{n\to\infty} \int_E f_n dm = \int_E f dm.$$

To prove dominated convergence theorem, we need the following strengthened version of Fatou's lemma:

LEMMA 4.25. *Let $E \in \mathcal{M}$. If (f_n) is a sequence of a.e. nonnegative measurable functions and $f = \liminf_{n\to\infty} f_n$ a.e., then*

$$\liminf_{n\to\infty} \int_E f_n dm \geq \int_E f dm.$$

PROOF. From the assumptions, we know there exists measurable $B \subseteq \mathbb{R}$ such that $m(B) = 0$, $f_n \geq 0$ pointwise in B^c for all $n \in \mathbb{N}$, and $f = \liminf_{n\to\infty} f_n$ pointwise in B^c. So, applying Fatou's lemma (Theorem 4.11), we find

$$\liminf_{n\to\infty} \int_E f_n dm = \liminf_{n\to\infty} \int_{E\cap B^c} f_n dm \geq \int_{E\cap B^c} f dm = \int_E f dm.$$

(Here, you may also refer to Theorem 4.2 (iv) and the last assertion in the Remark after Definition 4.4.) □

PROOF OF THEOREM 4.24. Suppose for the moment that $f_n \geq 0$ a.e. on E. Fatou's lemma (Lemma 4.25) gives

$$\int_E f dm \leq \liminf_{n\to\infty} \int_E f_n dm.$$

Therefore, it is sufficient to show that

$$\limsup_{n\to\infty} \int_E f_n dm \leq \int_E f dm. \tag{4.2}$$

Fatou's lemma (Lemma 4.25) applied to $g - f_n$ gives

$$\int_E \lim_{n\to\infty} (g - f_n) dm \leq \liminf_{n\to\infty} \int_E (g - f_n) dm.$$

On the left, we have

$$\int_E \lim_{n\to\infty} (g - f_n) dm = \int_E (g - f) dm = \int_E g dm - \int_E f dm.$$

On the right

$$\liminf_{n\to\infty} \int_E (g - f_n) dm = \liminf_{n\to\infty} \left(\int_E g dm - \int_E f_n dm \right) = \int_E g dm - \limsup_{n\to\infty} \int_E f_n dm,$$

where we have used the elementary fact that

$$\liminf_{n\to\infty} (-a_n) = -\limsup_{n\to\infty} a_n.$$

Putting these together, we get

$$\int_E g dm - \int_E f dm \leq \int_E g dm - \limsup_{n\to\infty} \int_E f_n dm.$$

Finally, subtract $\int_E g dm$ (which is finite) and multiply by -1 to arrive at (4.2).

Now consider a general, not necessarily a.e. nonnegative sequence (f_n). Since by the hypothesis

$$-g(x) \leq f_n(x) \leq g(x) \quad \text{a.e. on} \quad E$$

we have

$$0 \leq f_n(x) + g(x) \leq 2g(x) \quad \text{a.e. on} \quad E,$$

we can apply the result proved for a.e. nonnegative functions to the sequence $f_n + g$ (the function $2g$ is of course integrable). □

EXAMPLE 4.26. *Consider*

$$f_n(x) = \frac{n \sin x}{1 + n^2 x^{1/2}}$$

for $x \in (0,1)$. *Clearly* $f_n(x) \to 0$ *for all* $x \in (0,1)$. *To conclude that* $\lim_{n\to\infty} \int_0^1 f_n dm = 0$, *we need an integrable dominating function. Let us evaluate:*

$$\left| \frac{n \sin x}{1 + n^2 x^{1/2}} \right| \leq \frac{n}{1 + n^2 x^{1/2}} \leq \frac{n}{n^2 x^{1/2}} = \frac{1}{n x^{1/2}} \leq \frac{1}{x^{1/2}}.$$

We shall show that the Lebesgue and Riemann (or improper) integrals of a function coincide if the latter exists, and the function $\frac{1}{x^{1/2}}$ *is integrable on* $(0,1)$ *as*

$$\int_0^1 \frac{1}{x^{1/2}} dx = \lim_{\varepsilon \to 0^+} \int_\varepsilon^1 \frac{1}{x^{1/2}} dx = \lim_{\varepsilon \to 0^+} \left[2x^{1/2} \right]_\varepsilon^1 = 2.$$

The following facts will be useful later.

PROPOSITION 4.27. *Suppose* f *is integrable and define*

$$g_n = f \mathbf{1}_{[-n,n]}, \quad h_n = \max(\min(f, n), -n)$$

(both truncate f *in some way: the* g_n *vanish outside a bounded interval, while the* h_n *are bounded). Then* $\int_{\mathbb{R}} |f - g_n| dm \to 0$, $\int_{\mathbb{R}} |f - h_n| dm \to 0$ *as* $n \to \infty$.

PROOF. If $n \to \infty$, then $g_n = f \mathbf{1}_{[-n,n]} \to f$ pointwise and hence $f - g_n \to 0$ pointwise. The convergence is dominated: $|f - g_n| = |f| \mathbf{1}_{(-\infty,-n) \cup (n,\infty)} \leq |f|$ and by the dominated convergence theorem we have $\int_{\mathbb{R}} |f - g_n| dm \to 0$. Similarly, we can get the second convergence. □

PROPOSITION 4.28. *For a sequence of nonnegative measurable functions* f_n, *we have*

$$\int_{\mathbb{R}} \sum_{n=1}^{\infty} f_n dm = \sum_{n=1}^{\infty} \int_{\mathbb{R}} f_n dm. \tag{4.3}$$

PROOF. The sequence $\sum_{k=1}^n f_k$ is nondecreasing ($f_k \geq 0$) and tends to $\sum_{k=1}^\infty f_k$ (note that the limit could be ∞ at some points $x \in \mathbb{R}$). So the monotone convergence theorem (and the Remark after it) gives

$$\begin{aligned} \int_{\mathbb{R}} \sum_{k=1}^{\infty} f_k dm &= \int_{\mathbb{R}} \lim_{n\to\infty} \sum_{k=1}^{n} f_k dm \\ &= \lim_{n\to\infty} \int_{\mathbb{R}} \sum_{k=1}^{n} f_k dm = \lim_{n\to\infty} \sum_{k=1}^{n} \int_{\mathbb{R}} f_k dm = \sum_{k=1}^{\infty} \int_{\mathbb{R}} f_k dm \end{aligned}$$

as required. (Here, we have used the result in Step 2 of the proof of Theorem 4.18.) □

As the right hand side of (4.3) may not be finite, $\sum_{n=1}^\infty f_n$ need not be integrable. However:

THEOREM 4.29 (Beppo-Levi). *Let* (f_k) *be a sequence of measurable functions satisfying*

$$\sum_{k=1}^{\infty} \int_{\mathbb{R}} |f_k| dm \quad \textit{is} \quad \textit{finite}.$$

Then, the series $\sum_{k=1}^{\infty} f_k(x)$ converges for almost all x, its sum is integrable, and

$$\int_{\mathbb{R}} \sum_{k=1}^{\infty} f_k dm = \sum_{k=1}^{\infty} \int_{\mathbb{R}} f_k dm.$$

PROOF. The function $\varphi(x) = \sum_{k=1}^{\infty} |f_k(x)|$ is a summation of nonnegative measurable functions, so by Proposition 4.28

$$\int_{\mathbb{R}} \varphi dm = \sum_{k=1}^{\infty} \int_{\mathbb{R}} |f_k| dm.$$

This is finite, so φ is integrable. Moreover, it follows from Proposition 4.22 (iii) that φ is finite a.e. Hence the series $\sum_{k=1}^{\infty} |f_k(x)|$ converges a.e. and so the series $\sum_{k=1}^{\infty} f_k(x)$ converges a.e. Let $f(x) = \sum_{k=1}^{\infty} f_k(x)$. For all partial sums, we have

$$\left| \sum_{k=1}^{n} f_k(x) \right| \leq \varphi(x) \quad \text{pointwise},$$

so we can apply the dominated convergence theorem to find

$$\begin{aligned} \int_{\mathbb{R}} f dm &= \int_{\mathbb{R}} \lim_{n\to\infty} \sum_{k=1}^{n} f_k dm \\ &= \lim_{n\to\infty} \int_{\mathbb{R}} \sum_{k=1}^{n} f_k dm = \lim_{n\to\infty} \sum_{k=1}^{n} \int_{\mathbb{R}} f_k dm = \sum_{k=1}^{\infty} \int_{\mathbb{R}} f_k dm \end{aligned}$$

as required. □

EXAMPLE 4.30. *Recalling that $\sum_{k=1}^{\infty} k x^{k-1} = \frac{1}{(1-x)^2}$ for $x \in (0,1)$ (you may need Theorem 6.5.7 in* [**1**] *to show this), we can use the Beppo-Levi theorem to evaluate the integral $\int_0^1 \left(\frac{\log x}{1-x}\right)^2 dx$. Firstly let $f_n(x) = n x^{n-1} (\log x)^2$ for $n \in \mathbb{N}$, $x \in (0,1)$, so that $f_n \geq 0$, f_n is continuous, hence measurable. Secondly, for all $n \in \mathbb{N}$, we evaluate*

$$\begin{aligned} \int_0^1 n x^{n-1} (\log x)^2 dx &= \left[x^n (\log x)^2\right]_0^1 - 2 \int_0^1 x^{n-1} \log x dx \\ &= -\frac{2}{n} \left[x^n \log x\right]_0^1 + \frac{2}{n} \int_0^1 x^{n-1} dx = \frac{2}{n^2}. \end{aligned}$$

So, $\sum_{k=1}^{\infty} \int_0^1 f_k dm = \sum_{k=1}^{\infty} \frac{2}{k^2} = \frac{\pi^2}{3}$, see Examples 7.11 in [**11**]. *By the Beppo-Levi, we infer $\left(\frac{\log x}{1-x}\right)^2 = \sum_{k=1}^{\infty} f_k$ is integrable and $\int_0^1 \left(\frac{\log x}{1-x}\right)^2 dx = \frac{\pi^2}{3}$.*

4.5. Relation to the Riemann integral

Our prime motivation for introducing the Lebesgue integral has been to provide a sound theoretical foundation for the twin concepts of measure and integral, and to serve as the model upon which an abstract theory of measure spaces can be built. Such a general theory has many applications, a principal one being the mathematical foundations of the theory of probability. At the same time, Lebesgue integration has greater scope and more flexibility in dealing with limit operations than does its Riemann counterpart.

To link the theory with the convenient techniques of elementary calculus, we need to take two further steps: to prove the fundamental theorem of calculus and to show that the

Lebesgue and Riemann integrals conincide whenever the latter exits. In the process, we shall find necessary and sufficient conditions for the existence of the Riemann integral.

We first prove the following form of the fundamental theorem:

PROPOSITION 4.31 (Fundamental theorem of calculus). *If $f : [a, b] \to \mathbb{R}$ is continuous, then f is integrable and the function F given by $F(x) = \int_a^x f dm$ is differentiable for $x \in (a, b)$, with derivative $F' = f$.*

PROOF. Continuous functions are measurable, and f is bounded on $[a, b]$, hence $f \in \mathcal{L}^1([a, b])$. Fix $a < x < x + h < b$, then $F(x + h) - F(x) = \int_x^{x+h} f dm$, since

$$\int_a^{x+h} f dm = \int_{[a,x]\cup(x,x+h]} f dm = \int_a^x f dm + \int_x^{x+h} f dm.$$

Using Proposition 4.22, we get

$$Ah \le \int_x^{x+h} f dm \le Bh,$$

where $A = \inf\{f(t) : t \in [x, x + h]\}$ and $B = \sup\{f(t) : t \in [x, x + h]\}$. Both extrema are attained, as f is continuous, so we can find t_1, t_2 in $[x, x + h]$ with $A = f(t_1)$ and $B = f(t_2)$. Thus,

$$f(t_1) \le \frac{1}{h} \int_x^{x+h} f dm \le f(t_2).$$

Letting $h \to 0^+$, the continuity of f ensures that $F'_+(x) = f(x)$. For $h \to 0^-$, we can use a similar argument to get $F'_-(x) = f(x)$. Therefore, F is differentiable at x and $F'(x) = f(x)$. □

Next, we show that Lebesgue's theory extends that of Riemann:

THEOREM 4.32. *Let $f : [a, b] \to \mathbb{R}$ be bounded.*

(i) f is Riemann integrable if and only if f is a.e. continuous with respect to Lebesgue measure on $[a, b]$.

(ii) Riemann integrable functions on $[a, b]$ are integrable with respect to Lebesgue measure on $[a, b]$ and the integrals are the same.

PROOF. We need to recall some notations and facts. Any partition

$$P = \{a_i : a = a_0 < a_1 < \cdots < a_n = b\}$$

of the interval $[a, b]$, with $\Delta_i = a_i - a_{i-1}$ $(i = 1, 2, \ldots, n)$ and with M_i (resp. m_i) the supremum (resp. infimum) of f on $I_i = [a_{i-1}, a_i]$, induces upper and lower Riemann sums $U_P = \sum_{i=1}^n M_i \Delta_i$ and $L_P = \sum_{i=1}^n m_i \Delta_i$. But these are just the Lebesgue integrals of the (signed) simple functions $u_P = \sum_{i=1}^n M_i \mathbf{1}_{I_i}$ and $l_P = \sum_{i=1}^n m_i \mathbf{1}_{I_i}$, by definition of the integral for such functions.

Choose a sequence of partitions (P_n) such that each P_{n+1} refines P_n and the length of the largest subinterval in P_n goes to 0; writing u_n for u_{P_n} and l_n for l_{P_n} we have $l_n \le f \le u_n$ a.e. for all n. Apply this on the measure space $([a, b], \mathcal{M}_{[a,b]}, m)$ where $m = m_{[a,b]}$ denotes Lebesgue measure restricted to $[a, b]$. Then $u = \inf_n u_n$ and $l = \sup_n l_n$ are measurable functions, and both sequences are monotone, since

$$l_1 \le l_2 \le \cdots \le f \le \cdots \le u_2 \le u_1 \quad \text{(a.e.)}. \tag{4.4}$$

(Note that we may have $f < l_n$ at some partition points of P_n.) Thus $u = \lim_{n\to\infty} u_n$ (pointwise), $l = \lim_{n\to\infty} l_n$ (pointwise) and all functions in (4.4) are bounded a.e. on $[a, b]$

by $M = \sup\{|f(x)| : x \in [a,b]\}$, which is integrable on $[a,b]$. By dominated convergence theorem, we conclude that

$$\lim_{n\to\infty} U_n = \lim_{n\to\infty} \int_a^b u_n dm = \int_a^b u dm, \quad \lim_{n\to\infty} L_n = \lim_{n\to\infty} \int_a^b l_n dm = \int_a^b l dm$$

and the limit functions u, l are Lebesgue integrable.

Now suppose that x is not an endpoint of any of the intervals in the partitions (P_n), which excludes only countably many points of $[a,b]$. Then we have:

$$f \text{ is continuous at } x \text{ if and only if } u(x) = f(x) = l(x). \tag{4.5}$$

This follows from the definition of continuity, since the length of each subinterval approaches 0 and so the variation of f over the intervals containing x approaches 0 if and only if f is continuous at x.

If f is Riemann integrable, the Riemann integral $\int_a^b f(x)dx$ is defined as the common value of $\lim_{n\to\infty} U_n = \int_a^b u dm$ and $\lim_n L_n = \int_a^b l dm$ whenever these limits are equal.

To prove (i), assume first that f is Riemann integrable, so that the upper and lower integrals coincide: $\int_a^b u dm = \int_a^b l dm$. But $l \le f \le u$ a.e., hence $\int_a^b (u-l) dm = 0$ means that $u = l = f$ a.e. by Theorem 4.8. Therefore, f is continuous a.e. by the above characterization of continuity of f at x (see (4.5)), which only excludes a further null set of partition points.

Conversely, if f is a.e. continuous, then $u = f = l$ a.e. and u, l are Lebesgue measurable, hence so is f. But f is also bounded by hypothesis, so it is Lebesgue integrable over $[a,b]$. As the integrands are a.e. equal, the integrals coincide:

$$\int_a^b l dm = \int_a^b f dm = \int_a^b u dm. \tag{4.6}$$

Since $\lim_{n\to\infty} U_n = \lim_{n\to\infty} L_n$, f is by definition also Riemann integrable which proves (i).

To prove (ii), note simply that if f is Riemann integrable, (i) shows that f is a.e. continuous, hence measurable, and then (4.6) shows that its Lebesgue integral coincides with the two outer integrals, hence with its Riemann integral. □

EXAMPLE 4.33. *Riemann function defined on* $[0,1]$ *by*

$$f(x) = \begin{cases} \frac{1}{\min\{n\in\mathbb{N} : nx \in \mathbb{Z}\}}, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

is a.e. continuous (see Page 125 of [**11**]*), hence Riemann integrable, and its Riemann integral equals its Lebesgue integral, which is 0, since* f *is zero outside the null set* $\mathbb{Q}$*.*

EXAMPLE 4.34 (Improper Riemann integrals). *Dealing with improper Riemann integrals involves an additional limit operation, and we define such an integral by*

$$\int_{-\infty}^{\infty} f(x)dx := \lim_{a\to-\infty,\, b\to\infty} \int_a^b f(x)dx$$

whenever the double limit exists. (Other cases of 'improper integrals' are discussed later.)

Now suppose for the function $f : \mathbb{R} \to \mathbb{R}$ *its improper Riemann integral exists. Then the Riemann integral* $\int_a^b f(x)dx$ *exists for each bounded interval* $[a,b]$*, so that* f *is a.e. continuous on each* $[a,b]$*, and thus on* $\mathbb{R}$*. The converse is false by considering the example: the function* f *which takes the value 1 on* $[n, n+1)$ *when* n *is even, and* -1 *when* n *is odd,*

is a.e. continuous (and thus Lebesgue measurable on $\mathbb{R}$) but clearly the above limits fail to exist.

If f is nonnegative, we have the following:

THEOREM 4.35. *If $f \geq 0$ and the above improper Riemann integral of f exists, then the Lebesgue integral $\int_{\mathbb{R}} f dm$ always exists and equals the improper integral.*

PROOF. Note that the sequence (f_n) with $f_n = f\mathbf{1}_{[-n,n]}$ (f_n is a.e. continuous, so Lebesgue measurable) increases monotonically to f, hence f is Lebesgue measurable. Since f_n is Riemann integrable on $[-n, n]$, the integrals coincide there, i.e.

$$\int_{\mathbb{R}} f_n dm = \int_{-n}^{n} f(x)dx$$

for each n. By hypothesis the double limit

$$\lim_{n\to\infty} \int_{-n}^{n} f(x)dx = \int_{-\infty}^{\infty} f(x)dx$$

exists. On the other hand, we have

$$\lim_{n\to\infty} \int_{\mathbb{R}} f_n dm = \int_{\mathbb{R}} f dm$$

by monotone convergence theorem, and so

$$\int_{\mathbb{R}} f dm = \int_{-\infty}^{\infty} f(x)dx$$

as required. □

REMARK. A second kind of improper Riemann integral is designed to handle functions which have asymptotes on a bounded interval, such as $f(x) = \frac{1}{x}$ on $(0, 1)$. For such cases, we can define

$$\int_a^b f(x)dx = \lim_{\varepsilon\to 0^+} \int_{a+\varepsilon}^{b} f(x)dx$$

when the limit exists. A similar result as Theorem 4.35 can be obtained.

4.6. Approximation of measurable functions

First, we will show a given measurable function f can be approximated by a sequence of special simple functions called *step functions*, eg. $g = \sum_{n=1}^{N} c_n \mathbf{1}_{I_n}$, where I_n are intervals.

THEOREM 4.36. *If $f \in \mathcal{L}^1([a, b])$ and $\varepsilon > 0$ is given, then there exists a step function h such that $\int_a^b |f - h|dm < \varepsilon$.*

PROOF. First, assume additionally that $f \geq 0$. Note that $\int_a^b f dm$ is finite and well-defined as

$$\sup\left\{\int_a^b \varphi dm : 0 \leq \varphi \leq f,\ \varphi \text{ simple}\right\}.$$

Since $f \geq \varphi$ we have $|f - \varphi| = f - \varphi$, so we can find a simple function φ satisfying

$$\int_a^b |f - \varphi|dm = \int_a^b f dm - \int_a^b \varphi dm < \frac{\varepsilon}{2}.$$

It then remains to approximate an arbitrary simple function φ which vanishes off $[a,b]$ by a step function h. The finite range $\{a_1, a_2, \ldots, a_n\}$ of the function φ partitions $[a,b]$, yielding disjoint measurable sets $E_i = \varphi^{-1}(\{a_i\}) \cap [a,b]$ such that $\cup_{i=1}^n E_i = [a,b]$. We now approximate each E_i by intervals: note that since φ is simple, $M = \sup\{\varphi(x) : x \in [a,b]\} < \infty$. By Theorem 2.16, we can find open sets O_i such that $E_i \subseteq O_i$ and $m(O_i \setminus E_i) < \frac{\varepsilon}{4nM}$ for all $1 \leq i \leq n$. Since each E_i has finite measure, so do the O_i. Observing that $O_i = \uplus_{j=1}^\infty I_{ij}$, where I_{ij} are open intervals, we have $m(O_i) = \sum_{j=1}^\infty m(I_{ij}) < \infty$. As the series converges, we can find k_i such that $m(O_i) - m(\cup_{j=1}^{k_i} I_{ij}) < \frac{\varepsilon}{4nM}$. Thus with $G_i = \cup_{j=1}^{k_i} I_{ij}$ we have $\int_a^b |\mathbf{1}_{E_i} - \mathbf{1}_{G_i}| dm = m(E_i \triangle G_i) < \frac{\varepsilon}{2nM}$ for each $1 \leq i \leq n$. So set $h = \sum_{i=1}^n a_i \mathbf{1}_{G_i}$. This step function satisfies

$$\int_a^b |\varphi - h| dm = \int_a^b \left| \sum_{i=1}^n a_i (\mathbf{1}_{E_i} - \mathbf{1}_{G_i}) \right| dm \leq \sum_{i=1}^n M \int_a^b |\mathbf{1}_{E_i} - \mathbf{1}_{G_i}| dm < \frac{\varepsilon}{2}$$

and hence $\int_a^b |f - h| dm < \varepsilon$.

The extension to general f is clear: f^+ and f^- can be approximated to within $\frac{\varepsilon}{2}$ by step functions h_1 and h_2, so with $h = h_1 - h_2$ we obtain

$$\int_a^b |f - h| dm \leq \int_a^b |f^+ - h_1| dm + \int_a^b |f^- - h_2| dm < \varepsilon$$

which completes the proof. □

With f and h as above, we can reorder the intervals $I_{ij} \cap (a,b)$ (this will guarantee the continuous function g constructed later to be zero at the two end points a, b) into a single finite sequence $(J_m)_{1 \leq m \leq n}$ with $J_m = (c_m, d_m)$ and $h = \sum_{m=1}^n a_m \mathbf{1}_{J_m}$. We may assume that $\ell(J_m) = d_m - c_m > \frac{\varepsilon'}{2}$, and approximate $\mathbf{1}_{J_m}$ by a continuous function g_m by setting $g_m = 1$ on the slightly smaller interval $(c_m + \frac{\varepsilon'}{4}, d_m - \frac{\varepsilon'}{4})$ and 0 outside J_m, while extending linearly in between. It is obvious that g_m is continuous and $\int_a^b |\mathbf{1}_{J_m} - g_m| dm < \frac{\varepsilon'}{2}$. Repeating for each J_m and taking $\varepsilon' < \frac{\varepsilon}{nK}$, where $K = \max_{1 \leq m \leq n} |a_m|$, shows that the continuous function $g = \sum_{m=1}^n a_m g_m$ satisfies $\int_a^b |h - g| dm < \frac{\varepsilon}{2}$. Combining this inequality with Theorem 4.36 yields:

THEOREM 4.37. *Given $f \in \mathcal{L}^1(\mathbb{R})$ and $\varepsilon > 0$, we can find a continuous function g, vanishing outside some finite interval such that $\int_{\mathbb{R}} |f - g| dm < \varepsilon$.*

PROOF. The preceding argument has verified this when f vanishes outside some finite interval $[a,b]$. For $f \in \mathcal{L}^1(\mathbb{R})$, choose n large enough to ensure that

$$\int_{\{|x| \geq n\}} |f| dm < \frac{\varepsilon}{2}$$

(which we can do as $\int_{\mathbb{R}} |f| dm$ is finite, see Proposition 4.27), and simultaneously choose a continuous g which vanishes outside $(-n, n)$ and satisfies $\int_{-n}^n |f - g| dm < \frac{\varepsilon}{2}$. Therefore,

$$\int_{\mathbb{R}} |f-g| dm = \int_{\{|x| \geq n\}} |f-g| dm + \int_{-n}^n |f-g| dm = \int_{\{|x| \geq n\}} |f| dm + \int_{-n}^n |f-g| dm < \varepsilon$$

as desired. □

REMARK. Using Theorem 4.36 and the argument for the proof of Theorem 4.37, we can show: given $f \in \mathcal{L}^1(\mathbb{R})$ and $\varepsilon > 0$, there exists a step function h, vanishing outside some finite interval such that $\int_{\mathbb{R}} |f - h| dm < \varepsilon$.

The well-known *Riemann-Lebesgue lemma*, which is very useful in the discussion of Fourier series, is easily deduced from the above approximation theorems:

LEMMA 4.38 (Riemann-Lebesgue). *Suppose* $f \in \mathcal{L}^1(\mathbb{R})$. *Then the sequence* $s_k = \int_{-\infty}^{\infty} f(x)\sin(kx)dx$ *and* $c_k = \int_{-\infty}^{\infty} f(x)\cos(kx)dx$ *both converge to* 0 *as* $k \to \infty$.

PROOF. We prove this for (s_k) leaving the other similar case to the reader. For simplicity of notation, write $\int$ for $\int_{-\infty}^{\infty}$. The transformation $x = y + \frac{\pi}{k}$ shows that

$$s_k = \int f\left(y + \frac{\pi}{k}\right)\sin(ky + \pi)dy = -\int f\left(y + \frac{\pi}{k}\right)\sin(ky)dy.$$

Since $|\sin x| \leq 1$,

$$\int \left|f(x) - f\left(x + \frac{\pi}{k}\right)\right| dx \geq \left|\int \left(f(x) - f\left(x + \frac{\pi}{k}\right)\right)\sin(kx)dx\right| = 2|s_k|.$$

Therefore, it suffices to prove that $\int |f(x) - f(x+h)|dx \to 0$ when $h \to 0$. This is most easily done by approximating f by a continuous g which vanishes outside some finite interval $[a, b]$, and such that $\int |f - g|dm < \frac{\varepsilon}{3}$ for a given $\varepsilon > 0$. For $|h| < 1$, the continuous function $g_h(x) = g(x + h)$ vanishes off $[a - 1, b + 1]$ and

$$\begin{aligned}&\int |f(x+h) - f(x)|dm\\ &\leq \int |f(x+h) - g(x+h)|dm + \int |g(x+h) - g(x)|dm + \int |g(x) - f(x)|dm.\end{aligned}$$

The first and last integrals on the right are less than $\frac{\varepsilon}{3}$, while the integrand of the second can be made less than $\frac{\varepsilon}{3(b-a+2)}$ whenever $|h| < \delta$, by an appropriate choice of $\delta > 0$, as g is (uniformly) continuous. Since g vanishes outside $[a - 1, b + 1]$, the second integral is also less than $\frac{\varepsilon}{3}$. Thus if $|h| < \delta$, $\int |f(x + h) - f(x)|dm < \varepsilon$. This proves that $\lim_{k\to\infty} \int f(x)\sin(kx)dx = 0$. □

4.7. Finishing remarks

This chapter is prepared mainly based on Chapter 4 of [**5**]. For the proof of Proposition 4.15, I have consulted that of Theorem 1.74 in [**8**].

CHAPTER 5

Spaces of integrable functions

Until now, we have treated the points of the measure space $(\mathbb{R}, \mathcal{M}, m)$ as the basic objects and regarded measurable or integrable functions as mappings associating real numbers with them. We now alter our point of view a little, by treating an integrable function as a 'point' in a function space, or, more precisely, as an element of a *normed vector space*. For this, we need some extra structure on the space of functions we deal with, and we need to come to terms with the fact that the measure and integral cannot distinguish between functions which are almost everywhere equal.

It is worth noting that the spaces of functions we shall discuss are all *infinite-dimensional* vector spaces: this can be seen already by considering the vector space $C([a,b],\mathbb{R})$ of real-valued continuous functions defined on $[a,b]$ and noting that a polynomial function of degree n cannot be represented as a linear combination of polynomials of lower degree.

5.1. The space L^1

First, we recall the definition of a general concept of 'distance' between points of a set:

DEFINITION 5.1. *Let X be any set. The function $d : X \times X \to \mathbb{R}$ is a metric on X (and (X,d) is called a metric space) if it satisfies:*

(i) $d(x,y) \geq 0$ *for all* $x, y \in X$*;*
(ii) $d(x,y) = 0$ *if and only if* $x = y$*;*
(iii) $d(y,x) = d(x,y)$ *for all* $x, y \in X$*;*
(iv) $d(x,z) \leq d(x,y) + d(y,z)$ *for all* $x, y, z \in X$*.*

Moreover, we say a sequence (x_n) converges to x in (X,d) if it satisfies $\lim_{n\to\infty} d(x_n, x) = 0$*.*

The property (iv) is known as the *triangle inequality* and generalizes the well-known inequality of that name for vectors in $\mathbb{R}^n$. When X is a vector space (as will be the case in almost all our examples), then there is a simple way to generate a metric by defining the distance between two vectors as the 'length' of their difference. For this, we require a further definition:

DEFINITION 5.2. *Let X be a vector space over $\mathbb{R}$ (or $\mathbb{C}$). The function $x \mapsto \|x\|$ from X into $\mathbb{R}$ is a norm on X (and $(X, \|\cdot\|)$ is called a normed vector space) if it satisfies:*

(i) $\|x\| \geq 0$ *for all* $x \in X$*;*
(ii) $\|x\| = 0$ *if and only if* $x = 0$*;*
(iii) $\|\alpha x\| = |\alpha|\,\|x\|$ *for all* $\alpha \in \mathbb{R}$ *(or* $\mathbb{C}$*),* $x \in X$*;*
(iv) $\|x + y\| \leq \|x\| + \|y\|$ *for all* $x, y \in X$*.*

Clearly, a norm $x \mapsto \|x\|$ on X induces a metric by setting $d(x,y) = \|x - y\|$. The triangle inequality follows once we observe that $\|x - z\| = \|(x - y) + (y - z)\|$ and apply (iv).

We naturally wish to use the integral to define the concept of distance between functions in $\mathcal{L}^1(E)$, for measurable $E \subseteq \mathbb{R}$. The presence of null sets in $(\mathbb{R}, \mathcal{M}, m)$ means that the integral cannot distinguish between the function that is identically 0 and one which is 0 a.e. The natural idea of defining the 'length' of the vector f as $\int_E |f| dm$ thus runs into trouble, since it would be possible for non-zero elements of $\mathcal{L}^1(E)$ to have 'zero length'.

The solution adopted is to identity functions which are a.e. equal, by defining an equivalence relation on $\mathcal{L}^1(E)$ and defining the length function for the resulting equivalence classes of functions, rather than for the functions themselves.

Thus we define

$$L^1(E) = \mathcal{L}^1(E)/\equiv$$

where the equivalence relation is given by:

$$f \equiv g \text{ if and only if } f(x) = g(x) \text{ for almost all } x \in E$$

(that is, $\{x \in E : f(x) \neq g(x)\}$ is null). Write $[f] \in L^1(E)$ for the equivalence class containing the function $f \in \mathcal{L}^1(E)$. Thus, $h \in [f]$ if and only if $h(x) = f(x)$ a.e.

Since $\mathcal{L}^1(E)$ is a vector space by Theorem 4.20, we now show that $L^1(E)$ is also a vector space. First, fix $f, g \in \mathcal{L}^1(E)$, and we claim $[f] + [g] \in L^1(E)$. Let $h \in [f] + [g]$ which means there exist $f_1 \in [f] \subseteq \mathcal{L}^1(E)$, $g_1 \in [g] \subseteq \mathcal{L}^1(E)$ such that $h = f_1 + g_1 \in \mathcal{L}^1(E)$. So, $f_1 = f$ and $g_1 = g$ a.e., which implies $h = f + g$ a.e. In other words, $h \in [f + g]$ with $f + g \in \mathcal{L}^1(E)$, so $[f] + [g] \in L^1(E)$. Similarly, we can show $a[f] \in L^1(E)$ for $a \in \mathbb{R}$ and $f \in \mathcal{L}^1(E)$. Hence $L^1(E)$ is a vector space.

Convention. By treating the equivalence class $[f]$ as if it were the function f, we implicitly identify two functions as soon as they are a.e. equal. With this convention, it will be clear that the 'length function' defined below is a genuine norm on $L^1(E)$.

We equip $L^1(E)$ with the norm

$$\|f\|_1 = \int_E |f| dm.$$

This is a norm on $L^1(E)$:

1. $\|f\|_1 = 0$ if and only if $f = 0$ a.e. (see Theorem 4.8), so that $f \equiv 0$ as an element of $L^1(E)$;
2. $\|cf\|_1 = \int_E |cf| dm = |c| \int_E |f| dm = |c| \cdot \|f\|_1$ ($c \in \mathbb{R}$);
3. $\|f + g\|_1 = \int_E |f + g| dm \leq \int_E |f| dm + \int_E |g| dm = \|f\|_1 + \|g\|_1$.

The most important feature of $L^1(E)$, from our perspective, is the fact that it is a *complete* normed vector space. The definition given below is stated for general normed vector spaces:

DEFINITION 5.3. *Let X be a vector space with norm $\|\cdot\|_X$. We say that a sequence (f_n) in X is Cauchy if*

$$\forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall n, m \geq N : \|f_n - f_m\|_X < \varepsilon.$$

If each Cauchy sequence is convergent to some element of X, then we say that X is complete. Moreover, the complete normed vector space $(X, \|\cdot\|)$ is called a Banach space.

EXAMPLE 5.4. *Let $f_n(x) = \frac{1}{x}\mathbf{1}_{[n,n+1]}(x)$ ($n \in \mathbb{N}$). Supposing that $n \neq m$, we evaluate*

$$\|f_n - f_m\|_1 = \int_0^\infty \frac{1}{x}|\mathbf{1}_{[n,n+1]} - \mathbf{1}_{[m,m+1]}|dx$$
$$= \int_n^{n+1} \frac{1}{x}dx + \int_m^{m+1} \frac{1}{x}dx = \log\frac{n+1}{n} + \log\frac{m+1}{m}.$$

For $\varepsilon > 0$, we can take N large enough such that $\log\frac{n+1}{n} < \frac{\varepsilon}{2}$ for all $n > N$. So (f_n) is a Cauchy sequence in $L^1(0,\infty)$. (When $E = (a,b)$, we write $L^1(a,b)$ for $L^1(E)$.) In addition, we have (f_n) converges to zero in $L^1(0,\infty)$.

The proof of the main result below makes essential use of the Beppo-Levi theorem (see Theorem 4.29) in order to transfer the main convergence question to that of series of real numbers; its role is essentially to provide the analogue of the fact that in $\mathbb{R}$ (and hence in $\mathbb{C}$) absolutely convergent series will always converge.

We digress briefly to recall how this property of series ensures completeness in $\mathbb{R}$: let (x_n) be a Cauchy sequence in $\mathbb{R}$, and extract a subsequence (x_{n_k}) such that $|x_n - x_{n_k}| < 2^{-k}$ for all $n \geq n_k$ as follows:

1. find $n_1 \in \mathbb{N}$ such that $|x_n - x_{n_1}| < 2^{-1}$ for all $n \geq n_1$;
2. find $n_2 > n_1$ such that $|x_n - x_{n_2}| < 2^{-2}$ for all $n \geq n_2$;
3. ...
4. find $n_k > n_{k-1}$ such that $|x_n - x_{n_k}| < 2^{-k}$ for all $n \geq n_k$.

The Cauchy property ensures each time that such n_k can be found. Now consider the telescoping series with partial sums

$$x_{n_1} + (x_{n_2} - x_{n_1}) + \cdots + (x_{n_k} - x_{n_{k-1}}) = x_{n_k}$$

which has

$$|x_{n_1}| + \sum_{i=2}^{k} |x_{n_i} - x_{n_{i-1}}| < |x_{n_1}| + \sum_{i=1}^{k-1} \frac{1}{2^i}.$$

Thus this series converges, in other words (x_{n_k}) converges in $\mathbb{R}$ and its limit is also that of the whole Cauchy sequence (x_n).

To apply the Beppo-Levi theorem below we therefore need to extract a 'rapidly convergent sequence' from the given Cauchy sequence in $L^1(E)$. This provides an a.e. limit for the original sequence, and the Fatou's lemma does the rest.

THEOREM 5.5. *The space $L^1(E)$ is complete.*

PROOF. Suppose (f_n) is a Cauchy sequence in $L^1(E)$. Let $\varepsilon = \frac{1}{2}$. There is $N_1 \in \mathbb{N}$ such that for $n \geq N_1$

$$\|f_n - f_{N_1}\|_1 < \frac{1}{2}.$$

Next, let $\varepsilon = \frac{1}{2^2}$, and for some $N_2 > N_1$ we have

$$\|f_n - f_{N_2}\|_1 < \frac{1}{2^2}$$

for $n \geq N_2$. In this way, we construct a subsequence (f_{N_k}) satisfying

$$\left\|f_{N_{n+1}} - f_{N_n}\right\|_1 < \frac{1}{2^n}$$

for all $n \in \mathbb{N}$. Hence the series $\sum_{n=1}^{\infty} \left\| f_{N_{n+1}} - f_{N_n} \right\|_1$ converges and by the Beppo-Levi theorem, the series

$$f_{N_1}(x) + \sum_{n=1}^{\infty} \left[f_{N_{n+1}}(x) - f_{N_n}(x) \right]$$

converges a.e.; denote the sum by $f(x)$. Since

$$f_{N_1}(x) + \sum_{n=1}^{k-1} \left[f_{N_{n+1}}(x) - f_{N_n}(x) \right] = f_{N_k}$$

and the left hand side converges to $f(x)$ a.e., $f_{N_k}(x)$ converges to $f(x)$ a.e.

We will show that $f \in L^1(E)$ and $\|f_n - f\|_1 \to 0$. Let $\varepsilon > 0$. The Cauchy condition gives an $N \in \mathbb{N}$ such that

$$\forall\, n, m \geq N : \ \|f_n - f_m\|_1 < \varepsilon.$$

By Fatou's lemma, we have

(5.1)

$$\|f - f_m\|_1 = \int_E |f - f_m| dm \leq \liminf_{k\to\infty} \int_E |f_{N_k} - f_m| dm = \liminf_{k\to\infty} \|f_{N_k} - f_m\|_1 \leq \varepsilon.$$

So $f - f_m \in L^1(E)$ which implies $f = (f - f_m) + f_m \in L^1(E)$, and (5.1) also gives $\|f - f_m\|_1 \to 0$. □

Applying the proof idea above, we can also get the following interesting result:

PROPOSITION 5.6. *Let (f_n) be a (function) sequence converging to f in $L^1(E)$. Then, (f_n) has a subsequence (f_{n_k}) such that f_{n_k} converges to f a.e. on E.*

PROOF. As $f_n \to f$ in $L^1(E)$, (f_n) is also a Cauchy sequence in $L^1(E)$. Then, we can follow the proof of Theorem 5.5 to get a subsequence (f_{n_k}) such that f_{n_k} converges to some $\tilde{f} \in L^1(E)$ a.e. on E. Since $L^1(E)$ is a metric space, the limit of $(f_{n_k}) \subseteq (f_n)$ must be unique and so $\tilde{f} = f$. This implies f_{n_k} converges to f a.e. on E as desired. □

REMARK. Proposition 5.6 is wrong if we replace the subsequence by the whole sequence in the statement. Indeed, consider the following famous example called 'Typewriter sequence'. Let $E = [0, 1]$, $f \equiv 0$ and f_n be defined by the formula:

$$f_n := \mathbf{1}_{\left[\frac{n-2^k}{2^k}, \frac{n-2^k+1}{2^k}\right]}$$

whenever $k \geq 0$ and $2^k \leq n < 2^{k+1}$. This is a sequence of indicator functions of intervals of decreasing length, marching across the unit interval [0,1] over and over again. Then, $f_n \to f$ in $L^1([0, 1])$, but not convergent pointwise a.e.

REMARK (Integration of complex functions). Let $E \in \mathcal{M}$. Suppose a complex function $f : E \to \mathbb{C}$ satisfies $f = \mathrm{Re}f + i\mathrm{Im}f$ with $\mathrm{Re}f, \mathrm{Im}f : E \to \mathbb{R}$ to be measurable. We say f is integrable on E if and only if both $\mathrm{Re}f$ and $\mathrm{Im}f$ are integrable on E. Moreover, we set

$$\int_E f dm = \int_E \mathrm{Re}f dm + i \int_E \mathrm{Im}f dm.$$

$\mathcal{L}^1(E, \mathbb{C})$ denotes the complex vector space containing all the complex integrable functions on E. Similarly, we can identify two complex measurable functions if their real and

imaginary parts are equal a.e. respectively. In this way, we can define the complex normed vector space $L^1(E,\mathbb{C})$ with norm

$$\|f\|_1 = \int_E |f| dm = \int_E \sqrt{(\mathrm{Re} f)^2 + (\mathrm{Im} f)^2} dm.$$

Moreover, it is not difficult to verify the linearity of the integral in this case, i.e.

$$\int_E (cf+g)dm = c\int_E f dm + \int_E g dm$$

holds for all $c \in \mathbb{C}$ and $f, g \in L^1(E,\mathbb{C})$. Finally, the following estimate still holds in the complex situation:

$$\left|\int_E f dm\right| \leq \int_E |f| dm.$$

Indeed, we know $\left|\int_E f dm\right|$ is real and it is equal to $e^{i\theta}\int_E f dm = \int_E e^{i\theta} f dm$ for some $\theta \in [0, 2\pi)$. From the definition of complex integration, we infer

$$\left|\int_E f dm\right| = \int_E e^{i\theta} f dm = \int_E \mathrm{Re}\left(e^{i\theta} f\right) dm \leq \int_E |f| dm$$

as desired.

5.2. The Hilbert space L^2

The space we now introduce plays a special role in the theory. It provides the closest analogue of the Euclidean space $\mathbb{R}^n$ among the spaces of functions, and its geometry is closely modelled on that of $\mathbb{R}^n$. It is possible, via the integral, to induce the norm via an inner product, which in turn provides a concept of orthogonality (and hence 'angles') between functions. This gives L^2 many pleasant properties, such as a 'Pythagoras theorem' and the concept of orthogonal projections, which plays vital role in many applications.

To define the norm, and hence the space $L^2(E)$ for a given measurable set $E \subseteq \mathbb{R}$, let

$$\|f\|_2 = \left(\int_E |f|^2 dm\right)^{\frac{1}{2}}$$

and *define* $\mathcal{L}^2(E)$ as the set of measurable functions for which this quantity is finite. (Note that, as for L^1, we require nonnegative integrands; it is essential that the integral is nonnegative in order for the square root to make sense. Although we always have $f^2(x) = (f(x))^2 \geq 0$ when $f(x)$ is real, the modulus is needed to include in the case of complex valued functions $f : E \to \mathbb{C}$.)

We introduce $L^2(E)$ as the set of equivalence classes of elements of $\mathcal{L}^2(E)$, under the equivalence relation $f \equiv g$ if and only $f = g$ a.e., exactly as for $L^1(E)$, and continue the convention of treating the equivalence classes as functions. If $f : E \to \mathbb{C}$ satisfies $\int_E |f|^2 dm < \infty$, we write $f \in L^2(E,\mathbb{C})$.

It is straightforward to prove that $L^2(E)$ is a vector space: clearly, for $a \in \mathbb{R}$, $|af|^2$ is integrable if $|f|^2$ is, while

$$|f+g|^2 \leq (|f|+|g|)^2 \leq 2^2 \max\{|f|^2, |g|^2\} \leq 4(|f|^2 + |g|^2)$$

shows that $L^2(E)$ is closed under addition.

We provide a simple proof that the map $f \mapsto \|f\|_2$ is a norm: to see that it satisfies the triangle inequality requires a little work. We state and prove the result for the general case of $L^2(E,\mathbb{C})$.

THEOREM 5.7 (The Cauchy-Schwarz inequality). *If $f, g \in L^2(E, \mathbb{C})$, then $fg \in L^1(E, \mathbb{C})$ and*

$$\left| \int_E f\bar{g} dm \right| \leq \|fg\|_1 \leq \|f\|_2 \|g\|_2$$

where $\bar{g}$ denotes the complex conjugate of g.

PROOF. Replacing f, g by $|f|, |g|$, we may assume that f, g are nonnegative since

$$\left| \int_E f\bar{g} dm \right| \leq \int_E |f\bar{g}| dm = \int_E |f||g| dm.$$

As we do not know in advance that $\int_E fg dm$ is finite, we shall first restrict attention to bounded measurable functions with bounded supports by setting $f_n = \min\{f, n\}\mathbf{1}_{E_n}$ and $g_n = \min\{g, n\}\mathbf{1}_{E_n}$, where $E_n = E \cap [-n, n]$.

For any $t \in \mathbb{R}$, we have

$$0 \leq \int_E (f_n + tg_n)^2 dm = \int_E f_n^2 dm + 2t \int_E f_n g_n dm + t^2 \int_E g_n^2 dm.$$

The right hand side is a quadratic in t which does not have two distinct solutions, so the discriminant is nonpositive. Thus, for all $n \in \mathbb{N}$, we have

$$\begin{aligned} \left(2 \int_E f_n g_n dm \right)^2 &\leq 4 \left(\int_E f_n^2 dm \right) \left(\int_E g_n^2 dm \right) \\ &\leq 4 \left(\int_E |f|^2 dm \right) \left(\int_E |g|^2 dm \right) = 4 \|f\|_2^2 \|g\|_2^2 . \end{aligned}$$

As $f_n \nearrow f$ and $g_n \nearrow g$ on E, monotone convergence theorem now yields

$$\|fg\|_1 = \int_E fg dm \leq \|f\|_2 \|g\|_2$$

as desired. □

The triangle inequality for the norm on $L^2(E, \mathbb{C})$ follows at once. Indeed, we need to show that $\|f + g\|_2 \leq \|f\|_2 + \|g\|_2$ for $f, g \in L^2(E, \mathbb{C})$. Note that

$$\|f + g\|_2^2 = \int_E |f + g|^2 dm = \int_E (f + g)\overline{(f + g)} dm = \int_E (f + g)(\bar{f} + \bar{g}) dm.$$

The latter integral is

$$\int_E |f|^2 dm + \int_E (f\bar{g} + \bar{f}g) dm + \int_E |g|^2 dm$$

which is dominated by $(\|f\|_2 + \|g\|_2)^2$ since the Cauchy-Schwarz inequality gives

$$\int_E (f\bar{g} + \bar{f}g) dm \leq 2 \int_E |fg| dm \leq 2 \|f\|_2 \|g\|_2 .$$

The triangle inequality follows.

The other properties are immediate:

(i) clearly $\|f\|_2 = 0$ means that $|f|^2 = 0$ a.e., consequently $f = 0$ a.e.;
(ii) for $a \in \mathbb{C}$, $\|af\|_2 = (\int_E |af|^2 dm)^{\frac{1}{2}} = |a| \|f\|_2 .$

Thus, the map $f \mapsto \|f\|_2$ is a norm on $L^2(E, \mathbb{C})$.

The proof that $L^2(E)$ is complete under this norm is similar to that for $L^1(E)$, and will be given in Theorem 5.27 below for arbitrary L^p spaces ($1 < p < \infty$).

In general, without restriction of the domain set E, neither $L^1 \subseteq L^2$ nor $L^2 \subseteq L^1$. To see this, consider $E = [1, \infty)$, $f(x) = \frac{1}{x}$. Then $f \in L^2(E)$ but $f \notin L^1(E)$. Next set $F = (0, 1)$, $g(x) = \frac{1}{\sqrt{x}}$. Now $g \in L^1(F)$ but $g \notin L^2(F)$.

For finite measure spaces, we do have a useful inclusion:

PROPOSITION 5.8. *If the set D has finite measure, i.e. $m(D) < \infty$, then $L^2(D) \subseteq L^1(D)$ (and $L^2(D, \mathbb{C}) \subseteq L^1(D, \mathbb{C})$).*

PROOF. Suppose that $f \in L^2(D)$ (or $f \in L^2(D, \mathbb{C})$). By the Cauchy-Schwarz inequality, we infer

$$\int_D |f| dm = \int_D |f| \cdot 1 dm \leq \|f\|_2 \|1\|_2 = \|f\|_2 m(D)^{\frac{1}{2}} < \infty$$

and this implies $f \in L^1(D)$ (or $f \in L^1(D, \mathbb{C})$). □

Before exploring the geometry induced on L^2 by its norm, we consider examples of sequences in L^2 to provide a little practice in determining which are Cauchy sequences for the L^2 norm, and compare this with their behaviour as elements of L^1.

EXAMPLE 5.9. *We show that the sequence $f_n = \frac{1}{x}\mathbf{1}_{[n,n+1]}$ is Cauchy in $L^2(0, \infty)$. For $n \neq m$, we evaluate*

$$\begin{aligned}\|f_n - f_m\|_2^2 &= \int_0^\infty \frac{1}{x^2} |\mathbf{1}_{[n,n+1]} - \mathbf{1}_{[m,m+1]}|^2 dx \\ &= \int_n^{n+1} \frac{1}{x^2} dx + \int_m^{m+1} \frac{1}{x^2} dx = \left(\frac{1}{n} - \frac{1}{n+1}\right) + \left(\frac{1}{m} - \frac{1}{m+1}\right) < \frac{1}{n} + \frac{1}{m}\end{aligned}$$

and for $\varepsilon > 0$ let N be large enough such that $\frac{1}{N} < \frac{\varepsilon^2}{2}$. Then $\|f_n - f_m\|_2 < \varepsilon$ whenever $m, n \geq N$ and (f_n) is Cauchy in $L^2(0, \infty)$. In addition, we have (f_n) converges to zero in $L^2(0, \infty)$.

We are ready for the additional structure specific (among the Lebesgue function spaces) to L^2:

$$\forall\, f, g \in L^2(E, \mathbb{C}) :\ (f, g) = \int_E f \bar{g} dm$$

defines an *inner product*, which induces the L^2-norm:

$$\sqrt{(f, f)} = \left(\int_E f \bar{f} dm\right)^{\frac{1}{2}} = \left(\int_E |f|^2 dm\right)^{\frac{1}{2}} = \|f\|_2 .$$

To explain what this means, we verify the following properties and all of which follow easily from the integration theory we have developed:

PROPOSITION 5.10. *Linearity (in the first argument), i.e. for all $f, g, h \in L^2(E, \mathbb{C})$ and $c \in \mathbb{C}$*

$$(cf + g, h) = c(f, h) + (g, h).$$

Conjugate symmetry, i.e. for all $f, g \in L^2(E, \mathbb{C})$

$$(f, g) = \overline{(g, f)}.$$

Positive definiteness, i.e. for all $f \in L^2(E, \mathbb{C})$

$$(f, f) \geq 0, \quad (f, f) = 0 \Leftrightarrow f = 0.$$

PROOF. We verify them one by one. Using the linearity of the integral, we have

$$(cf+g,h)=\int_E (cf+g)\bar{h}dm = c\int_E f\bar{h}dm + \int_E g\bar{h}dm = c(f,h)+(g,h)$$

and linearity in the first argument follows. For conjugate symmetry, we find

$$(f,g)=\int_E f\bar{g}dm = \int_E \overline{\bar{f}g}dm = \overline{\int_E g\bar{f}dm} = \overline{(g,f)}.$$

For all $f\in L^2(E,\mathbb{C})$, it is obvious that

$$(f,f)=\int_E |f|^2 dm \geq 0.$$

Moreover, the middle integral is equal to zero if and only if $|f|=0$ a.e. or $f=0$ in $L^2(E,\mathbb{C})$. □

As an immediate consequence, we get conjugate linearity with respect to the second argument

$$(f,cg+h)=\bar{c}(f,g)+(f,h).$$

Of course, if $f,g\in L^2$ are real valued, the inner product is real and also linear in the second argument.

We introduce the following important definition, which will be familiar from the finite dimensional context of $\mathbb{R}^n$:

DEFINITION 5.11. *An inner product on a vector space H over $\mathbb{C}$ is a map $(\cdot,\cdot): H\times H\to\mathbb{C}$ which satisfies the three conditions listed in Proposition 5.10 with $L^2(E,\mathbb{C})$ replaced by H. The pair $(H,(\cdot,\cdot))$ is called an inner product space.*

REMARK. Following the proof of Theorem 5.7, we can show the following Cauchy-Schwarz inequality is true in the general inner product space $(H,(\cdot,\cdot))$:

$$|(f,g)|\leq\sqrt{(f,f)(g,g)} \quad \text{for all} \quad f,g\in H.$$

By using this and the properties in Proposition 5.10, we can show as in the L^2 case that $\|\cdot\|=\sqrt{(\cdot,\cdot)}$ is indeed a norm on H.

EXAMPLE 5.12. *The usual scalar product in $\mathbb{R}^n$ makes this space into a real inner product space, and $\mathbb{C}^n$ equipped with $(z,w)=\sum_{i=1}^n z_i\overline{w_i}$ is a complex one.*

Proposition 5.10 shows that $L^2(E,\mathbb{C})$ is a (complex) inner product space. With the obvious simplifications in the definitions, the vector space $L^2(E,\mathbb{R})=L^2(E)$ is a real inner product space, i.e. with $\mathbb{R}$ as the set of scalars.

The following identities are immediate consequences of the above definitions.

PROPOSITION 5.13. *Let $(H,(\cdot,\cdot))$ be a complex inner product space, with induced norm $\|\cdot\|$. The following identities hold for all $h_1,h_2\in H$:*

(i) Parallelogram law:

$$\|h_1+h_2\|^2+\|h_1-h_2\|^2=2\left(\|h_1\|^2+\|h_2\|^2\right);$$

(ii) Polarization identity:

$$4(h_1,h_2)=\|h_1+h_2\|^2-\|h_1-h_2\|^2+i\left(\|h_1+ih_2\|^2-\|h_1-ih_2\|^2\right).$$

PROOF. We verify them one by one. For the first one, we evaluate

$$\|h_1+h_2\|^2=(h_1+h_2,h_1+h_2)=(h_1,h_1)+(h_1,h_2)+(h_2,h_1)+(h_2,h_2),$$
$$\|h_1-h_2\|^2=(h_1-h_2,h_1-h_2)=(h_1,h_1)-(h_1,h_2)-(h_2,h_1)+(h_2,h_2).$$

Adding together leads to the parallelogram law. For the second one, we subtract the above two equations:

$$\|h_1+h_2\|^2-\|h_1-h_2\|^2=2(h_1,h_2)+2(h_2,h_1). \tag{5.2}$$

Replacing h_2 in (5.2) by ih_2, we get

$$\|h_1+ih_2\|^2-\|h_1-ih_2\|^2=2(h_1,ih_2)+2(ih_2,h_1)=-2i(h_1,h_2)+2i(h_2,h_1).$$

Multiplying the above equation by i and adding it to (5.2) yield the polarization identity. □

REMARK. The identities in Proposition 5.13 are useful in checking that certain norms *cannot* be induced by inner products. The parallelogram law serves to characterize inner product norms: it can be proved that in the class of normed spaces, those whose norms are induced by inner products are precisely those for which the parallelogram law holds, and the inner product is then recovered from the norm via the polarization identity, see Pages 39-40 of [**17**].

Recall that for vectors in $\mathbb{R}^n$ we have the important notion of orthogonality, which means that the scalar product of two vectors is zero. This extends to any inner product space, and we say the elements $f,g\in(H,(\cdot,\cdot))$ are *orthogonal* if $(f,g)=0$.

EXAMPLE 5.14. *Consider $L^2(\mathbb{R})$ with the usual definition of inner product. If $f=\mathbf{1}_{[0,1]}$, then $(f,g)=0$ if and only if $\int_0^1 g(x)dm=0$, for example if $g(x)=x-\frac{1}{2}$.*

In fact, in any inner product space we can define the *angle* between the elements g,h by setting

$$\cos\theta=\frac{(g,h)}{\|g\|\,\|h\|}.$$

Note that by the Cauchy-Schwarz inequality this quantity lies in $[-1,1]$ and that $(g,h)=0$ means $\cos\theta=0$, i.e. θ is an odd multiple of $\frac{\pi}{2}$. It is therefore natural to say that g is *orthogonal to* h if $(g,h)=0$.

Orthogonality of vectors in a complex inner product space $(H,(\cdot,\cdot))$ provides a way of formulating Pythagoras' theorem in H: since $\|g+h\|^2=(g+h,g+h)=(g,g)+(h,h)+(g,h)+(h,g)=\|g\|^2+\|h\|^2+(g,h)+(h,g)$, we see at once that if g and h are orthogonal in H then $\|g+h\|^2=\|g\|^2+\|h\|^2$.

Now restrict attention to the case where $(H,(\cdot,\cdot))$ is complete in the inner product norm $\|\cdot\|$, recalling that this means (see Definition 5.3) that if (h_n) is a Cauchy sequence in H then there exists $h\in H$ such that $\lim_{n\to\infty}\|h_n-h\|=0$. Moreover, we call H a *Hilbert space* if this holds. In addition, if K is a closed subspace of H, then the above condition also holds for (h_n) in K and yields $h\in K$. Now, we prove the existence of orthogonal projection of a vector in H onto the subspace K.

THEOREM 5.15. *Let K be a closed subspace of the Hilbert space H. For each $h\in H$, we can find a unique $h'\in K$ such that $h''=h-h'$ is orthogonal to every element of K (or $h''\perp k$ for all $k\in K$ in the short notation). Equivalently, $\|h-h'\|=\inf\{\|h-k\|:k\in K\}$.*

PROOF. Firstly, we show the uniqueness of h'. Suppose there are $h_1, h_2 \in K$ such that $h - h_1 \perp k$ and $h - h_2 \perp k$ for all $k \in K$. In particular, as $h_1 - h_2 \in K$, we have

$$(h - h_1, h_1 - h_2) = 0 \quad \text{and} \quad (h - h_2, h_1 - h_2) = 0.$$

Subtracting two equations above, we have $-\|h_1 - h_2\|^2 = (h_2 - h_1, h_1 - h_2) = 0$ which implies $h_1 = h_2$. So, we have the uniqueness.

Secondly, we show the two conditions defining h' are equivalent. Assume that $h' \in K$ has been found so that $h'' = h - h'$ is orthogonal to every $k \in K$. Given $k \in K$, as $h' - k \in K$ and $(h'', h' - k) = 0$, Pythagoras' theorem implies

$$\|h - k\|^2 = \|(h - h') + (h' - k)\|^2 = \|h''\|^2 + \|h' - k\|^2 > \|h''\|^2$$

unless $k = h'$. Hence, $\|h''\| = \|h - h'\| = \inf\{\|h - k\| : k \in K\} =: \delta_K$. Conversely, having found $h' \in K$ such that $\|h - h'\| = \delta_K$, then for any real t and $k \in K$, we have $h' + tk \in K$ and

$$\|h - (h' + tk)\|^2 \geq \|h - h'\|^2.$$

Writing $h'' = h - h'$, we have $\|h'' - tk\|^2 \geq \|h''\|^2$ which simplifies to

$$t^2 \|k\|^2 - t[(h'', k) + (k, h'')] \geq 0. \tag{5.3}$$

Dividing both sides by $t > 0$ and letting $t \to 0^+$, and we get

$$-2\text{Re}\{(h'', k)\} = -[(h'', k) + (k, h'')] \geq 0 \quad \text{or} \quad \text{Re}\{(h'', k)\} \leq 0.$$

Similarly, dividing both sides of (5.3) by $t < 0$ and letting $t \to 0^-$, and we get

$$-2\text{Re}\{(h'', k)\} = -[(h'', k) + (k, h'')] \leq 0 \quad \text{or} \quad \text{Re}\{(h'', k)\} \geq 0.$$

Hence, $\text{Re}\{(h'', k)\} = 0$. Now, we set $t = ib$ with $b \in \mathbb{R}$. As $h' + ibk \in K$ for any $k \in K$, we can follow the above procedure to deduce

$$b^2 \|k\|^2 - 2b\text{Im}\{(h'', k)\} = b^2 \|k\|^2 + bi[(h'', k) - (k, h'')] \geq 0.$$

We can analyze as above on b, and find $\text{Im}\{(h'', k)\} = 0$. Therefore, $(h'', k) = 0$ and $h'' \perp k$ for every $k \in K$.

Finally, we prove the assertions of the theorem (in fact the last one). To find $h' \in K$ with $\|h - h'\| = \delta_K$, first choose a sequence (k_n) in K such that $\|h - k_n\| \to \delta_K$ as $n \to \infty$. Then apply the parallelogram law (see Proposition 5.13 (i)) to the vectors $h_1 = h - \frac{1}{2}(k_m + k_n)$ and $h_2 = \frac{1}{2}(k_m - k_n)$. Note that $h_1 + h_2 = h - k_n$ and $h_1 - h_2 = h - k_m$. Hence the parallelogram law reads

$$\begin{aligned}\|h - k_n\|^2 + \|h - k_m\|^2 &= 2\left(\left\|h - \frac{1}{2}(k_m + k_n)\right\|^2 + \left\|\frac{1}{2}(k_m - k_n)\right\|^2\right) \\ &\geq 2\delta_K^2 + \frac{1}{2}\|k_m - k_n\|^2 \geq 2\delta_K^2\end{aligned}$$

where we have used the fact $\frac{1}{2}(k_m + k_n) \in K$ in the first inequality. Letting $m, n \to \infty$, the left hand side converges to $2\delta_K^2$, consequently, we must have (k_n) is Cauchy in K. As K is complete[1], it converges to an element h' of K. But since $\|h - k_n\| \to \delta_K$ while $\|k_n - h'\| \to 0$ as $n \to \infty$, we infer

$$\delta_K \leq \|h - h'\| \leq \|h - k_n\| + \|k_n - h'\| \to \delta_K$$

as $n \to \infty$, which implies $\|h - h'\| = \delta_K$ as desired. □

[1] A closed set of a complete space is complete.

In writing $h = h' + h''$, we have decomposed the vector $h \in H$ as the sum of two vectors, the first being its orthogonal projection onto K, while the second is orthogonal to all vectors in K. We say that h'' is *orthogonal to* K, and denote the set of all vectors orthogonal to K by $K^\perp$. This exhibits H as a *direct sum* $H = K \oplus K^\perp$ with each vector of the first factor being orthogonal to each vector in the second factor.

REMARK. The concept of orthogonality in an inner product space leads to consideration of *orthonormal sets*, i.e. families $\{e_\alpha\}$ in H that are mutually orthogonal and have norm 1. A natural question arises whether every element of H can be represented (or at least approximated) by linear combinations of the $\{e_\alpha\}$. In $L^2([-\pi, \pi])$, this leads, for example, to *Fourier series representations* of functions in the form $f(x) = \sum_{n=0}^{\infty}(f, \psi_n)\psi_n$, where the orthonormal functions are $\psi_0 = \frac{1}{\sqrt{2\pi}}$, $\psi_{2n}(x) = \frac{1}{\sqrt{\pi}}\cos nx$, $\psi_{2n-1}(x) = \frac{1}{\sqrt{\pi}}\sin nx$, and the series converges in L^2-norm. The completeness of L^2 is crucial in ensuring the existence of such an *orthonormal* basis ($\star$ see Section 5.4 in [**4**]).

To introduce the Riesz representation theorem in Hilbert spaces, we will need some background information from functional analysis. Let H be a Hilbert space. We say $L : H \to \mathbb{C}$ is a linear functional if it satisfies

$$L(ax + y) = aL(x) + L(y) \quad \text{for all } a \in \mathbb{C} \text{ and } x, y \in H.$$

Then, we have the following equivalent definitions for a linear functional to be continuous.

PROPOSITION 5.16. *Let H be a Hilbert space and $L : H \to \mathbb{C}$ a linear functional. The following statements are equivalent.*

(a) L is continuous.
(b) L is continuous at 0.
(c) L is continuous at some point.
(d) There is a constant $c > 0$ such that $|L(h)| \le c\,\|h\| = c\sqrt{(h,h)}$ for every $h \in H$.

REMARK. The continuity of the linear functional L at the point $\bar{h} \in H$ is defined as follows: for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|L(h - \bar{h})| = |L(h) - L(\bar{h})| < \varepsilon \quad \text{for all } h \in H \text{ satisfying } \left\|h - \bar{h}\right\| < \delta.$$

It is routine to show the following is also an equivalent definition: for all sequence (h_n) converging to $\bar{h}$ in H (i.e. $\lim_{n\to\infty} \left\|h_n - \bar{h}\right\| = 0$), we have

$$\lim_{n\to\infty} |L(h_n) - L(\bar{h})| = \lim_{n\to\infty} |L(h_n - \bar{h})| = 0.$$

Moreover, we say L is continuous (over H) if and only if L is continuous at every point of H.

PROOF OF PROPOSITION 5.16. It is clear that (a) $\Rightarrow$ (b) $\Rightarrow$ (c) and (d) $\Rightarrow$ (b). Let us show that (c) $\Rightarrow$ (a) and (b) $\Rightarrow$ (d).

(c) $\Rightarrow$ (a): Suppose L is continuous at h_0 and h is any point in H. If $h_n \to h$ in H, then $h_n - h + h_0 \to h_0$. By assumption,

$$\begin{aligned} L(h_0) &= \lim_{n\to\infty}[L(h_n - h + h_0)] \\ &= \lim_{n\to\infty}[L(h_n) - L(h) + L(h_0)] = \lim_{n\to\infty} L(h_n) - L(h) + L(h_0). \end{aligned}$$

Hence, $L(h) = \lim_{n\to\infty} L(h_n)$.

(b) $\Rightarrow$ (d): The definition of continuity at 0 implies that there exists $\delta > 0$ such that

$$\{h \in H : \|h\| < \delta\} \subseteq L^{-1}(\{\alpha \in \mathbb{C} : |\alpha| < 1\}).$$

That is, $\|h\| < \delta$ implies $|L(h)| < 1$. If h is an arbitrary element of H and $\varepsilon > 0$, then $\left\| \frac{\delta h}{\|h\|+\varepsilon} \right\| < \delta$. Hence,

$$1 > \left| L\left(\frac{\delta h}{\|h\| + \varepsilon} \right) \right| = \frac{\delta}{\|h\| + \varepsilon} |L(h)|;$$

thus,

$$|L(h)| < \frac{1}{\delta}(\|h\| + \varepsilon).$$

Letting $\varepsilon \to 0^+$, we see that (d) holds with $c = \frac{1}{\delta}$. □

DEFINITION 5.17. *A bounded linear functional L on H is a linear functional for which there is a constant $c > 0$ such that $|L(h)| \leq c\,\|h\|$ for all h in H. In light of the preceding proposition, a linear functional is bounded if and only if it is continuous.*

For a bounded linear functional $L : H \to \mathbb{C}$, define

$$\|L\| = \sup\{|L(h)| : \|h\| \leq 1\}.$$

Note that by definition, $\|L\| < \infty$; $\|L\|$ is called the norm of L.

PROPOSITION 5.18. *If L is a bounded linear functional, then*

$$\begin{aligned} \|L\| &= \sup\{|L(h)| : \|h\| = 1\} \\ &= \sup\left\{ \frac{|L(h)|}{\|h\|} : h \in H,\ h \neq 0 \right\} \\ &= \inf\{c > 0 : |L(h)| \leq c\,\|h\| \text{ for all } h \in H\}. \end{aligned}$$

Also, $|L(h)| \leq \|L\|\,\|h\|$ for every $h \in H$.

PROOF. Let $\alpha = \inf\{c > 0 : |L(h)| \leq c\,\|h\| \text{ for all } h \in H\}$. It will be shown that $\|L\| = \alpha$; the remaining equalities are left as an exercise. If $\varepsilon > 0$, then the definition of $\|L\|$ shows that

$$\left| L\left(\frac{h}{\|h\| + \varepsilon} \right) \right| \leq \|L\|.$$

Hence, $|L(h)| \leq \|L\|\,(\|h\|+\varepsilon)$. Letting $\varepsilon \to 0^+$ shows that $|L(h)| \leq \|L\|\,\|h\|$ holds for all $h \in H$. So, the definition of α shows that $\alpha \leq \|L\|$. On the other hand, if $|L(h)| \leq c\,\|h\|$ for all $h \in H$, then $\|L\| \leq c$. Hence, $\|L\| \leq \alpha$. □

Fix an $h_0 \in H$ and define $L : H \to \mathbb{C}$ by $L(h) = (h, h_0)$. It is easy to see that L is linear. Also, the Cauchy-Schwarz inequality (see the Remark after Definition 5.11) gives that

$$|L(h)| = |(h, h_0)| \leq \|h\|\,\|h_0\|.$$

So, L is bounded and $\|L\| \leq \|h_0\|$. In fact

$$L\left(\frac{h_0}{\|h_0\|} \right) = \left(\frac{h_0}{\|h_0\|}, h_0 \right) = \|h_0\|,$$

so that $\|L\| = \|h_0\|$. The main result below provides a converse to these observations.

THEOREM 5.19 (The Riesz representation theorem). *If $L : H \to \mathbb{C}$ is a bounded linear functional, then there is a unique vector $h_0 \in H$ such that $L(h) = (h, h_0)$ holds for every $h \in H$. Moreover, $\|L\| = \|h_0\|$.*

PROOF. Firstly, we show the uniqueness of h_0. Suppose there exists $h' \in H$ such that $L(h) = (h, h')$ holds for every $h \in H$. It follows that $(h, h' - h_0) = 0$ holds for every $h \in H$. Letting $h = h' - h_0$, we must have $h' = h_0$ which shows the uniqueness.

If $L = 0$ (i.e. $L(h) = 0$ for all $h \in H$), then we let $h_0 = 0$. Now, suppose $L \neq 0$, and set $K = \{h \in H : L(h) = 0\}$. Because L is continuous, K is a closed linear subspace of H. As $L \neq 0$, we have $K \subsetneqq H$, consequently, $H = K \oplus K^\perp$ and $K^\perp \neq \emptyset$ (see Theorem 5.15 and the paragraph after its proof). Moreover, there exists a vector $f_0 \in K^\perp$ such that $L(f_0) = 1$. If $h \in H$ and $\alpha = L(h)$, then $L(h - \alpha f_0) = L(h) - \alpha = 0$; so $h - L(h)f_0 \in K$. Thus,

$$0 = (h - L(h)f_0, f_0) = (h, f_0) - L(h) \|f_0\|^2 .$$

So, if $h_0 = \frac{f_0}{\|f_0\|^2}$, then $L(h) = (h, h_0)$ for all $h \in H$. The last assertion follows from the paragraph before the theorem. □

A direct consequence is the following

COROLLARY 5.20. *If $F : L^2(E, \mathbb{C}) \to \mathbb{C}$ is a bounded linear functional, then there is a unique $h_0 \in L^2(E, \mathbb{C})$ such that*

$$F(h) = \int_E h\overline{h_0} dm$$

holds for every $h \in L^2(E, \mathbb{C})$.

5.3. The L^p spaces: completeness

More generally, the space $L^p(E)$ is obtained when we integrate the pth power of $|f|$. For $p \in [1, \infty)$, we say that $f \in L^p$ (and similarly for $L^p(E)$ and $L^p(E, \mathbb{C})$) if $|f|^p$ is integrable (with the same convention of identifying f and g when they are equal a.e.). Some work will be required to check that L^p is a vector space and that the 'natural generalization' of the norm introduced for L^2 is in fact a norm. We shall need $p \geq 1$ to achieve this.

DEFINITION 5.21. *For each $p \in [1, \infty)$, we define (identifying the functions in a equivalence class)*

$$L^p(E) = \left\{ f \text{ measurable} : \int_E |f|^p dm \text{ is finite} \right\}$$

and the norm on L^p is defined by

$$\|f\|_p = \left(\int_E |f|^p dm \right)^{\frac{1}{p}} .$$

(With this in mind, we denoted the norm in $L^1(E)$ by $\|f\|_1$ and that in $L^2(E)$ by $\|f\|_2$.)

Recall Definition 3.13: for any measurable function $f : E \to \overline{\mathbb{R}}$

$$\operatorname{ess\,sup} f := \inf\{c \in \overline{\mathbb{R}} : f \leq c \text{ a.e.}\}.$$

More precisely, if $F = \{c \in \overline{\mathbb{R}} : m(f^{-1}((c, \infty])) = 0\}$, we set $\operatorname{ess\,sup} f = \inf F$. It is not hard to show that the infimum belongs to F, see the paragraph after Definition 3.13.

DEFINITION 5.22. *A measurable function f satisfying ess $\sup |f| < \infty$ is said to be essentially bounded and the set of all essentially bounded functions on E is denoted by $L^\infty(E)$ (again with the usual identification of functions with a.e. equivalence classes), with the norm $\|f\|_\infty = ess \sup |f|$.*

We shall need to justify the notation by showing that for each $p \in [1,\infty]$, $(L^p(E), \|\cdot\|_p)$ is a normed vector space.

First, we observe that $L^p(E)$ is a vector space for $1 \le p < \infty$. If f and g belong to $L^p(E)$, then they are measurable, hence cf and $f+g$ are also measurable for $c \in \mathbb{R}$. As $|cf(x)|^p = |c|^p |f(x)|^p$, it follows that

$$\|cf\|_p = \left(\int_E |cf(x)|^p dm\right)^{\frac{1}{p}} = |c| \left(\int_E |f(x)|^p dm\right)^{\frac{1}{p}} = |c| \, \|f\|_p .$$

Next, we have

$$|f(x)+g(x)|^p \le 2^p \max\{|f(x)|^p, |g(x)|^p\} \le 2^p (|f(x)|^p + |g(x)|^p)$$

and so $\|f+g\|_p$ is finite if $\|f\|_p$ and $\|g\|_p$ are finite. Moreover, if $\|f\|_p = 0$ then $|f(x)|^p = 0$ a.e. and so $f(x) = 0$ a.e. The converse is obvious.

The triangle inequality

$$\|f+g\|_p \le \|f\|_p + \|g\|_p$$

is by no means obvious for general $p \in [1,\infty)$: we need to derive a famous inequality called *Young's inequality*.

REMARK. Before tackling this, we observe that the case of $L^\infty(E)$ is rather easier: $|f+g| \le |f|+|g|$ pointwise, in combined with Proposition 3.14, implies that $\|f+g\|_\infty \le \|f\|_\infty + \|g\|_\infty$ and similarly $|af| = |a||f|$ ($a \in \mathbb{R}$) gives $\|af\|_\infty = |a| \, \|f\|_\infty$. Thus $L^\infty(E)$ is a vector space. Since $\|f\|_\infty = 0$ holds if and only if $f = 0$ a.e., it follows that $\|\cdot\|_\infty$ is a norm on $L^\infty(E)$.

LEMMA 5.23 (Young's inequality). *For any nonnegative real numbers x, y and all $\alpha, \beta \in (0,1)$ with $\alpha + \beta = 1$, we have*

$$x^\alpha y^\beta \le \alpha x + \beta y.$$

PROOF. If $x = 0$, the claim is obvious. So take $x > 0$. Consider $f(t) = 1 - \beta + \beta t - t^\beta$ for $t \ge 0$. We have $f'(t) = \beta - \beta t^{\beta-1} = \beta(1 - t^{\beta-1})$. Since $0 < \beta < 1$, $f'(t) < 0$ on $(0,1)$. So f decreases on $[0,1]$ while f increases on $[1,\infty)$ as $f'(t) > 0$ on $(1,\infty)$. So $f(1) = 0$ is the only minimum point of f on $[0,\infty)$, that is $f(t) \ge 0$ for $t \ge 0$. Now set $t = \frac{y}{x}$, then $1 - \beta + \beta\frac{y}{x} - \left(\frac{y}{x}\right)^\beta \ge 0$, that is, $\left(\frac{y}{x}\right)^\beta \le \alpha + \beta\frac{y}{x}$. Multiplying both sides by $x = x^{\alpha+\beta}$, we get $x^\alpha y^\beta \le \alpha x + \beta y$ as required. □

THEOREM 5.24 (Hölder's inequality). *If $p, q \in (1,\infty)$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$, then for $f \in L^p(E)$ and $g \in L^q(E)$, we have $fg \in L^1(E)$ and*

$$\|fg\|_1 \le \|f\|_p \, \|g\|_q ,$$

that is

$$\int_E |fg| dm \le \left(\int_E |f|^p dm\right)^{\frac{1}{p}} \left(\int_E |g|^q dm\right)^{\frac{1}{q}} .$$

PROOF. Step 1. Assume that $\|f\|_p = \|g\|_q = 1$, so we only need to show that $\|fg\|_1 \le 1$. We apply Young's inequality, see Lemma 5.23, with $\alpha = \frac{1}{p}$, $\beta = \frac{1}{q}$, $x = |f|^p$ and $y = |g|^q$:

$$|fg| = x^{\frac{1}{p}} y^{\frac{1}{q}} \le \frac{1}{p}x + \frac{1}{q}y = \frac{1}{p}|f|^p + \frac{1}{q}|g|^q.$$

Integrating both sides leads to

$$\int_E |fg|dm \leq \frac{1}{p}\int_E |f|^p dm + \frac{1}{q}\int_E |g|^q dm = \frac{1}{p} + \frac{1}{q} = 1$$

since $\int_E |f|^p dm = \int_E |g|^q dm = 1$. So we have $\|fg\|_1 \leq 1$ as desired.

Step 2. For general $f \in L^p(E)$ and $g \in L^q(E)$, we set $\|f\|_p = a$ and $\|g\|_q = b$ for some $a, b > 0$. (If either a or b is zero, then one of the functions is zero a.e. and the inequality is trivial.) Hence the functions $\tilde{f} = \frac{1}{a}f$ and $\tilde{g} = \frac{1}{b}g$ satisfy the assumption of Step 1, and so $\left\|\tilde{f}\tilde{g}\right\|_1 \leq \left\|\tilde{f}\right\|_p \|\tilde{g}\|_q$. This yields

$$\frac{1}{ab}\|fg\|_1 \leq \frac{1}{a}\|f\|_p \cdot \frac{1}{b}\|g\|_q$$

and after multiplying by ab the result is proved.

□

Letting $p = q = 2$ and recalling the definition of the scalar product in L^2, we obtain the following familiar special case of Hölder's inequality.

COROLLARY 5.25 (Cauchy-Schwarz inequality). *If $f, g \in L^2(E)$, then*

$$|(f, g)| \leq \|f\|_2 \|g\|_2 .$$

REMARK. Note that if $p = 1$ in Hölder's inequality, see Theorem 5.24, we have $q = \infty$ and the inequality is still true. Indeed, for $f \in L^1(E)$ and $g \in L^\infty(E)$, we find

$$\|fg\|_1 = \int_E |fg|dm \leq \|g\|_\infty \int_E |f|dm = \|f\|_1 \|g\|_\infty$$

where we have used the fact $|fg| \leq |f| \|g\|_\infty$ a.e. on E in the first inequality.

THEOREM 5.26 (Minkowski's inequality). *For each $p \in [1, \infty)$ and $f, g \in L^p(E)$, we have*

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p .$$

PROOF. Assume $1 < p < \infty$ (the case $p = 1$ was done earlier). We have

$$|f + g|^p = |f + g||f + g|^{p-1} \leq |f||f + g|^{p-1} + |g||f + g|^{p-1}$$

Taking $q \in (1, \infty)$ such that $\frac{1}{p} + \frac{1}{q} = 1$, in other words, $p + q = pq$, we obtain

$$\int_E |f + g|^{(p-1)q} dm = \int_E |f + g|^p dm < \infty$$

as L^p is a vector space. Hence $|f + g|^{p-1} \in L^q(E)$ and

$$\left\||f + g|^{p-1}\right\|_q = \left(\int_E |f + g|^p dm\right)^{\frac{1}{q}}.$$

We apply Hölder's inequality:

$$\begin{aligned}
\int_E |f + g|^p dm &\leq \int_E |f||f + g|^{p-1} dm + \int_E |g||f + g|^{p-1} dm \\
&\leq \|f\|_p \left(\int_E |f + g|^p dm\right)^{\frac{1}{q}} + \|g\|_p \left(\int_E |f + g|^p dm\right)^{\frac{1}{q}} \\
&= (\|f\|_p + \|g\|_p) \cdot A
\end{aligned}$$

with $A = \left(\int_E |f+g|^p dm\right)^{\frac{1}{q}}$. If $A = 0$, then $\|f+g\|_p = 0$ and there is nothing to prove. So suppose $A > 0$ and divide by A:

$$\|f+g\|_p = \left(\int_E |f+g|^p dm\right)^{1-\frac{1}{q}} = \frac{1}{A}\int_E |f+g|^p dm \leq \|f\|_p + \|g\|_p$$

as desired. □

Next we prove that $L^p(E)$ is an example of a complete normed vector space (i.e. a Banach space) for $p \in (1,\infty)$, i.e. every Cauchy sequence in $L^p(E)$ converges in norm to an element of $L^p(E)$. The proof is quite similar to the case $p = 1$.

THEOREM 5.27. *The space $L^p(E)$ is complete for $p \in (1,\infty)$.*

PROOF. Given a Cauchy sequence (f_n) (that is, $\|f_n - f_m\|_p \to 0$ as $n, m \to \infty$), we find a subsequence f_{n_k} with

$$\|f_{n_{k+1}} - f_{n_k}\|_p < \frac{1}{2^k}$$

for all $k \geq 1$ and we set

$$g_k = \sum_{i=1}^{k} |f_{n_{i+1}} - f_{n_i}|, \quad g = \lim_{k\to\infty} g_k = \sum_{i=1}^{\infty} |f_{n_{i+1}} - f_{n_i}|.$$

The Minkowski's inequality (or triangle inequality) yields

$$\|g_k\|_p \leq \sum_{i=1}^{k} \|f_{n_{i+1}} - f_{n_i}\|_p \leq \sum_{i=1}^{k} \frac{1}{2^i} \leq 1.$$

Now, we can apply Fatou's lemma to the nonnegative measurable functions g_k^p, $k \geq 1$:

$$\|g\|_p^p = \int_E \lim_{k\to\infty} g_k^p dm \leq \liminf_{k\to\infty} \int_E g_k^p dm \leq 1.$$

Hence, g is almost everywhere finite (see Proposition 4.22 (iii)) and $f_{n_1} + \sum_{i=1}^{\infty}(f_{n_{i+1}} - f_{n_i})$ converges absolutely almost everywhere, defining a measurable function f as its sum.

We need to show that $f \in L^p(E)$. Note first that $f = \lim_{k\to\infty} f_{n_k}$ a.e., and given $\varepsilon > 0$ we can find $N \in \mathbb{N}$ such that $\|f_n - f_m\|_p \leq \varepsilon$ for $m, n \geq N$. Applying Fatou's lemma to the sequence $(|f_{n_i} - f_m|^p)_{i=1}^{\infty}$, we have

$$\int_E |f - f_m|^p dm \leq \liminf_{i\to\infty} \int_E |f_{n_i} - f_m|^p dm \leq \varepsilon^p.$$

Hence $f - f_m \in L^p(E)$ and so $f = f_m + (f - f_m) \in L^p(E)$ and we have $\|f - f_m\|_p < \varepsilon$ for all $m \geq N$. Thus $f_m \to f$ in L^p-norm as $m \to \infty$ as required. □

THEOREM 5.28. *The space $L^\infty(E)$ is complete.*

PROOF. Let (f_n) be a Cauchy sequence in $L^\infty(E)$. For every $k \in \mathbb{N}$, there is $N_k \in \mathbb{N}$ such that $\|f_m - f_n\|_\infty < \frac{1}{k}$ for $m, n \geq N_k$. Hence, there is a null set $E_k \subseteq E$ such that

$$|f_m(x) - f_n(x)| < \frac{1}{k} \quad \text{for all } x \in E \setminus E_k \text{ and } m, n \geq N_k. \tag{5.4}$$

Then, we let $F = \cup_{k=1}^{\infty} E_k$ which is null and we see that for all $x \in E \setminus F$, the sequence $(f_n(x))$ is Cauchy in $\mathbb{R}$. Thus $f_n(x)$ converges to a number denoted $f(x)$ for all $x \in E \setminus F$. Letting $m \to \infty$ in (5.4), we obtain

$$|f(x) - f_n(x)| \leq \frac{1}{k} \quad \text{for all } x \in E \setminus F \text{ and } n \geq N_k.$$

We conclude that $f \in L^\infty(E)$ and $\|f - f_n\|_\infty \leq \frac{1}{k}$ for all $n \geq N_k$. Therefore, $f_n \to f$ in $L^\infty(E)$ as desired. □

We have the following relations between the L^p spaces for different p which generalize Proposition 5.8.

THEOREM 5.29. *If E has finite Lebesgue measure, then $L^q(E) \subseteq L^p(E)$ when $1 \leq p \leq q \leq \infty$.*

PROOF. Fix an arbitrary $f \in L^q(E)$. If $p = q$, the assertion is trivial. We assume $p < q$. If $q = \infty$, then

$$\int_E |f|^p dm \leq \|f\|_\infty^p \, m(E) < \infty$$

as $|f| \leq \|f\|_\infty$ a.e. on E. Now, we assume $p < q < \infty$. Since $s = \frac{q}{p} \in (1, \infty)$, we can apply Hölder's inequality (see Theorem 5.24) as follows:

$$\int_E |f|^p dm = \int_E |f|^p \cdot 1 dm \leq \left(\int_E |f|^{ps} \right)^{\frac{1}{s}} \left(\int_E 1^{s'} dm \right)^{\frac{1}{s'}} = \|f\|_q^p \, (m(E))^{\frac{1}{s'}} < \infty$$

where $s' \in (1, \infty)$ satisfies $\frac{1}{s} + \frac{1}{s'} = 1$. This completes the proof. □

5.4. Modes of convergence

Let E be a (Lebesgue) measurable set of $\mathbb{R}$. For a given sequence (f_n) in $L^p(E)$ with $p \in [1, \infty)$, we can express the statement '$f_n \to f$ as $n \to \infty$' in a number of distinct ways:

DEFINITION 5.30. *(1) $f_n \to f$ in $L^\infty(E)$: given $\varepsilon > 0$, there exists $N = N(\varepsilon) \in \mathbb{N}$ such that, for all $n \geq N$,*

$$\|f_n - f\|_\infty = \textit{ess}\sup_{x \in E}(|f_n(x) - f(x)|) < \varepsilon.$$

(Note that we need $f_n \in L^\infty(E)$ for the ess sup to be finite in general.)

(2) $f_n \to f$ pointwise a.e. on E: there is a null set $F \subseteq E$ such that $f_n \to f$ pointwise on $E \setminus F$.

(3) $f_n \to f$ in L^p-norm (in the pth mean): $\|f_n - f\|_p \to 0$ as $n \to \infty$, i.e., for given $\varepsilon > 0$, there exists $N = N(\varepsilon) \in \mathbb{N}$ such that

$$\|f_n - f\|_p = \left(\int_E |f_n - f|^p dm \right)^{\frac{1}{p}} < \varepsilon$$

for all $n \geq N$.

(4) $f_n \to f$ in measure: for each $\varepsilon > 0$,

$$m(\{x \in E : |f_n(x) - f(x)| > \varepsilon\}) \to 0$$

as $n \to \infty$.

Handling these different modes of convergence requires some care; at this point we have not even shown that they are all genuinely different. Clearly, pointwise a.e. limits are often easier to 'guess', however, we cannot always be certain that the limit function, if it exists, is again a member of $L^p(E)$. For mean convergence, however, this is ensured by the completeness of $L^p(E)$. Note that the conclusions of the dominated and monotone convergence theorems yield the mean convergence of a.e. convergent sequences (f_n), but only by imposing additional conditions on the (f_n). In order to understand the relation between a.e. convergence and convergence in measure, we first show the following

LEMMA 5.31. *Let $E \in \mathcal{M}$ with $m(E) < \infty$, and (f_n) be a sequence of measurable functions on E. The following conditions are equivalent:*
(a) $f_n \to 0$ pointwise a.e. on E;
(b) for each $\varepsilon > 0$,

$$\lim_{k\to\infty} m\left(\bigcup_{n=k}^{\infty}\{x \in E : |f_n(x)| \geq \varepsilon\}\right) = 0.$$

PROOF. Convergence a.e., expressed succinctly, means that

$$m(\{x \in E : \forall\, \varepsilon > 0,\, \exists\, N \in \mathbb{N},\, \forall\, n > N,\, |f_n(x)| < \varepsilon\}) = m(E).$$

Writing this set of full measure another way, we have

$$m\left(\bigcap_{\varepsilon>0}\bigcup_{N\in\mathbb{N}}\bigcap_{n\geq N}\{x \in E : |f_n(x)| < \varepsilon\}\right) = m(E).$$

The measure of the outer intersection (over all $\varepsilon > 0$) is less than the measure of any of its terms, but being already $m(E)$, it cannot increase, hence for all $\varepsilon > 0$

$$m\left(\bigcup_{N\in\mathbb{N}}\bigcap_{n\geq N}\{x \in E : |f_n(x)| < \varepsilon\}\right) = m(E).$$

We have a union of increasing sets so

$$\lim_{N\to\infty} m\left(\bigcap_{n\geq N}\{x \in E : |f_n(x)| < \varepsilon\}\right) = m(E),$$

thus

$$\lim_{N\to\infty}\left[m(E) - m\left(\bigcap_{n\geq N}\{x \in E : |f_n(x)| < \varepsilon\}\right)\right] = 0.$$

As $m(E) < \infty$, we have

$$\begin{aligned} m(E) - m\left(\bigcap_{n\geq N}\{x \in E : |f_n(x)| < \varepsilon\}\right) &= m\left(E \setminus \left[\bigcap_{n\geq N}\{x \in E : |f_n(x)| < \varepsilon\}\right]\right) \\ &= m\left(\bigcup_{n\geq N}\{x \in E : |f_n(x)| \geq \varepsilon\}\right) \end{aligned}$$

by De Morgan's law. Hence, (a) implies (b). Working backwards, these steps also prove the converse. Note that, in the converse direction, you may use the following equivalence

$$\bigcap_{\varepsilon>0}\bigcup_{N\in\mathbb{N}}\bigcap_{n\geq N}\{x \in E : |f_n(x)| < \varepsilon\} = \bigcap_{\varepsilon>0,\varepsilon\in\mathbb{Q}}\bigcup_{N\in\mathbb{N}}\bigcap_{n\geq N}\{x \in E : |f_n(x)| < \varepsilon\}.$$

□

The following result is a direct consequence of Lemma 5.31:

THEOREM 5.32. *Let $E \in \mathcal{M}$ with $m(E) < \infty$, and (f_n) be a sequence of measurable functions on E such that $f_n \to f$ pointwise a.e. on E. Then, $f_n \to f$ in measure.*

PROOF. Consider the difference $g_n = f_n - f$ and the problem reduces to the discussion of convergence of g_n to zero. We have

$$\lim_{k\to\infty} m\left(\bigcup_{n=k}^{\infty}\{x\in E : |g_n(x)|\geq\varepsilon\}\right) \geq \limsup_{k\to\infty} m\left(\{x\in E : |g_k(x)|\geq\varepsilon\}\right)$$
$$\geq \liminf_{k\to\infty} m\left(\{x\in E : |g_k(x)|\geq\varepsilon\}\right) \geq 0$$

and by Lemma 5.31 the limit on the left is zero hence so is that in the middle. □

For the relations between the convergence in Definition 5.30, we have the following

THEOREM 5.33. *Let $E\in\mathcal{M}$, and (f_n) be a sequence of measurable functions on E. The only valid implications are the following: (1)⇒(2), (1)⇒(4) and (3)⇒(4). If $m(E)<\infty$, then we also have (1)⇒(3) and (2)⇒(4).*

PROOF. (1)⇒(2): let $\varepsilon=\frac{1}{n}$ in (1), and we can find $N\in\mathbb{N}$ and full set E_n such that

$$|f_n(x)-f(x)|<\frac{1}{n}\quad\text{for all } x\in E_n.$$

So, $f_n\to f$ pointwise in the full set $\tilde{E}=\cap_{n=1}^{\infty}E_n$, i.e., $f_n\to f$ pointwise a.e. on E.

(1)⇒(4): it directly follows from the definition.

(3)⇒(4): applying Markov inequality (see the Remark after Theorem 4.3), we have

$$m(\{x\in E : |f_n(x)-f(x)|^p>\varepsilon\})\leq\frac{1}{\varepsilon}\int_E |f_n-f|^p dm,$$

which implies the desired result.

[(3) or (4)]⇏[(1) and (2)]: consider the 'Typewriter sequence' constructed in the Remark after Proposition 5.1.

[(2) or (4)]⇏[(1) and (3)]: consider the 'Escape to vertical infinity' sequence. Let $E=[0,1]$ and $f_n=n^{\frac{1}{p}}\mathbf{1}_{[0,\frac{1}{n}]}$. Clearly, we have $f_n\to 0$ a.e. and $f_n\to 0$ in measure, but $\|f_n\|_\infty=n^{\frac{1}{p}}$ and $\|f_n\|_p=1$ for all $n\in\mathbb{N}$.

(2)⇏(4): consider the 'Escape to horizontal infinity' sequence. Let $E=[0,\infty)$ and $f_n=\mathbf{1}_{[n,n+1]}$. Clearly, $f_n\to 0$ a.e. on E but $m(\{x\in E : |f_n(x)|>\varepsilon\})=1$ for $\varepsilon\in(0,1)$.

(1)⇏(3): consider the 'Escape to width infinity' sequence. Let $E=[0,\infty)$ and $f_n=n^{-\frac{1}{p}}\mathbf{1}_{[0,n]}$. Clearly, $f_n\to 0$ in $L^\infty(E)$ but $\|f_n\|_p=1$ for all $n\in\mathbb{N}$.

In summary, see Figure 5.1(A).

Now, suppose $m(E)<\infty$. The implication (2)⇒(4) follows from Theorem 5.32, while the implication (1)⇒(3) follows from Theorem 5.29 and also its proof. Therefore, in this case, the relations are shown in Figure 5.1(B). □

EXAMPLE 5.34. *We investigate $h_n(x)=x^n$ for convergence in each of these modes on $E=[0,1]$.*

The sequence converges everywhere to the function $h(x)=0$ for $x\in[0,1)$, $h(1)=1$, and so it also converges a.e. to zero.

It does not converge in L^∞ since $\|h_n\|_\infty=1$ for all $n\in\mathbb{N}$.

It converges in L^p for $p\geq 1$:

$$\int_0^1 |h_n(x)-h(x)|^p dx=\int_0^1 x^{pn}dx=\left[\frac{1}{pn+1}x^{pn+1}\right]_0^1\to 0.$$

This also implies $h_n\to 0$ in measure.

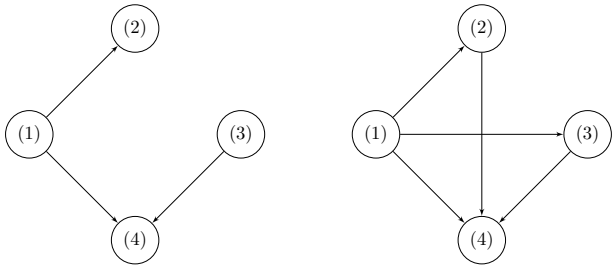


FIGURE 5.1. Relations between the convergence in Definition 5.30: (A) general case; (B) $m(E) < \infty$.

REMARK. There are still other modes of convergence which can be considered for sequences of measurable functions, and the relations between these and the above are quite complex in general.

5.5. Finishing remarks

This chapter is prepared mainly based on Chapter 5, Section 8.1 and Section 8.2.1 of [**5**]. For the proof of Theorem 5.28, I have consulted that of Theorem 4.8 in [**4**]. Moreover, I have also used the materials from Section I.3 in [**6**] to write the parts related to Riesz representation theorem in Section 5.2. For the last section, I have also used some nice examples from Section 1.5 in [**16**].

CHAPTER 6

Product measures

6.1. Multi-dimensional Lebesgue measure

In Chapter 2, we constructed Lebesgue measure on the real line. The basis for that was the notion of the length of an interval. Consider now the plane $\mathbb{R}^2$ in place of $\mathbb{R}$. Here instead of intervals we use rectangles:

$$R = I_1 \times I_2,$$

where I_1, I_2 are any intervals. The 'length' of a rectangle is its area

$$a(R) = \ell(I_1) \cdot \ell(I_2).$$

The concept of null set is introduced as in the one-dimensional case. As before, countable sets are null. It is worth noting that on the plane we have more sophisticated null sets such as, for example, a line segment or the graph of a function.

The whole construction goes through without change and the resulting measure is the Lebesgue measure m_2 on the plane defined on the $\sigma-$algebra generated by the rectangles.

A subtle point is the following: any set of the form $A \times \{a\}$, $A \subseteq \mathbb{R}$ and $a \in \mathbb{R}$, is null and hence Lebesgue measurable on the plane. An interesting case of this is when A is a non-measurable set on the real line.

Next, we consider $\mathbb{R}^3$. Instead of intervals, we use cubes, and the 'length' is its volume:

$$C = I_1 \times I_2 \times I_3, \quad v(C) = \ell(I_1) \cdot \ell(I_2) \cdot \ell(I_3).$$

Now surfaces are examples of null sets. Following the same construction, we obtain Lebesgue measure m_3 in $\mathbb{R}^3$.

Finally, we consider $\mathbb{R}^n$. 'Intervals' are now $n-$dimensional cubes:

$$\mathbf{I} = I_1 \times \cdots \times I_n$$

and generalized 'length' is given by

$$\ell(\mathbf{I}) = \ell(I_1) \cdot \cdots \cdot \ell(I_n).$$

An interesting example of a null set is a hyperplane, which means $(n-1)-$dimensional subspace of $\mathbb{R}^n$.

The above provides motivation for what follows. The multi-dimensional Lebesgue measures will emerge again from the considerations below where we will work with general measure spaces.

6.2. Product $\sigma-$algebras

Let $(\Omega_1, \mathcal{F}_1, P_1)$, $(\Omega_2, \mathcal{F}_2, P_2)$ be two measure spaces. Set

$$\Omega = \Omega_1 \times \Omega_2.$$

We want to define a measure P on Ω to 'agree' with the measures given on Ω_1, Ω_2. Before we construct P, we need to specify its domain, i.e. a $\sigma-$algebra on Ω.

DEFINITION 6.1. *Let $\mathcal{F}$ be the smallest $\sigma-$algebra of subsets of Ω containing the 'rectangles' $A_1 \times A_2$ for all $A_1 \in \mathcal{F}_1$, $A_2 \in \mathcal{F}_2$. We call $\mathcal{F}$ the product $\sigma-$algebra of $\mathcal{F}_1$ and $\mathcal{F}_2$. In other words, the product $\sigma-$algebra is generated by the family of sets ('rectangles')*

$$\mathcal{R} = \{A_1 \times A_2 : A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2\}.$$

The notation used for the product $\sigma-$algebra is $\mathcal{F} = \mathcal{F}_1 \times \mathcal{F}_2$.

There are many ways in which the same product $\sigma-$algebra may be generated, and each of them will prove useful in the sequel.

THEOREM 6.2. *(i) The product $\sigma-$algebra $\mathcal{F}_1 \times \mathcal{F}_2$ is generated by the family of sets ('cylinders' or 'strips')*

$$\mathcal{C} = \{A_1 \times \Omega_2 : A_1 \in \mathcal{F}_1\} \cup \{\Omega_1 \times A_2 : A_2 \in \mathcal{F}_2\}.$$

(ii) The product $\sigma-$algebra $\mathcal{F}$ is the smallest $\sigma-$algebra such that the projections

$$Pr_1 : \Omega \to \Omega_1, \quad Pr_1(w_1, w_2) = w_1$$
$$Pr_2 : \Omega \to \Omega_2, \quad Pr_2(w_1, w_2) = w_2$$

are measurable[1].

PROOF. We will write $\sigma(\mathcal{E})$ for the $\sigma-$algebra generated by a family $\mathcal{E}$.

Clearly $\mathcal{C}$ is contained in $\mathcal{R}$ hence the $\sigma-$algebra generated by $\mathcal{C}$ is smaller than the $\sigma-$algebra generated by $\mathcal{R}$, i.e. $\sigma(\mathcal{C}) \subseteq \sigma(\mathcal{R})$. On the other hand,

$$A_1 \times A_2 = (A_1 \times \Omega_2) \cap (\Omega_1 \times A_2),$$

hence the rectangles belong to the $\sigma-$algebra generated by cylinders: $\mathcal{R} \subseteq \sigma(\mathcal{C})$. This implies $\sigma(\mathcal{R}) \subseteq \sigma(\sigma(\mathcal{C})) = \sigma(\mathcal{C})$ which completes the proof of (i).

For (ii), note that

$$Pr_1^{-1}(A_1) = A_1 \times \Omega_2, \quad Pr_2^{-1}(A_2) = \Omega_1 \times A_2,$$

hence the projections are measurable by (i). But the smallest $\sigma-$algebra such that they are both measurable is the smallest $\sigma-$algebra containing the cylinders, which is $\mathcal{F}$ by (i) again. □

Consider a particular case: $\Omega_1 = \Omega_2 = \mathbb{R}$ and $\mathcal{F}_1 = \mathcal{F}_2 = \mathcal{B}$ are Borel $\sigma-$algebras on $\mathbb{R}$. Then, we have two ways of producing $\mathcal{F} = \mathcal{B} \times \mathcal{B}$ which is the Borel $\sigma-$algebra on the plane: we can use the family of products of Borel sets or the family of products of intervals. The following result shows that they give the same collection of subsets of $\mathbb{R}^2$.

PROPOSITION 6.3. *The $\sigma-$algebras generated by*

$$\mathcal{R} = \{B_1 \times B_2 : B_1, B_2 \in \mathcal{B}\},$$

$$\mathcal{I} = \{I_1 \times I_2 : I_1, I_2 \text{ are intervals}\}$$

are the same.

[1]For a function $g : (\Omega_1, \mathcal{F}_1, P_1) \to (\Omega_2, \mathcal{F}_2, P_2)$, we say it is measurable if for all $E \in \mathcal{F}_2$ we have $g^{-1}(E) \in \mathcal{F}_1$.

PROOF. Denote by $\mathcal{F}_{\mathcal{R}}$ the $\sigma-$algebra generated by the Borel 'rectangles' $\mathcal{R} = \{B_1 \times B_2 : B_1, B_2 \in \mathcal{B}\}$, and by $\mathcal{F}_{\mathcal{I}}$ the $\sigma-$algebra generated by the true rectangles $\mathcal{I} = \{I_1 \times I_2 : I_1, I_2 \text{ are intervals}\}$.

Since $\mathcal{I} \subseteq \mathcal{R}$, we have $\mathcal{F}_{\mathcal{I}} \subseteq \mathcal{F}_{\mathcal{R}}$. To show the inverse inclusion, we show that Borel cylinders $B_1 \times \Omega_2$ and $\Omega_1 \times B_2$ are in $\mathcal{F}_{\mathcal{I}}$. Let us set $\mathcal{D} = \{A \subseteq \Omega_1 : A \times \Omega_2 \in \mathcal{F}_{\mathcal{I}}\}$, and note that this is a $\sigma-$algebra containing all the intervals. Hence, $\mathcal{B} \subseteq \mathcal{D}$, and so $B_1 \times \Omega_2 \in \mathcal{F}_{\mathcal{I}}$. Similarly, we can show $\Omega_1 \times B_2 \in \mathcal{F}_{\mathcal{I}}$. It follows that $\mathcal{R} \subseteq \mathcal{F}_{\mathcal{I}}$, and so $\mathcal{F}_{\mathcal{R}} \subseteq \mathcal{F}_{\mathcal{I}}$. Therefore, $\mathcal{F}_{\mathcal{R}} = \mathcal{F}_{\mathcal{I}}$ as desired. □

We may easily generalize to n factors: suppose we are given n measure spaces $(\Omega_i, \mathcal{F}_i, P_i)$, $i = 1, \ldots, n$, then the product $\sigma-$algebra on $\Omega = \Omega_1 \times \cdots \times \Omega_n$ is the $\sigma-$algebra generated by the sets

$$\{A_1 \times \cdots \times A_n : A_i \in \mathcal{F}_i \text{ for all } 1 \leq i \leq n\}.$$

6.3. Construction of the product measure

Let $(\Omega_1, \mathcal{F}_1, P_1)$, $(\Omega_2, \mathcal{F}_2, P_2)$ be arbitrary measure spaces. We shall construct a measure P on $\Omega = \Omega_1 \times \Omega_2$ which is determined by P_1, P_2 in a natural way. A technical assumption is needed here: P_1 and P_2 are taken to be $\sigma-$finite, that is, there is a sequence of measurable sets A_n with $P_1(A_n)$ finite and $\cup_{n=1}^{\infty} A_n = \Omega_1$ (and the same for P_2, Ω_2). This is true for probability measures which have total measure 1 and also for Lebesgue measure (in the latter case set $A_n = [-n, n]$). For simplicity, we assume that P_1, P_2 are finite. (The extension to the case of $\sigma-$finite is obtained by a routine limit passage $n \to \infty$ in the results obtained for the restrictions of the measures to $B_n = \cup_{i=1}^{n} A_i$.)

The motivation provided by construction of multi-dimensional Lebesgue measures gives the following natural condition on P:

$$P(A_1 \times A_2) = P_1(A_1)P_2(A_2) \tag{6.1}$$

for $A_1 \in \mathcal{F}_1$ and $A_2 \in \mathcal{F}_2$.

We want to generalize (6.1) to all sets from the product $\sigma-$algebra $\mathcal{F} = \mathcal{F}_1 \times \mathcal{F}_2$. To do this, we introduce the notion of a *section* of a subset A of $\Omega_1 \times \Omega_2$: for $w_2 \in \Omega_2$,

$$A_{w_2} := \{w_1 \in \Omega_1 : (w_1, w_2) \in A\} \subseteq \Omega_1.$$

A similar construction is carried out for $w_1 \in \Omega_1$:

$$A_{w_1} := \{w_2 \in \Omega_2 : (w_1, w_2) \in A\} \subseteq \Omega_2.$$

See Figure 6.1.

THEOREM 6.4. *If A is in the product $\sigma-$algebra $\mathcal{F}$, then for each w_2, $A_{w_2} \in \mathcal{F}_1$, and for each w_1, $A_{w_1} \in \mathcal{F}_2$.*

PROOF. Let

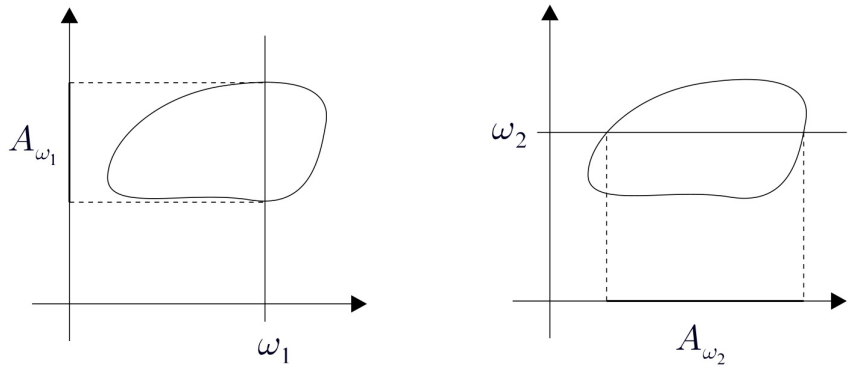
$$\mathcal{G} = \{A \in \mathcal{F} : \text{for all } w_2,\ A_{w_2} \in \mathcal{F}_1\}.$$

If we show that $\mathcal{G}$ is a $\sigma-$algebra containing rectangles, then $\mathcal{G} = \mathcal{F}$ since $\mathcal{F}$ is the smallest $\sigma-$algebra with this property.

If $A = A_1 \times A_2$ with $A_1 \in \mathcal{F}_1$ and $A_2 \subseteq \Omega_2$, then

$$A_{w_2} = \begin{cases} A_1, & \text{if } w_2 \in A_2, \\ \emptyset, & \text{if } w_2 \notin A_2 \end{cases}$$

is in $\mathcal{F}_1$ so the rectangles are in $\mathcal{G}$.



(A) w_2 section of A (B) w_1 section of A

FIGURE 6.1. A_{w_1} and A_{w_2}

If $A \in \mathcal{G}$, then $\Omega \setminus A \in \mathcal{G}$ since

$$\begin{aligned}(\Omega \setminus A)_{w_2} &= \{w_1 : (w_1, w_2) \in \Omega \setminus A\} \\ &= \Omega_1 \setminus \{w_1 : (w_1, w_2) \in A\} = \Omega_1 \setminus A_{w_2} \in \mathcal{F}_1.\end{aligned}$$

Finally, let $A_n \in \mathcal{G}$ for all $n \in \mathbb{N}$. Since

$$\left(\bigcup_{n=1}^{\infty} A_n\right)_{w_2} = \bigcup_{n=1}^{\infty} (A_n)_{w_2} \in \mathcal{F}_1,$$

the union $\cup_{n=1}^{\infty} A_n$ is also in $\mathcal{G}$. The proof for A_{w_1} is exactly the same. □

If $A = A_1 \times A_2$ with $A_1 \in \mathcal{F}_1$ and $A_2 \in \mathcal{F}_2$, then the function $w_2 \mapsto P_1(A_{w_2})$ is a step function:

$$P_1(A_{w_2}) = \begin{cases} P_1(A_1), & \text{if } w_2 \in A_2, \\ 0, & \text{if } w_2 \notin A_2, \end{cases}$$

and hence we have

$$P(A) = P_1(A_1)P_2(A_2) = \int_{\Omega_2} P_1(A_{w_2}) dP_2(w_2).$$

This motivates the general formula: for any $A \in \mathcal{F}$, we set

$$P(A) = \int_{\Omega_2} P_1(A_{w_2}) dP_2(w_2). \tag{6.2}$$

We call P the *product measure* and will denote it by $P = P_1 \times P_2$. We already know that $P_1(A_{w_2})$ makes sense since $A_{w_2} \in \mathcal{F}_1$ as shown before. For the integral to make sense, we need the following:

THEOREM 6.5. *Suppose that P_1, P_2 are finite. If $A \in \mathcal{F}$, then the functions*

$$w_2 \mapsto P_1(A_{w_2}), \quad w_1 \mapsto P_2(A_{w_1})$$

are measurable with respect to $\mathcal{F}_2, \mathcal{F}_1$ respectively, and

$$\int_{\Omega_2} P_1(A_{w_2}) dP_2(w_2) = \int_{\Omega_1} P_2(A_{w_1}) dP_1(w_1). \tag{6.3}$$

As an immediate consequence of Theorem 6.5, the definition in (6.2) is true for all $A \in \mathcal{F}$.

Before we prove this theorem, we allow ourselves a digression concerning the ways in which $\sigma-$algebras can be produced. This will greatly simplify several proofs. Given a family of sets $\mathcal{A}$, let $\mathcal{F}$ be the smallest $\sigma-$algebra containing $\mathcal{A}$. Suppose that $\mathcal{A}$ is an algebra, i.e. it contains the whole space and is closed with respect to finite union and differences: $A, B \in \mathcal{A}$ implies $A \cup B, A \setminus B \in \mathcal{A}$. Then, we have an alternative way of characterizing $\sigma-$algebras by means of so-called monotone classes.

DEFINITION 6.6. *A monotone class is a family of sets closed under countable unions of increasing sets and countable intersections of decreasing sets. That is, $\mathcal{G}$ is a monotone class if*

$$A_1 \subseteq A_2 \subseteq \ldots, \quad A_i \in \mathcal{G} \quad \Rightarrow \quad \bigcup_{i=1}^{\infty} A_i \in \mathcal{G},$$

$$A_1 \supseteq A_2 \supseteq \ldots, \quad A_i \in \mathcal{G} \quad \Rightarrow \quad \bigcap_{i=1}^{\infty} A_i \in \mathcal{G}.$$

LEMMA 6.7 (Monotone class theorem). *The smallest monotone class $\mathcal{G}_{\mathcal{A}}$ containing an algebra $\mathcal{A}$ coincides with the $\sigma-$algebra $\mathcal{F}_{\mathcal{A}}$ generated by $\mathcal{A}$.*

PROOF. A $\sigma-$algebra is a monotone class, so $\mathcal{G}_{\mathcal{A}} \subseteq \mathcal{F}_{\mathcal{A}}$ (since $\mathcal{G}_{\mathcal{A}}$ is the smallest monotone class containing $\mathcal{A}$).

To prove the converse, we first show that $\mathcal{G}_{\mathcal{A}}$ is an algebra. Set the whole space to be E. The family of sets

$$\mathcal{C} = \{A : A^c \in \mathcal{G}_A\}$$

contains $\mathcal{A}$ since $A \in \mathcal{A}$ implies $E \setminus A = A^c \in \mathcal{A} \subseteq \mathcal{G}_{\mathcal{A}}$. We observe that $\mathcal{C}$ is a monotone class. Suppose (A_n) satisfies $A_1 \subseteq A_2 \subseteq \ldots$ and $A_i^c \in \mathcal{G}_{\mathcal{A}}$ for all $i \in \mathbb{N}$. We have to show that $(\cup_{i=1}^{\infty} A_i)^c \in \mathcal{G}_{\mathcal{A}}$. As $A_1^c \supseteq A_2^c \supseteq \ldots$, it follows that $\cap_{i=1}^{\infty} A_i^c \in \mathcal{G}_{\mathcal{A}}$ because $\mathcal{G}_{\mathcal{A}}$ is a monotone class. By de Morgan's law, $\cap_{i=1}^{\infty} A_i^c = (\cup_{i=1}^{\infty} A_i)^c$ and so $\cup_{i=1}^{\infty} A_i \in \mathcal{C}$. The proof for the intersection is similar: $A_1 \supseteq A_2 \supseteq \ldots$ implies $A_1^c \subseteq A_2^c \subseteq \ldots$, hence $\cup_{i=1}^{\infty} A_i^c \in \mathcal{G}_{\mathcal{A}}$ and it follows that $(\cap_{i=1}^{\infty} A_i)^c = \cup_{i=1}^{\infty} A_i^c$ also belongs to $\mathcal{G}_{\mathcal{A}}$. We conclude that $\mathcal{G}_{\mathcal{A}} \subseteq \mathcal{C}$ and so $\mathcal{G}_{\mathcal{A}}$ is closed with respect to taking complements.

Now consider unions. First fix $A \in \mathcal{A}$ and consider

$$\{B : A \cup B \in \mathcal{G}_{\mathcal{A}}\}.$$

This family contains $\mathcal{A}$ (if $B \in \mathcal{A}$, then $A \cup B \in \mathcal{A} \subseteq \mathcal{G}_{\mathcal{A}}$) and we will show it is a monotone class. Indeed, let (B_n) satisfy $B_1 \subseteq B_2 \subseteq \ldots$ and $A \cup B_i \in \mathcal{G}_{\mathcal{A}}$ for all $i \in \mathbb{N}$. We have $A \cup B_1 \subseteq A \cup B_2 \subseteq \ldots$, hence $\cup_{i=1}^{\infty}(A \cup B_i) \in \mathcal{G}_{\mathcal{A}}$ (as $\mathcal{G}_{\mathcal{A}}$ is a monotone class) and it follows that $A \cup (\cup_{i=1}^{\infty} B_i) \in \mathcal{G}_{\mathcal{A}}$. Similar arguments work for the intersection of a deceasing sequence of sets. So, for this fixed A, we have

$$\mathcal{G}_{\mathcal{A}} \subseteq \{B : A \cup B \in \mathcal{G}_{\mathcal{A}}\}.$$

This means that for $A \in \mathcal{A}$ and $B \in \mathcal{G}_{\mathcal{A}}$ we have $A \cup B \in \mathcal{G}_{\mathcal{A}}$.

Now take arbitrary $A \in \mathcal{G}_{\mathcal{A}}$. By what we have just observed,

$$\mathcal{A} \subseteq \{B : A \cup B \in \mathcal{G}_{\mathcal{A}}\}$$

and by the same argument as before, the latter family is a monotone class. So

$$\mathcal{G}_{\mathcal{A}} \subseteq \{B : A \cup B \in \mathcal{G}_{\mathcal{A}}\}$$

holds for general $A \in \mathcal{G}_{\mathcal{A}}$, which completes the proof that $\mathcal{G}_{\mathcal{A}}$ is an algebra.

Next, we intend to show $\mathcal{G}_{\mathcal{A}}$ is in fact a $\sigma-$algebra. Indeed, let (A_i) be a sequence in $\mathcal{G}_{\mathcal{A}}$. By setting $B_n = \cup_{i=1}^{n} A_i$, we have $B_1 \subseteq B_2 \subseteq \dots$ and $B_i \in \mathcal{G}_{\mathcal{A}}$ for all $i \in \mathbb{N}$ (by the algebra property of $\mathcal{G}_{\mathcal{A}}$). Moreover, $\cup_{i=1}^{\infty} A_i = \cup_{i=1}^{\infty} B_i \in \mathcal{G}_A$ since $\mathcal{G}_{\mathcal{A}}$ is a monotone class.

Therefore $\mathcal{G}_{\mathcal{A}}$ is a $\sigma-$algebra containing $\mathcal{A}$ so it is bigger than the $\sigma-$algebra generated by $\mathcal{A}$, i.e. $\mathcal{F}_{\mathcal{A}} \subseteq \mathcal{G}_{\mathcal{A}}$, which completes the proof. □

The family $\mathcal{R}$ of rectangles introduced before is not an algebra, and so we take $\mathcal{A}$ to be the family of all unions of finite disjoint rectangles in the following proof.

PROOF OF THEOREM 6.5. Set

$$\mathcal{G} = \Bigg\{ A : w_2 \mapsto P_1(A_{w_2}),\ w_1 \mapsto P_2(A_{w_1}) \text{ are measurable and } \int_{\Omega_2} P_1(A_{w_2}) dP_2(w_2) = \int_{\Omega_1} P_2(A_{w_1}) dP_1(w_1) \Bigg\}.$$

This idea of the proof is: first we show that $\mathcal{R} \subseteq \mathcal{G}$, then $\mathcal{A} \subseteq \mathcal{G}$, and finally we show that $\mathcal{G}$ is a monotone class; by Lemma 6.7, $\mathcal{G} \supseteq \mathcal{F}$ which means that the claim of the theorem holds for all sets from $\mathcal{F}$.

If A is a rectangle, $A = A_1 \times A_2$, then as we noticed before, $w_2 \mapsto P_1(A_{w_2})$, $w_1 \mapsto P_2(A_{w_1})$ are indicator functions multiplied by some constants and (6.3) holds with both sides being equal to $P_1(A_1)P_2(A_2)$.

Next let $A = (A_1 \times A_2) \cup (B_1 \times B_2)$ be the union of disjoint rectangles. Disjoint means that either $A_1 \cap B_1 = \emptyset$ or $A_2 \cap B_2 = \emptyset$. Assume the former, for example. Then

$$A_{w_2} = \begin{cases} A_1 \cup B_1, & \text{if } w_2 \in A_2 \cap B_2, \\ A_1, & \text{if } w_2 \in A_2 \setminus B_2, \\ B_1, & \text{if } w_2 \in B_2 \setminus A_2, \\ \emptyset, & \text{if } w_2 \in (A_2 \cup B_2)^c, \end{cases}$$

and

$$\begin{aligned} &\int_{\Omega_2} P_1(A_{w_2}) dP_2(w_2) \\ &= [P_1(A_1) + P_1(B_1)] P_2(A_2 \cap B_2) + P_1(A_1) P_2(A_2 \setminus B_2) + P_1(B_1) P_2(B_2 \setminus A_2) \\ &= P_1(A_1)[P_2(A_2 \cap B_2) + P_2(A_2 \setminus B_2)] + P_1(B_1)[P_2(A_2 \cap B_2) + P_2(B_2 \setminus A_2)] \\ &= P_1(A_1)P_2(A_2) + P_1(B_1)P_2(B_2). \end{aligned}$$

On the other hand,

$$A_{w_1} = \begin{cases} A_2, & \text{if } w_1 \in A_1, \\ B_2, & \text{if } w_1 \in B_1, \\ \emptyset, & \text{if } w_1 \in (A_1 \cup B_1)^c, \end{cases}$$

and

$$\int_{\Omega_1} P_2(A_{w_1}) dP_1(w_1) = P_1(A_1)P_2(A_2) + P_1(B_1)P_2(B_2)$$

which is the same as the integral above.

The general case of finitely many rectangles can be proved in the same way. We skip this argument hoping that presenting it in detail for two rectangles is sufficient to guide the reader in the general case. It remains true that the functions $w_2 \mapsto P_1(A_{w_2})$,

$w_1 \mapsto P_2(A_{w_1})$ are simple functions, and so the verification of (6.3) is just simple algebra. (Instead, we may verify this by an induction argument, exercise!)

It remains to verify that $\mathcal{G}$ is a monotone class. Let (A_n) be sets from $\mathcal{G}$ satisfying $A_1 \subseteq A_2 \subseteq \ldots$, hence the functions $w_2 \mapsto P_1((A_i)_{w_2})$, $w_1 \mapsto P_2((A_i)_{w_1})$ are measurable. These two functions increase with i since the sections $(A_i)_{w_2}$, $(A_i)_{w_1}$ are increasing. Letting $i \to \infty$, we have

$$P_1((A_i)_{w_2}) \to P_1\left(\bigcup_{i=1}^{\infty}(A_i)_{w_2}\right) = P_1\left(\left(\bigcup_{i=1}^{\infty} A_i\right)_{w_2}\right)$$

by (a generalized version of) Theorem 2.17 and so the function $w_2 \mapsto P_1((\cup_{i=1}^{\infty}A_i)_{w_2})$ is measurable. The same argument shows that the function $w_1 \mapsto P_2((\cup_{i=1}^{\infty}A_i)_{w_1})$ is measurable. The equality (6.3) holds for each $i \in \mathbb{N}$ and by the monotone convergence theorem it is preserved in the limit. Thus (6.3) holds for unions $\cup_{i=1}^{\infty}A_i$.

For intersections, the argument is similar. The sequences we consider are decreasing; the functions $w_2 \mapsto P_1((\cap_{i=1}^{\infty}A_i)_{w_2})$, $w_1 \mapsto P_2((\cap_{i=1}^{\infty}A_i)_{w_1})$ are measurable as they are the limits of measurable functions and (6.3) holds by the dominated convergence theorem since $P_1(\Omega_1)$, $P_2(\Omega_2)$ are finite. □

THEOREM 6.8. *Suppose that P_1, P_2 are finite measures. The set function P given by (6.2) is countably additive. Any other measure coinciding with P on rectangles is equal to P on the product $\sigma-$algebra.*

PROOF. Let (A_i) be a sequence in $\mathcal{F}$ which is pairwise disjoint. Then $(A_i)_{w_2}$ are also pairwise disjoint and

$$P\left(\bigcup_{i=1}^{\infty}A_i\right) = \int_{\Omega_2} P_1\left(\left(\bigcup_{i=1}^{\infty}A_i\right)_{w_2}\right)dP_2(w_2) = \int_{\Omega_2} P_1\left(\bigcup_{i=1}^{\infty}(A_i)_{w_2}\right)dP_2(w_2)$$
$$= \int_{\Omega_2}\sum_{i=1}^{\infty}P_1((A_i)_{w_2})dP_2(w_2) = \sum_{i=1}^{\infty}\int_{\Omega_2}P_1((A_i)_{w_2})dP_2(w_2) = \sum_{i=1}^{\infty}P(A_i)$$

where we have employed the fact that the section of the union is the union of the sections and (a generalized version of) Proposition 4.28.

For uniqueness, let Q be a measure defined on the product $\sigma-$algebra $\mathcal{F}$ such that $P(A_1 \times A_2) = Q(A_1 \times A_2)$ for all $A_1 \in \mathcal{F}_1$ and $A_2 \in \mathcal{F}_2$. Set

$$\mathcal{H} = \{A \subseteq \Omega_1 \times \Omega_2 : A \in \mathcal{F},\ P(A) = Q(A)\}.$$

It contains unions of finite disjoint rectangles by the additivity of both P and Q; in other words, $\mathcal{H}$ contains the algebra $\mathcal{A}$.

It remains to show that it is a monotone class since then it contains $\mathcal{F}$ by Lemma 6.7. This is quite straightforward using again the fact that P and Q are measures. Suppose (A_i) is a sequence in $\mathcal{H}$ satisfying $A_1 \subseteq A_2 \subseteq \ldots$. Then $P(A_i) = Q(A_i)$ for all $i \in \mathbb{N}$, and it follows that $P(\cup_{i=1}^{\infty}A_i) = \lim_{i\to\infty}P(A_i) = \lim_{i\to\infty}Q(A_i) = Q(\cup_{i=1}^{\infty}A_i)$ which means that $\mathcal{H}$ is closed with respect to increasing unions. The argument for decreasing intersections is exactly the same: if $A_1 \supseteq A_2 \supseteq \ldots$, then $P(\cap_{i=1}^{\infty}A_i) = \lim_{i\to\infty}P(A_i) = \lim_{i\to\infty}Q(A_i) = Q(\cap_{i=1}^{\infty}A_i)$ as $P(\Omega) = P_1(\Omega_2)P_2(\Omega_2) = Q(\Omega)$ is finite. □

REMARK. The uniqueness part of the proof of Theorem 6.8 illustrates an important technique: in order to show that two measures on a $\sigma-$algebra coincide, it suffices to

prove that they coincide on the generating sets of that $\sigma-$algebra by an application of the monotone class theorem.

The completion of the product $\sigma-$algebra $\mathcal{M} \times \mathcal{M}$ built from the $\sigma-$algebra of Lebesgue measurable sets is the $\sigma-$algebra $\mathcal{M}_2$ on which m_2 is defined.

We can show that two-dimensional Lebesgue measure coincides with the completion of the product of one-dimensional ones: $m_2 = c(m \times m)$. First, they agree on rectangles built from intervals. As a consequence, they agree on the $\sigma-$algebra generated by such rectangles, which is the Borel $\sigma-$algebra on the plane. The completion of the Borel $\sigma-$algebra gives the $\sigma-$algebra of Lebesgue measurable sets in the same way as in the one-dimensional case.

6.4. Fubini's theorem

We wish to integrate functions defined on the product of the spaces $(\Omega_1, \mathcal{F}_1, P_1)$, $(\Omega_2, \mathcal{F}_2, P_2)$ by exploiting the integration with respect to the measures P_1, P_2 individually.

We tackle the issue of measurability first.

THEOREM 6.9. *If a nonnegative function $f : \Omega_1 \times \Omega_2 \to \mathbb{R}$ is measurable with respect to $\mathcal{F}_1 \times \mathcal{F}_2$, then for each $w_1 \in \Omega_1$ the function (which we shall call a section of f) $w_2 \mapsto f(w_1, w_2)$ is $\mathcal{F}_2-$measurable, and for each $w_2 \in \Omega_2$ the section $w_1 \mapsto f(w_1, w_2)$ is $\mathcal{F}_1-$measurable.*

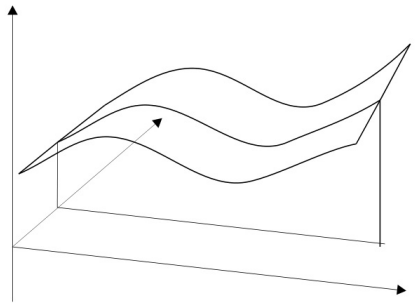


FIGURE 6.2. Section of f.

PROOF. First, we approximate f by a sequence of increasing simple functions (s_n) by using (a generalized version of) Proposition 4.15, i.e. $s_n \nearrow f$. From Theorem 6.4, we infer the sections of simple measurable functions are simple and measurable. Finally, it is clear that the sections of s_n converge to that of f, and so the section of f is measurable since measurability is preserved in the limit. □

COROLLARY 6.10. *Let f be as in Theorem 6.9. The functions*

$$w_1 \mapsto \int_{\Omega_2} f(w_1, w_2) dP_2(w_2), \quad w_2 \mapsto \int_{\Omega_1} f(w_1, w_2) dP_1(w_1)$$

are $\mathcal{F}_1, \mathcal{F}_2-$measurable, respectively.

PROOF. The integrals may be taken (being possibly infinite) due to measurability of the integrated functions, see Theorem 6.9. If f is a simple function, i.e. $f = \sum_{i=1}^{n} c_i \mathbf{1}_{E_i}$ with $c_i \geq 0$ and $E_i \in \mathcal{F}_1 \times \mathcal{F}_2$ for all $1 \leq i \leq n$, we have

$$\begin{aligned}\int_{\Omega_2} f(w_1, w_2) dP_2(w_2) &= \int_{\Omega_2} \sum_{i=1}^{n} c_i \mathbf{1}_{E_i}(w_1, w_2) dP_2(w_2) \\ &= \sum_{i=1}^{n} c_i \int_{\Omega_2} \mathbf{1}_{E_i}(w_1, w_2) dP_2(w_2) = \sum_{i=1}^{n} c_i P_2\left((E_i)_{w_1}\right).\end{aligned}$$

By Theorem 6.5, this implies the first function in the Corollary is $\mathcal{F}_1-$measurable. For the general f, as in the proof of Theorem 6.9, we approximate f by a sequence of increasing simple functions (s_n). By the monotone convergence theorem, the first function in the Corollary is the pointwise limit of the corresponding integrals of s_n which are $\mathcal{F}_1-$measurable, so it is $\mathcal{F}_1-$measurable. Similarly, we can show the second function in the Corollary is $\mathcal{F}_2-$measurable. □

THEOREM 6.11 (Tonelli's theorem). *Let f be a nonnegative measurable function defined on $\Omega_1 \times \Omega_2$. Then*

$$\begin{aligned}&\int_{\Omega_1 \times \Omega_2} f(w_1, w_2) d(P_1 \times P_2)(w_1, w_2) \\ &= \int_{\Omega_1} \left(\int_{\Omega_2} f(w_1, w_2) dP_2(w_2) \right) dP_1(w_1) = \int_{\Omega_2} \left(\int_{\Omega_1} f(w_1, w_2) dP_1(w_1) \right) dP_2(w_2).\end{aligned} \tag{6.4}$$

PROOF. For the indicator function of $A \in \mathcal{F} = \mathcal{F}_1 \times \mathcal{F}_2$, we can use Theorem 6.5 and the definition of product measure $P = P_1 \times P_2$ (see (6.2)) to verify (6.4). Then, by additivity of the integral, the formula is true for simple functions. Monotone approximation of general f by simple functions allows us to extend this formula to the general case. □

THEOREM 6.12 (Fubini's theorem). *If $f \in L^1(\Omega_1 \times \Omega_2)$, then*

$$\int_{\Omega_2} |f(w_1, w_2)| dP_2(w_2) < \infty \text{ for a.e. } w_1 \in \Omega_1,$$
$$\int_{\Omega_1} |f(w_1, w_2)| dP_1(w_1) < \infty \text{ for a.e. } w_2 \in \Omega_2.$$

Moreover, the functions

$$w_1 \mapsto \int_{\Omega_2} f(w_1, w_2) dP_2(w_2), \quad w_2 \mapsto \int_{\Omega_1} f(w_1, w_2) dP_1(w_1)$$

are in $L^1(\Omega_1), L^1(\Omega_2)$, respectively. Finally, (6.4) holds: in concise form, it reads

$$\int_{\Omega_1 \times \Omega_2} f d(P_1 \times P_2) = \int_{\Omega_1} \left(\int_{\Omega_2} f dP_2 \right) dP_1 = \int_{\Omega_2} \left(\int_{\Omega_1} f dP_1 \right) dP_2.$$

PROOF. Observing that $|f| \in L^1(\Omega_1 \times \Omega_2)$, then the first assertion follows from Tonelli's theorem (see Theorem 6.11) and Proposition 4.22 (iii). For the second assertion, we only concern the first function and it is in $L^1(\Omega_1)$ by the estimate

$$\int_{\Omega_1} \left| \int_{\Omega_2} f(w_1, w_2) dP_2(w_2) \right| dP_1(w_1) \leq \int_{\Omega_1} \left(\int_{\Omega_2} |f(w_1, w_2)| dP_2(w_2) \right) dP_1(w_1)$$

and Tonelli's theorem. Using the decomposition $f = f^+ - f^-$ and Tonelli's theorem, we can show the last formula. In particular, we may also need the following observation: if $f \in L^1$, then $f^+, f^- \in L^1$ and so all the integrals involved with f^+, f^- are finite. □

REMARK. The whole procedure may be extended to the product of an arbitrary finite number of spaces. In particular, we have a method of constructing $n-$dimensional Lebesgue measure as the completion of the product of n copies of one-dimensional Lebesgue measure.

EXAMPLE 6.13. *Let $\Omega_1 = \Omega_2 = (0,1)$, $P_1 = P_2 = m_{(0,1)}$, and*

$$f(x,y) = \begin{cases} \frac{1}{x^2}, & \text{if } 0 < y < x < 1, \\ 0, & \text{if } 0 < x \leq y < 1. \end{cases}$$

We shall see that the integral of f over the square is infinite. For this, we take a (non-negative) simple function dominated by f and compute its integral. Let $\varphi_k(x,y) = k$ if $f(x,y) \in [k+1, \infty)$ and $\varphi_k(x,y) = n$ if $f(x,y) \in [n, n+1)$ for each integer n satisfying $0 \leq n \leq k$. Then $\varphi_k(x,y) = n$ if $1 > x > y > 0$ and $x \in (\frac{1}{\sqrt{n+1}}, \frac{1}{\sqrt{n}}]$. The areas of this set is $\frac{1}{2}(\frac{1}{n} - \frac{1}{n+1})$ and

$$\int_{(0,1)^2} f dm_2 \geq \lim_{k\to\infty} \sum_{n=1}^{k} n \cdot \frac{1}{2}\left(\frac{1}{n} - \frac{1}{n+1}\right) = \lim_{k\to\infty} \sum_{n=1}^{k} \frac{1}{2(n+1)} = \infty.$$

Moreover, the function

$$g(x,y) = \begin{cases} \frac{1}{x^2}, & \text{if } 0 < y < x < 1, \\ -\frac{1}{y^2}, & \text{if } 0 < x \leq y < 1 \end{cases}$$

is not integrable since the integral of g^+ is infinite (the same is true for the integral of g^-).

The following is a useful proposition.

PROPOSITION 6.14. *Let $f : \mathbb{R} \to \mathbb{R}$ be measurable and positive. Let A_f denote the set of all points in the upper half-plane below the graph of f, i.e.*

$$A_f = \{(x,y) \in \mathbb{R}^2 : 0 \leq y < f(x)\}.$$

Then, A_f is m_2-measurable and $m_2(A_f) = \int_{\mathbb{R}} f dm$.

PROOF. If f is simple, then $f = \sum_{k=1}^{n} c_k \mathbf{1}_{A_k}$ with $c_k \geq 0$ and $A_k \in \mathcal{M}$ for all $1 \leq k \leq n$. Let $R_k = A_k \times [0, c_k)$ and the union of such rectangles is in fact A_f which is m_2-measurable. For general f, let (s_n) be a increasing sequence of simple functions converging to f pointwise. Then, A_{s_n} is m_2-measurable for all $n \in \mathbb{N}$, and so $A_f = \cup_{n=1}^{\infty} A_{s_n}$ is also m_2-measurable.

For the second claim, take a y section of A_f which is the interval $[0, f(x))$, and its measure is $f(x)$. By the definition of the product measure and Theorem 6.5, $m_2(A_f) = \int_{\mathbb{R}} f dm$. □

6.5. Finishing remarks

This chapter is prepared mainly based on Chapter 6 of [**5**]. Another useful resource could be [**2**].

CHAPTER 7

Differentiation theory

7.1. The Radon-Nikodym theorem

We start with a definition for the general case of two given measures.

DEFINITION 7.1. *Let Ω be a set and let $\mathcal{F}$ be a $\sigma-$algebra of its subsets. (The pair $(\Omega, \mathcal{F})$ is a measurable space.) Suppose that ν and μ are measures on $(\Omega, \mathcal{F})$. We say that ν is absolutely continuous with respect to μ if $\mu(A) = 0$ implies $\nu(A) = 0$ for $A \in \mathcal{F}$. We write this as $\nu \ll \mu$.*

EXAMPLE 7.2. *Let $f : \mathbb{R} \to \mathbb{R}$ be measurable and non-negative. Then, $A \mapsto \int_A f dm$ is a measure over the measurable space $(\mathbb{R}, \mathcal{M})$. Indeed, denote $\mu(A) = \int_A f dm$. The goal is to show*

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i)$$

for pairwise disjoint $(E_i)_{i=1}^{\infty}$. To this end, consider the sequence $g_n = f\mathbf{1}_{\cup_{i=1}^{n} E_i}$ and note that $g_n \nearrow g$, where $g = f\mathbf{1}_{\cup_{i=1}^{\infty} E_i}$. Now,

$$\int_{\mathbb{R}} g dm = \int_{\cup_{i=1}^{\infty} E_i} f dm = \mu\left(\bigcup_{i=1}^{\infty} E_i\right),$$

$$\int_{\mathbb{R}} g_n dm = \int_{\cup_{i=1}^{n} E_i} f dm = \sum_{i=1}^{n} \int_{E_i} f dm = \sum_{i=1}^{n} \mu(E_i),$$

and the monotone convergence theorem (see Theorem 4.13) completes the proof. Moreover, we can use Theorem 4.7 (iv) to show that $\mu \ll m$.

The following reformulation of the definition, which is not needed for the main result we will prove, but serves to bring the relationship between ν and μ a little 'closer to home' and is useful in many applications:

PROPOSITION 7.3. *Let ν and μ be finite measures on the measurable space $(\Omega, \mathcal{F})$. Then, $\nu \ll \mu$ if and only if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that for $F \in \mathcal{F}$, $\mu(F) < \delta$ implies $\nu(F) < \varepsilon$.*

PROOF. Suppose the $(\varepsilon, \delta)-$condition fails. We can then find $\varepsilon > 0$ and sets (F_n) such that for all $n \geq 1$, $\mu(F_n) < \frac{1}{2^n}$ but $\nu(F_n) > \varepsilon$. Denoting $A = \cap_{n \geq 1}(\cup_{i \geq n} F_i)$, we have $\mu(A) \leq \mu(\cup_{i \geq n} F_i) \leq \sum_{i=n}^{\infty} \frac{1}{2^i} = \frac{1}{2^{n-1}}$ for every $n \geq 1$. Thus, $\mu(A) = 0$. But $\nu(F_n) \geq \varepsilon$ for every n, hence $\nu(E_n) \geq \varepsilon$, where $E_n = \cup_{i \geq n} F_i$. The sequence $(E_n)_{n=1}^{\infty}$ decreases, so that as ν is finite, Theorem 2.17 (ii) gives $\nu(A) = \lim_{n \to \infty} \nu(E_n) \geq \varepsilon$. Thus, ν is not absolutely continuous with respect to μ. Conversely, if the $(\varepsilon, \delta)-$condition holds, and $\mu(F) = 0$, then $\mu(F) < \delta$ for any $\delta > 0$, and so, for every given $\varepsilon > 0$, $\nu(F) < \varepsilon$. Hence, $\nu(F) = 0$. So, $\nu \ll \mu$. $\square$

We generalise from the special case of Lebesgue measure: if μ is any measure on $(\Omega, \mathcal{F})$ and $f : \Omega \to \mathbb{R}$ is a measurable function for which $\int_\Omega f d\mu$ exists, then $\nu(F) = \int_F f d\mu$ for all $F \in \mathcal{F}$ defines a measure $\nu \ll \mu$. (Note that we employ the convention $0 \times \infty = 0$.)

For $\sigma-$finite measures, the following key result asserts the converse:

THEOREM 7.4 (Radon-Nikodym). *Given two $\sigma-$finite measures ν, μ on a measurable space $(\Omega, \mathcal{F})$, with $\nu \ll \mu$, then there is a non-negative measurable function $h : \Omega \to \mathbb{R}$ such that $\nu(F) = \int_F h d\mu$ for every $F \in \mathcal{F}$. The function h is unique up to $\mu-$null sets: if g also satisfies $\nu(F) = \int_F g d\mu$ for all $F \in \mathcal{F}$, then $g = h$ a.e. with respect to μ.*

It is instructive to begin with a special case: suppose that $\mu(\Omega) = 1$, i.e., μ is a probability measure. We say that the measure μ *dominates* ν when $0 \leq \nu(F) \leq \mu(F)$ for every $F \in \mathcal{F}$. This obviously implies $\nu \ll \mu$. In this simplified situation, we shall construct the required function h explicitly. First, we generalise the idea of partitions and their refinements, which we used to construct the Riemann integral, to measurable subsets in $(\Omega, \mathcal{F})$.

DEFINITION 7.5. *Let $(\Omega, \mathcal{F})$ be a measurable space. A finite (measurable) partition of Ω is a finite collection of disjoint subsets $\mathcal{P} = (A_i)_{i \leq n}$ in $\mathcal{F}$ whose union is Ω. The finite partition $\mathcal{P}'$ is a refinement of $\mathcal{P}$ if each set in $\mathcal{P}$ is a disjoint union of sets in $\mathcal{P}'$.*

The following is a simplified 'Radon-Nikodym theorem' for dominated measures:

THEOREM 7.6. *Suppose that $\mu(\Omega) = 1$ and $0 \leq \nu(F) \leq \mu(F)$ for every $F \in \mathcal{F}$. Then, there exists a non-negative $\mathcal{F}-$measurable function h on Ω such that $\nu(F) = \int_F h d\mu$ for all $F \in \mathcal{F}$.*

We shall prove this in three steps: in Step 1, we define the required function $h_\mathcal{P}$ for sets in a finite partition $\mathcal{P}$ and compare the functions $h_{\mathcal{P}_1}$ and $h_{\mathcal{P}_2}$ when the partitions $\mathcal{P}_2$ refines $\mathcal{P}_1$. This enables us to show that the integrals $\int_\Omega h^2_\mathcal{P} d\mu$ are non-decreasing if we take successive refinements. Since they are also bounded above (by 1), $c = \sup \int_\Omega h^2_\mathcal{P} d\mu$ exists in $\mathbb{R}$. In Step 2, we then construct the desired function h by a careful limit argument, using the convergence theorems of Chapter 4. In Step 3, we show that h has the desired properties.

PROOF OF THEOREM 7.6. Step 1: **The function $h_\mathcal{P}$ for a finite partition.** Suppose that $0 \leq \nu(F) \leq \mu(F)$ for every $F \in \mathcal{F}$. Let $\mathcal{P} = \{A_1, A_2, \ldots, A_k\}$ be a finite partition of Ω such that each $A_i \in \mathcal{F}$. Definite the simple function $h_\mathcal{P} : \Omega \to \mathbb{R}$ by setting

$$h_\mathcal{P}(w) = c_i = \frac{\nu(A_i)}{\mu(A_i)} \text{ for } w \in A_i \text{ when } \mu(A_i) > 0, \text{ and } h_\mathcal{P}(w) = 0 \text{ otherwise.}$$

Since $h_\mathcal{P}$ is constant on each 'atom' A_i, $\nu(A_i) = \int_{A_i} h_\mathcal{P} d\mu$. Then, $h_\mathcal{P}$ has the following properties:

(i) For each finite partition $\mathcal{P}$ of Ω, $0 \leq h_\mathcal{P}(w) \leq 1$ for all $w \in \Omega$.
(ii) If $A = \cup_{j \in J} A_j$ for an index set $J \subseteq \{1, 2, \ldots, k\}$, then $\nu(A) = \int_A h_\mathcal{P} d\mu$. Thus, $\nu(\Omega) = \int_\Omega h_\mathcal{P} d\mu$.
(iii) If $\mathcal{P}_1$ and $\mathcal{P}_2$ are finite partitions of Ω and $\mathcal{P}_2$ refines $\mathcal{P}_1$, then, with $h_n = h_{\mathcal{P}_n}$ $(n = 1, 2)$, we have
  (a) for all $A \in \mathcal{P}_1$, $\int_A h_1 d\mu = \nu(A) = \int_A h_2 d\mu$,
  (b) for all $A \in \mathcal{P}_1$, $\int_A h_1 h_2 d\mu = \int_A h_1^2 d\mu$.

(iv) $\int_\Omega (h_2^2 - h_1^2)d\mu = \int_\Omega (h_2 - h_1)^2 d\mu$ and therefore

$$\int_\Omega h_2^2 d\mu = \int_\Omega h_1^2 d\mu + \int_\Omega (h_2 - h_1)^2 d\mu \geq \int_\Omega h_1^2 d\mu.$$

We now prove these assertions in turn.

(i) This is trivial by construction of $h_\mathcal{P}$, since μ dominates ν.

(ii) Let $A = \cup_{j\in J} A_j$ for some index set $J \subseteq \{1, 2, \ldots, k\}$. Since the $\{A_j\}$ are disjoint and $\nu(A_j) = 0$ whenever $\mu(A_j) = 0$, we have

$$\begin{aligned}\nu(A) = \sum_{j\in J} \nu(A_j) &= \sum_{j\in J,\, \mu(A_j)>0} \frac{\nu(A_j)}{\mu(A_j)} \mu(A_j) \\ &= \sum_{j\in J,\, \mu(A_j)>0} c_j \mu(A_j) = \sum_{j\in J} \int_{A_j} h_\mathcal{P} d\mu = \int_A h_\mathcal{P} d\mu.\end{aligned}$$

In particular, since $\mathcal{P}$ partitions Ω, this holds for $A = \Omega$.

(iii) (a) With the $\mathcal{P}_n, h_n$ as above ($n = 1, 2$), we can write $A = \cup_{j\in J} B_j$ for each $A \in \mathcal{P}_1$, where J is a finite index set and $B_j \in \mathcal{P}_2$. The sets B_j are pairwise disjoint, and again $\nu(B_j) = 0$ when $\mu(B_j) = 0$, so that

$$\begin{aligned}\int_A h_1 d\mu = \nu(A) = \sum_{j\in J} \nu(B_j) &= \sum_{j\in J,\, \mu(B_j)>0} \frac{\nu(B_j)}{\mu(B_j)} \mu(B_j) \\ &= \sum_{j\in J} \int_{B_j} h_2 d\mu = \int_A h_2 d\mu.\end{aligned}$$

(b) With A as in part (a) and $\mu(A) > 0$, note that $h_1 = \frac{\nu(A)}{\mu(A)}$ is constant on A, so that

$$\int_A h_1 h_2 d\mu = \frac{\nu(A)}{\mu(A)} \int_A h_2 d\mu = \frac{(\nu(A))^2}{\mu(A)} = \int_A \left(\frac{\nu(A)}{\mu(A)}\right)^2 d\mu = \int_A h_1^2 d\mu.$$

(iv) By (iii) (b), $\int_A h_1(h_2 - h_1)d\mu = 0$ for every $A \in \mathcal{P}_1$. Since the $A_i \in \mathcal{P}_1$ partition Ω, we also have

$$\int_\Omega h_1(h_2 - h_1)d\mu = \sum_{i=1}^k \int_{A_i} h_1(h_2 - h_1)d\mu = 0.$$

Hence,

$$\begin{aligned}\int_\Omega (h_2 - h_1)^2 d\mu &= \int_\Omega (h_2^2 - 2h_1 h_2 + h_1^2) d\mu \\ &= \int_\Omega \left[h_2^2 - 2h_1(h_2 - h_1) - h_1^2\right] d\mu = \int_\Omega \left(h_2^2 - h_1^2\right) d\mu,\end{aligned}$$

and thus

$$\int_\Omega h_2^2 d\mu = \int_\Omega h_1^2 d\mu + \int_\Omega (h_2 - h_1)^2 d\mu \geq \int_\Omega h_1^2 d\mu.$$

Step 2: **Passage to the limit – construction of h.** In Step 1, we showed that the integrals $\int_\Omega h_\mathcal{P}^2 d\mu$ are non-decreasing over successive refinements of a finite partition of Ω. Moreover, by (i) above, each function $h_\mathcal{P}$ satisfies $0 \leq h_\mathcal{P}(w) \leq 1$ for all $w \in \Omega$. Thus, setting $c = \sup \int_\Omega h_\mathcal{P}^2 d\mu$, where the supremum is taken over all finite partitions of Ω, we have $0 \leq c \leq 1$. (Here, we use the assumption that $\mu(\Omega) = 1$.)

For each $n \geq 1$, let $\mathcal{P}_n$ be a finite measurable partition of Ω such that $\int_\Omega h^2_{\mathcal{P}_n} d\mu > c - \frac{1}{4^n}$. Let $\mathcal{Q}_n$ be the smallest common refinement of the partitions $\mathcal{P}_1, \mathcal{P}_2, \ldots, \mathcal{P}_n$. For each n, $\mathcal{Q}_n$ refines $\mathcal{P}_n$ by construction, and $\mathcal{Q}_{n+1}$ refines $\mathcal{Q}_n$ since each $\mathcal{Q}_k$ consists of all intersections $A_1 \cap A_2 \cap \cdots \cap A_k$, where $A_i \in \mathcal{P}_i$, $i \leq k$. Hence, each set in $\mathcal{Q}_n$ is a disjoint union of sets in $\mathcal{Q}_{n+1}$. We therefore have the inequalities:

$$c - \frac{1}{4^n} < \int_\Omega h^2_{\mathcal{P}_n} d\mu \leq \int_\Omega h^2_{\mathcal{Q}_n} d\mu \leq \int_\Omega h^2_{\mathcal{Q}_{n+1}} d\mu \leq c.$$

Using the identity proved in Step 1 (iv), we now have

$$\int_\Omega (h_{\mathcal{Q}_{n+1}} - h_{\mathcal{Q}_n})^2 d\mu = \int_\Omega (h^2_{\mathcal{Q}_{n+1}} - h^2_{\mathcal{Q}_n}) d\mu < \frac{1}{4^n}.$$

The Cauchy-Schwarz inequality applied with $f = |h_{\mathcal{Q}_{n+1}} - h_{\mathcal{Q}_n}|$ and $g \equiv 1$, then yields for each $n \geq 1$,

$$\int_\Omega |h_{\mathcal{Q}_{n+1}} - h_{\mathcal{Q}_n}| d\mu < \frac{1}{2^n}.$$

By the Beppo-Levi Theorem, since $\sum_{n\geq 1} \int_\Omega |h_{\mathcal{Q}_{n+1}} - h_{\mathcal{Q}_n}| d\mu$ is finite, we conclude that the series converges $\mu-$a.e., so that the limit function

$$h = h_{\mathcal{P}_1} + \sum_{n\geq 1} (h_{\mathcal{Q}_{n+1}} - h_{\mathcal{Q}_n}) = \lim_{n\to\infty} h_{\mathcal{Q}_n}$$

(noting that $\mathcal{Q}_1 = \mathcal{P}_1$) is well-defined $\mu-$a.e. We complete the construction by setting $h = 0$ on the exceptional $\mu-$null set.

Step 3: **Verification of the properties of h.** By Step 1 (i), it follows that $0 \leq h(w) \leq 1$, and it is clear from its construction that h is $\mathcal{F}-$measurable.

We need to show that $\nu(F) = \int_F h d\mu$ for every $F \in \mathcal{F}$. Fix any such measurable set F and let $n \geq 1$. Define $\mathcal{R}_n$ as the smallest common refinement of the two partitions $\mathcal{Q}_n$ (defined as in Step 2) and $\{F, F^c\}$. Since F is a finite disjoint union of sets in $\mathcal{R}_n$, we have $\nu(F) = \int_F h_{\mathcal{R}_n} d\mu$ from Step 1 (ii).

By Step 2, $c - \frac{1}{4^n} < \int_\Omega h^2_{\mathcal{Q}_n} d\mu \leq \int_\Omega h^2_{\mathcal{R}_n} d\mu \leq c$, so, as before, we can conclude that $\int_\Omega (h_{\mathcal{R}_n} - h_{\mathcal{Q}_n})^2 d\mu < \frac{1}{4^n}$, and using the Cauchy-Schwarz inequality once more, this time with $g = \mathbf{1}_F$, we have

$$\left| \int_F (h_{\mathcal{R}_n} - h_{\mathcal{Q}_n}) d\mu \right| \leq \int_F |h_{\mathcal{R}_n} - h_{\mathcal{Q}_n}| d\mu < \frac{1}{2^n}.$$

For all n, $\nu(F) = \int_F h_{\mathcal{R}_n} d\mu = \int_F (h_{\mathcal{R}_n} - h_{\mathcal{Q}_n}) d\mu + \int_F h_{\mathcal{Q}_n} d\mu$. The first integral on the right converges to 0 as $n \to \infty$, while the second converges to $\int_F h d\mu$ by dominated convergence theorem (since for all $n \geq 1$, $0 \leq h_{\mathcal{Q}_n} \leq 1$ and $\mu(\Omega)$ is finite). Thus, we have verified that $\nu(F) = \int_F h d\mu$ as required.

□

It is straightforward to check that the assumption $\mu(\Omega) = 1$ is not essential since for any finite positive measure μ we can repeat the above arguments using $\frac{\mu}{\mu(\Omega)}$ instead of μ. We write the function h defined above as $\frac{d\nu}{d\mu}$ and call it the *Radon-Nikodym derivative* of ν with respect to μ. The next observation is an easy application of the general procedure highlighted in the Remark before Theorem 4.18:

PROPOSITION 7.7. *If μ and φ are finite measures with $0 \leq \mu \leq \varphi$, and if $h_\mu = \frac{d\mu}{d\varphi}$ is constructed as above, then for any non-negative $\mathcal{F}-$measurable function g on Ω we have*

$$\int_\Omega g d\mu = \int_\Omega g h_\mu d\varphi.$$

The same identity holds for any $g \in \mathcal{L}^1(\mu) := \{g \text{ is } \mathcal{F}-\text{measurable} : \int_\Omega |g| d\mu < \infty\}$.

PROOF. We proceed as in the Remark before Theorem 4.18. Begin with g as the indicator function $\mathbf{1}_G$ for $G \in \mathcal{F}$. Then, we have $\int_\Omega g d\mu = \mu(G) = \int_G h_\mu d\varphi$ by construction of h_μ. Next, let $g = \sum_{i=1}^n a_i \mathbf{1}_{G_i}$ for sets $G_1, G_2, \ldots, G_n$ in $\mathcal{F}$ and reals $a_1, a_2, \ldots, a_n$; then linearity of the integrals yields

$$\int_\Omega g d\mu = \sum_{i=1}^n a_i \mu(G_i) = \sum_{i=1}^n a_i \left(\int_{G_i} h_\mu d\mu \right) = \sum_{i=1}^n a_i \left(\int_\Omega \mathbf{1}_{G_i} h_\mu d\varphi \right) = \int_\Omega g h_\mu d\varphi.$$

Finally, any $\mathcal{F}-$measurable non-negative function g is approximated from below by an increasing sequence of $\mathcal{F}-$simple functions $g_n = \sum_{i=1}^{N_n} a_i \mathbf{1}_{G_i}$, and its integral $\int_\Omega g d\mu$ is the limit of the increasing sequence $\left(\int_\Omega g_n d\mu \right)_{n=1}^\infty$. (We may apply a variant of Theorem 4.13 and Proposition 4.15.) But since $0 \leq h_\mu \leq 1$ by construction, $(g_n h_\mu)$ increases to $g h_\mu$ pointwise, hence the sequence $(\int_\Omega g_n h_\mu d\varphi)_{n=1}^\infty$ also increases to $\int_\Omega g h_\mu d\varphi$ by monotone convergence theorem, so the limits are equal. For integrable $g = g^+ - g^-$, we can apply the above to each of g^+, g^- separately, and then use linearity. □

For finite measures, we can now prove the general result announced earlier:

THEOREM 7.8 (Radon-Nikodym). *Let ν and μ be finite measures on the measurable space $(\Omega, \mathcal{F})$ and suppose that $\nu \ll \mu$. Then, there is a non-negative $\mathcal{F}-$measurable function h on Ω such that $\nu(A) = \int_A h d\mu$ for all $A \in \mathcal{F}$.*

PROOF. Let $\varphi = \nu + \mu$. Then, φ is a positive finite measure which dominates both ν and μ. Hence, the Radon-Nikodym derivatives $h_\nu = \frac{d\nu}{d\varphi}$ and $h_\mu = \frac{d\mu}{d\varphi}$ are well-defined by the earlier constructions. Consider the sets $F = \{h_\mu > 0\}$ and $G = \{h_\mu = 0\}$ in $\mathcal{F}$. Clearly, $\mu(G) = \int_G h_\mu d\varphi = 0$, hence also $\nu(G) = 0$, since $\nu \ll \mu$. Define $h = \frac{h_\nu}{h_\mu} \mathbf{1}_F$, and let $A \in \mathcal{F}$, $A \subseteq F$. By Proposition 7.7 with $h\mathbf{1}_A$ instead of g, we have

$$\nu(A) = \int_A h_\nu d\varphi = \int_A h h_\mu d\varphi = \int_A h d\mu$$

as required. Since μ and ν are both null on G, this proves the theorem. □

You can now easily complete the picture for $\sigma-$finite measures and verify that the function h is 'essentially unique':

PROPOSITION 7.9. *The Radon-Nikodym theorem remains valid if the measures ν and μ are $\sigma-$finite: for any two such measures with $\nu \ll \mu$, we can find a finite-valued non-negative measurable function h on Ω such that $\nu(F) = \int_F h d\mu$ for all $F \in \mathcal{F}$. The function h so defined is unique up to $\mu-$null sets, i.e., if $g : \Omega \to \mathbb{R}^+$ also satisfies $\nu(F) = \int_\Omega g d\mu$ for all $F \in \mathcal{F}$ then $g = h$ a.e. with respect to μ.*

PROOF. Suppose $\Omega = \cup_{n \geq 1} A_n = \cup_{m \geq 1} B_m$, where $A_n, B_m \in \mathcal{F}$ and $\nu(A_n), \mu(B_m)$ are finite for all $n, m \in \mathbb{N}$. If $\cup_{n \geq 1} A_n = \Omega$ and the A_n are not disjoint, replace them by E_n, where $E_1 = A_1$, $E_n = A_n \setminus (\cup_{i=1}^{n-1} E_i)$, $n > 1$. The same can be done for the B_m and hence we can take both sequences as disjoint. Now, $\Omega = \cup_{n,m \geq 1} (A_n \cap B_m)$ is also a disjoint union, and ν, μ are both finite on each $A_n \cap B_m$. Re-order these sets into

a single sequence $(C_n)_{n\geq 1}$ and fix $n \geq 1$. Restricting both measures to the $\sigma-$algebra $\mathcal{F}_n = \{F \cap C_n : F \in \mathcal{F}\}$ yields them as finite measures on $(\Omega, \mathcal{F}_n)$, so that the Radon-Nikodym theorem (see Theorem 7.8) applies, and provides a non-negative $\mathcal{F}_n-$measurable function h_n such that $\nu(E) = \int_E h_n d\mu$ for each $E \in \mathcal{F}_n$. But any set $F \in \mathcal{F}$ has the form $F = \cup_{n=1}^{\infty} F_n$ for $F_n \in \mathcal{F}_n$, so we can define h by setting $h = h_n$ for every $n \geq 1$. Now, $\nu(F) = \sum_{n=1}^{\infty} \int_{F_n} h_n d\mu = \int_F h d\mu$. (Keep in mind that each h_n is only defined over C_n and (c_n) is pairwise disjoint.) The uniqueness is clear: if g has the same properties as h, then $\int_F (h-g)d\mu = 0$ for each $F \in \mathcal{F}$, so $h - g = 0$ a.e. with respect to μ by a variant of Theorem 4.21 (we may need to decompose Ω into finite pieces if necessary). □

Radon-Nikodym derivatives of measures obey simple combination rules which follow from the uniqueness property. We illustrate this with the sum and composition of two Radon-Nikodym derivatives.

PROPOSITION 7.10. *Assume we are given $\sigma-$finite measures λ, ν, μ satisfying $\lambda \ll \mu$ and $\nu \ll \mu$ with Radon-Nikodym derivatives $\frac{d\lambda}{d\mu}$ and $\frac{d\nu}{d\mu}$, respectively.*

(i) With $\varphi = \lambda + \nu$, we have $\frac{d\varphi}{d\mu} = \frac{d\lambda}{d\mu} + \frac{d\nu}{d\mu}$ a.e. with respect to μ;

(ii) If $\lambda \ll \nu$, then $\frac{d\lambda}{d\mu} = \frac{d\lambda}{d\nu}\frac{d\nu}{d\mu}$ a.e. with respect to μ.

PROOF. (i) This is trivial, since $\varphi = \lambda + \nu$ is $\sigma-$finite and absolutely continuous with respect to μ, and we have, for $F \in \mathcal{F}$:

$$\int_F \frac{d\varphi}{d\mu} d\mu = \varphi(F) = (\lambda+\nu)(F) = \lambda(F) + \nu(F) = \int_F \left(\frac{d\lambda}{d\mu} + \frac{d\nu}{d\mu}\right) d\mu.$$

The integrands on the right and left extremes are thus equal μ-a.e. (see Proposition 7.9), so the result follows.

(ii) Write $\frac{d\lambda}{d\nu} = g$ and $\frac{d\nu}{d\mu} = h$, where g, h are non-negative $\mathcal{F}-$measurable functions. We need to show that, for $F \in \mathcal{F}$

$$\lambda(F) = \int_F gh d\mu.$$

First, consider this when g is replaced by a simple function of the form $\eta = \sum_{i=1}^n c_i \mathbf{1}_{E_i}$. Then, we obtain:

$$\int_F \eta d\nu = \sum_{i=1}^n c_i \nu(F \cap E_i) = \sum_{i=1}^n c_i \int_{F\cap E_i} h d\mu = \int_F \eta h d\mu.$$

Now, let (η_n) be a sequence of simple functions increasing pointwise to g. Then, by monotone convergence theorem:

$$\lambda(F) = \int_F g d\nu = \lim_{n\to\infty} \int_F \eta_n d\nu = \lim_{n\to\infty} \int_F \eta_n h d\mu = \int_F gh d\mu,$$

since $(\eta_n h)$ increases to gh. This proves the result.

□

Given a pair of $\sigma-$finite measures λ, μ on $(\Omega, \mathcal{F})$, it is natural to ask whether we can identify the sets for which $\mu(E) = 0$ implies $\lambda(E) = 0$. This would mean that we can split the mass of λ into two pieces, one being represented by a $\mu-$integral, and the other 'concentrated' on $\mu-$null sets, i.e., away from the mass of μ. We turn this idea of 'separating' the masses of two measures into the following

DEFINITION 7.11. *If there is a set $E \in \mathcal{F}$ such that $\lambda(F) = \lambda(E \cap F)$ for every $F \in \mathcal{F}$, then λ is concentrated on E. If two measures μ, ν are concentrated on disjoint subsets of Ω, we say that they are mutually singular and write $\mu \perp \nu$.*

Observe that λ is concentrated on E if and only if for all $F \in \mathcal{F}$, $F \cap E = \emptyset$ implies $\lambda(F) = 0$. Indeed, if λ is concentrated on E and $E \cap F = \emptyset$, then $\lambda(F) = \lambda(E \cap F) = 0$. Conversely, if for all $F \in \mathcal{F}$, $F \cap E = \emptyset$ implies $\lambda(F) = 0$, consider $\lambda(F) = \lambda(F \cap E) + \lambda(F \setminus E)$. Since $(F \setminus E) \cap E = \emptyset$, we must have $\lambda(F \setminus E) = 0$, so $\lambda(F) = \lambda(F \cap E)$. We gather some simple facts about mutually singular measures:

PROPOSITION 7.12. *If $\mu, \nu, \lambda_1, \lambda_2$ are measures on a $\sigma-$algebra $\mathcal{F}$, the following are true:*

(i) If $\lambda_1 \perp \mu$ and $\lambda_2 \perp \mu$, then also $(\lambda_1 + \lambda_2) \perp \mu$.
(ii) If $\lambda_1 \ll \mu$ and $\lambda_2 \perp \mu$, then $\lambda_2 \perp \lambda_1$.
(iii) If $\nu \ll \mu$ and $\nu \perp \mu$, then $\nu = 0$.

PROOF. (i) For $i = 1, 2$, let A_i, B_i be disjoint sets with λ_i concentrated on A_i and μ concentrated on B_i. It follows that $\lambda_1 + \lambda_2$ is concentrated on $A_1 \cup A_2$, while μ is concentrated on $B_1 \cap B_2$. But $A_1 \cup A_2$ is disjoint from $B_1 \cap B_2$, hence the measures $\lambda_1 + \lambda_2$ and μ are mutually singular.

(ii) Choose a set E for which $\mu(E) = 0$ while λ_2 is concentrated on E. Let $F \subseteq E$, so that $\mu(F) = 0$ and hence $\lambda_1(F) = 0$ (since $\lambda_1 \ll \mu$). This shows that λ_1 is concentrated on E^c, hence λ_1 and λ_2 are mutually singular.

(iii) Applying (ii) with $\lambda_1 = \lambda_2 = \nu$ shows that $\nu \perp \nu$, which can only happen when $\nu = 0$.

□

The next result shows that a unique 'mass splitting' of a $\sigma-$finite measure relative to another is always possible:

THEOREM 7.13 (Lebesgue decomposition). *Let λ, μ be $\sigma-$finite measures on $(\Omega, \mathcal{F})$. Then, λ can be expressed uniquely as a sum of two measures, $\lambda = \lambda_a + \lambda_s$ where $\lambda_a \ll \mu$ and $\lambda_s \perp \mu$.*

PROOF. Existence: We consider finite measures; the extension to the $\sigma-$finite case is routine. Since $0 \leq \lambda \leq \lambda + \mu = \varphi$, i.e., φ dominates λ, there is $0 \leq h \leq 1$ such that $\lambda(E) = \int_E h d\varphi$ for all measurable E. Let $A = \{w \in \Omega : h(w) < 1\}$ and $B = \{w \in \Omega : h(w) = 1\}$. Set $\lambda_a(E) = \lambda(A \cap E)$ and $\lambda_s(E) = \lambda(B \cap E)$ for every $E \in \mathcal{F}$.

Now, if $E \subseteq A$ and $\mu(E) = 0$, then $\lambda(E) = \int_E h d\varphi = \int_E h d\lambda$, so that $\int_E (1 - h) d\lambda = 0$. But $h < 1$ on A, hence also on E. Therefore, we must have $\lambda(E) = 0$. Hence, if $E \in \mathcal{F}$ and $\mu(E) = 0$, $\lambda_a(E) = \lambda(A \cap E) = 0$ as $A \cap E \subseteq A$. So, $\lambda_a \ll \lambda$. On the other hand, if $E \subseteq B$, we obtain $\lambda(E) = \int_E h d\varphi = \int_E 1 d(\lambda + \mu) = \lambda(E) + \mu(E)$, so that $\mu(E) = 0$. As $A = B^c$, we have shown that $\mu(E) = 0$ whenever $E \cap A = \emptyset$, so that μ is concentrated on A. Since λ_s is concentrated on B, this shows that λ_s and μ are mutually singular.

Uniqueness is left as an exercise. The theorem is proved. □

Combining this with the Radon-Nikodym theorem, we can describe the structure of λ with respect to μ as 'basis measure':

COROLLARY 7.14. *With $\mu, \lambda, \lambda_a, \lambda_s$ as in Theorem 7.13, there is a $\mu-$a.e. unique non-negative measurable function h such that $\lambda(E) = \int_E h d\mu + \lambda_s(E)$ for every $E \in \mathcal{F}$.*

REMARK. This result is reminiscent of the structure theory of finite-dimensional vector spaces: if $x \in \mathbb{R}^n$ and $m < n$, we can write $x = y + z$, where $y = \sum_{i=1}^m y_i e_i$ is the orthogonal projection onto $\mathbb{R}^m$ and z is orthogonal to this subspace. We also exploited similar ideas for Hilbert space. In this sense, the measure μ has the role of a 'basis' providing the 'linear combination' which describes the projection of the measure λ onto a subspace of the space of measures on Ω.

7.2. $\star$ Lebesgue's theorem of differentiation

A closed bounded interval $[c, d]$ is said to be non-degenerate provided $c < d$.

DEFINITION 7.15. *A collection $\mathcal{F}$ of closed, bounded, non-degenerate intervals is said to cover a set $E \subseteq \mathbb{R}$ in the sense of Vitali provided for each point x in E and $\varepsilon > 0$, there is an interval I in $\mathcal{F}$ that contains x and has $\ell(I) < \varepsilon$.*

LEMMA 7.16 (The Vitali covering lemma). *Let E be a set of finite (Lebesgue) outer measure and $\mathcal{F}$ a collection of closed, bounded, non-degenerate intervals that covers E in the sense of Vitali. Then, for each $\varepsilon > 0$, there is a finite disjoint subcollection $\{I_k\}_{k=1}^n$ of $\mathcal{F}$ for which*

$$m^* \left(E \setminus (\cup_{k=1}^n I_k)\right) < \varepsilon. \tag{7.1}$$

PROOF. Since $m^*(E) < \infty$, there is an open set O containing E for which $m(O) < \infty$ by the outer regularity of Lebesgue measure (see Theorem 2.16 (i)). Because $\mathcal{F}$ is a Vitali covering of E, we may assume that each interval in $\mathcal{F}$ is contained in O. By the countable additivity of monotonicity of measure,

$$\text{if } \{I_k\}_{k=1}^\infty \subseteq \mathcal{F} \text{ is disjoint, then } \sum_{k=1}^\infty \ell(I_k) \leq m(O) < \infty. \tag{7.2}$$

Moreover, since each I_k is closed and $\mathcal{F}$ is a Vitali covering of E,

$$\text{if } \{I_k\}_{k=1}^n \subseteq \mathcal{F}, \text{ then } E \setminus (\cup_{k=1}^n I_k) \subseteq \cup_{I \in \mathcal{F}_n} I, \tag{7.3}$$

where $\mathcal{F}_n = \{I \in \mathcal{F} : I \cap (\cup_{k=1}^n I_k) = \emptyset\}$. If there is a finite disjoint subcollection of $\mathcal{F}$ that covers E, the proof is complete. Otherwise, we inductively choose a disjoint countable subcollection $\{I_k\}_{k=1}^\infty$ of $\mathcal{F}$ which has the following property:

$$E \setminus (\cup_{k=1}^n I_k) \subseteq \cup_{k=n+1}^\infty 5 * I_k \text{ for all } n \in \mathbb{N}, \tag{7.4}$$

where, for a closed, bounded interval I, $5 * I$ denotes the closed interval that has the same midpoint as I and 5 times its length. To begin this selection, let I_1 be any interval in $\mathcal{F}$. Suppose n is a natural number and the finite disjoint subcollection $\{I_k\}_{k=1}^n$ of $\mathcal{F}$ has been chosen. Since $E \setminus (\cup_{k=1}^n I_k) \neq \emptyset$, the collection $\mathcal{F}_n$ defined in the line below (7.3) is non-empty. Moreover, the supremum, s_n, of the lengths of the intervals in $\mathcal{F}_n$ is finite since $m(O)$ is an upper bound for these lengths. Choose I_{n+1} to be an interval in $\mathcal{F}_n$ for which $\ell(I_{n+1}) > \frac{s_n}{2}$. This inductively defines $\{I_k\}_{k=1}^\infty$, a countable disjoint subcollection of $\mathcal{F}$ such that for each n,

$$\ell(I_{n+1}) > \frac{\ell(I)}{2} \text{ if } I \in \mathcal{F} \text{ and } I \cap (\cup_{k=1}^n I_k) = \emptyset. \tag{7.5}$$

We infer from (7.2) that $(\ell(I_k)) \to 0$. Fix a natural number n. To verify the inclusion (7.4), let x belong to $E \setminus (\cup_{k=1}^n I_k)$. We infer from (7.3) that there is an $I \in \mathcal{F}$ which contains x and is disjoint from $\cup_{k=1}^n I_k$. Now, I must have non-empty intersection with some I_k, for otherwise, by (7.5), $\ell(I_k) > \frac{\ell(I)}{2}$ for all $k > n$, which contradicts the convergence of

$(\ell(I_k))$ to 0. Let N be the first natural number for which $I \cap I_N \neq \emptyset$. Then, $N > n$. Since $I \cap (\cup_{k=1}^{N-1} I_k) = \emptyset$, we infer from (7.5) that $\ell(I_N) > \frac{\ell(I)}{2}$. Since x belongs to I and $I \cap I_N \neq \emptyset$, the distance form x to the midpoint of I_N is at most $\ell(I) + \frac{\ell(I_N)}{2}$ and hence, since $\ell(I) < 2\ell(I_N)$, the distance from x to the midpoint of I_N is less than $\frac{5}{2}\ell(I_N)$. This means that x belongs to $5 * I_N$. Thus,

$$x \in 5 * I_N \subseteq \cup_{k=n+1}^{\infty} 5 * I_k.$$

We have established the inclusion (7.4).

Let $\varepsilon > 0$. We infer from (7.2) that there is $n \in \mathbb{N}$ for which $\sum_{k=n+1}^{\infty} \ell(I_k) < \frac{\varepsilon}{5}$. This choice of n, together with the inclusion (7.4) and the monotonicity and countable additivity of measure, established (7.1). □

For a real-valued function f and an interior point x of its domain, the *upper derivative* of f at x, $\overline{D}f(x)$ and the *lower derivative* of f at x, $\underline{D}f(x)$ are defined as follows:

$$\overline{D}f(x) = \lim_{h \to 0^+} \left[\sup_{0<|t|\leq h} \frac{f(x+t) - f(x)}{t} \right];$$
$$\underline{D}f(x) = \lim_{h \to 0^+} \left[\inf_{0<|t|\leq h} \frac{f(x+t) - f(x)}{t} \right].$$

We have $\overline{D}f(x) \geq \underline{D}f(x)$. If $\overline{D}f(x)$ equals to $\underline{D}f(x)$ and is finite, we say that f is *differentiable* at x and define $f'(x)$ to be the common value of the upper and lower derivatives.

The mean value theorem of calculus (see for example Theorem 12.50 in [**11**]) tells us that if a function f is continuous on the closed, bounded interval $[c, d]$ and differentiable on its interior (c, d) with $f' \geq \alpha$ on (c, d), then

$$\alpha(d - c) \leq [f(d) - f(c)].$$

The proof of the following generalization of this inequality, inequality (7.6), is a nice illustration of the fruitful interplay between the Vitali covering lemma and monotonicity properties of functions.

LEMMA 7.17. *Let f be an increasing function on the closed, bounded interval $[a, b]$. Then, for each $\alpha > 0$,*

$$m^*(\{x \in (a, b) : \overline{D}f(x) \geq \alpha\}) \leq \frac{1}{\alpha}[f(b) - f(a)] \tag{7.6}$$

and

$$m^*(\{x \in (a, b) : \overline{D}f(x) = \infty\}) = 0. \tag{7.7}$$

PROOF. Let $\alpha > 0$. Define $E_\alpha = \{x \in (a, b) : \overline{D}f(x) \geq \alpha\}$. Choose $\alpha' \in (0, \alpha)$. Let $\mathcal{F}$ be the collection of closed, bounded, non-degenerate intervals $[c, d]$ contained in (a, b) for which $f(d) - f(c) \geq \alpha'(d - c)$. Since $\overline{D}f \geq \alpha$ on E_α, $\mathcal{F}$ is a Vitali covering of E_α. The Vitali covering lemma tells us that there is a finite disjoint subcollection $\{[c_k, d_k]\}_{k=1}^n$ of $\mathcal{F}$ for which

$$m^* \left(E_\alpha \setminus (\cup_{k=1}^n [c_k, d_k]) \right) < \varepsilon.$$

Since $E_\alpha \subseteq (\cup_{k=1}^n [c_k, d_k]) \cup (E_\alpha \setminus (\cup_{k=1}^n [c_k, d_k]))$, by the finite subadditivity of outer measure, the preceding inequality and the choice of the intervals $[c_k, d_k]$,

$$m^*(E_\alpha) < \sum_{k=1}^n (d_k - c_k) + \varepsilon \leq \frac{1}{\alpha'} \sum_{k=1}^n [f(d_k) - f(c_k)] + \varepsilon.$$

However, the function f is increasing on $[a,b]$ and $\{[c_k,d_k]\}_{k=1}^n$ is a disjoint collection of subintervals of $[a,b]$. Therefore,

$$\sum_{k=1}^{n}[f(d_k)-f(c_k)] \le f(b)-f(a).$$

Thus, for each $\varepsilon>0$ and each $\alpha'\in(0,\alpha)$,

$$m^*(E_\alpha) \le \frac{1}{\alpha'}[f(b)-f(a)]+\varepsilon.$$

This proves (7.6). For each natural number n, $\{x\in(a,b) : \overline{D}f(x)=\infty\}\subseteq E_n$ and therefore

$$m^*(\{x\in(a,b):\overline{D}f(x)=\infty\}) \le m^*(E_n) \le \frac{1}{n}[f(b)-f(a)].$$

Letting n tend to infinity proves (7.7). □

THEOREM 7.18 (Lebesgue's theorem). *If the function f is monotone on the open interval (a,b), then it is differentiable almost everywhere on (a,b).*

PROOF. Assume f is increasing. Furthermore, assume (a,b) is bounded. Otherwise, express (a,b) as the union of an ascending sequence of open, bounded intervals and use the continuity of Lebesgue measure. The set of points x in (a,b) at which $\overline{D}f(x)>\underline{D}f(x)$ is the union of the sets

$$E_{\alpha,\beta}=\{x\in(a,b):\overline{D}f(x)>\alpha>\beta>\underline{D}f(x)\}$$

where α and β are positive rational numbers. Hence, since this is a countable collection, by the countable subadditivity of outer measure, it suffices to prove that each $E_{\alpha,\beta}$ has outer measure zero. Fix rational numbers α,β with $\alpha>\beta$ and set $E=E_{\alpha,\beta}$. Let $\varepsilon>0$. Choose an open set O for which

$$E\subseteq O\subseteq(a,b) \text{ and } m(O)<m^*(E)+\varepsilon. \tag{7.8}$$

Let $\mathcal{F}$ be the collection of closed, bounded, non-degenerate intervals $[c,d]$ contained in O for which $f(d)-f(c)<\beta(d-c)$. Since $\underline{D}f<\beta$ on E, $\mathcal{F}$ is a Vatali covering of E. The Vitali covering lemma tells us that there is a finite disjoint subcollection $\{[c_k,d_k]\}_{k=1}^n$ of $\mathcal{F}$ for which

$$m^*\left(E\setminus(\cup_{k=1}^n[c_k,d_k])\right)<\varepsilon. \tag{7.9}$$

By the choice of the intervals $[c_k,d_k]$, the inclusion of the union of the disjoint collection intervals $\{[c_k,d_k]\}_{k=1}^n$ in O and (7.8),

$$\sum_{k=1}^{n}[f(d_k)-f(c_k)]<\beta\left[\sum_{k=1}^{n}(d_k-c_k)\right]\le\beta m(O)\le\beta[m^*(E)+\varepsilon]. \tag{7.10}$$

For $1\le k\le n$, we infer from the preceding lemma, applied to the restriction of f to $[c_k,d_k]$, that

$$m^*(E\cap(c_k,d_k))\le\frac{1}{\alpha}[f(d_k)-f(c_k)].$$

Therefore, by (7.9),

$$m^*(E)\le\sum_{k=1}^{n}m^*(E\cap(c_k,d_k))+\varepsilon\le\frac{1}{\alpha}\left[\sum_{k=1}^{n}[f(d_k)-f(c_k)]\right]+\varepsilon. \tag{7.11}$$

We infer from (7.10) and (7.11) that

$$m^*(E) \leq \frac{\beta}{\alpha} m^*(E) + \frac{\beta}{\alpha}\varepsilon + \varepsilon \text{ for all } \varepsilon > 0.$$

Therefore, since $0 \leq m^*(E) < \infty$ and $\frac{\beta}{\alpha} < 1$, $m^*(E) = 0$. □

Lebesgue's theorem is the best possible in the sense that if E is a set of measure zero contained in the open interval (a, b), there is an increasing function on (a, b) that fails to be differentiable at each point of E (see Problem 9 in Page 98 of [**13**]).

Let f be integrable over the closed, bounded interval $[a, b]$. Extend f to take the value $f(b)$ on $(b, b + 1]$. For $0 < h \leq 1$, define the *divided difference function* $\text{Diff}_h f$ and *average value function* $\text{Av}_h f$ of $[a, b]$ by

$$\text{Diff}_h f(x) = \frac{f(x+h) - f(x)}{h} \text{ and } \text{Av}_h f(x) = \frac{1}{h}\int_{[x,x+h]} f dm \text{ for all } x \in [a, b].$$

By a change of variables in the integral (see Question 6 of Tutorial 7 if necessary) and cancellation, for all $a \leq u < v \leq b$,

$$\int_{[u,v]} \text{Diff}_h f dm = \text{Av}_h f(v) - \text{Av}_h f(u). \tag{7.12}$$

COROLLARY 7.19. *Let f be an increasing function on the closed, bounded interval $[a, b]$. Then, f' is integrable over $[a, b]$ and*

$$\int_{[a,b]} f' dm \leq f(b) - f(a). \tag{7.13}$$

PROOF. Since f is increasing on $[a, b + 1]$, it is measurable (see Question 1 of Tutorial 4 if necessary) and therefore the divided difference functions are also measurable. Lebesgue's theorem tells us that f is differentiable a.e. on (a, b). Therefore, $(\text{Diff}_{\frac{1}{n}} f)$ is a sequence of non-negative measurable functions that converges pointwise a.e. on $[a, b]$ to f'. According to Fatou's lemma (see Theorem 4.11),

$$\int_{[a,b]} f' dm \leq \liminf_{n\to\infty} \int_{[a,b]} \text{Diff}_{\frac{1}{n}} f dm. \tag{7.14}$$

By the change of variable formula (7.12), for each natural number n, since f is increasing,

$$\int_{[a,b]} \text{Diff}_{\frac{1}{n}} f dm = n\int_{[b,b+\frac{1}{n}]} f dm - n\int_{[a,a+\frac{1}{n}]} f dm$$
$$= f(b) - n\int_{[a,a+\frac{1}{n}]} f dm \leq f(b) - f(a).$$

Thus,

$$\limsup_{n\to\infty} \int_{[a,b]} \text{Diff}_{\frac{1}{n}} f dm \leq f(b) - f(a). \tag{7.15}$$

This inequality (7.13) follows from the inequalities (7.14) and (7.15). □

REMARK. The integral in (7.13) is independent of the values taken by f at the end-points. On the other hand, the right-hand side of this equality holds for the extension of any increasing extension of f on the open, bounded interval (a, b) to its closure $[a, b]$. Therefore, a tighter form of inequality (7.13) is

$$\int_{[a,b]} f' dm \leq \sup_{x\in(a,b)} f(x) - \inf_{x\in(a,b)} f(x).$$

The right-hand side of this inequality equals $f(b) - f(a)$ if and only if f is continuous at the endpoints. However, even if f is increasing and continuous on $[a, b]$, inequality (7.13) may be strict. It is strict for the Cantor-Lebesgue function F on $[0, 1]$ (see the Remark before Section 2.2) since $F(1) - F(0) = 1$ while F' vanishes a.e. on $(0, 1)$. However, we know the inequality in (7.13) will become equality if f is only assumed to be absolutely continuous (see Theorem 12 in Page 107 of [**13**] for details).

Lebesgue's theorem tells us that a monotone function on an open interval is differentiable a.e. Therefore, the *difference* of two increasing functions on an open interval is also differentiable a.e. We now provide a characterization of the class of functions on a closed, bounded interval that may be expressed as the difference of increasing functions, which shows that this class is surprisingly large: it includes, for instance, all Lipschitz functions.

Let f be a real-valued function defined on the closed, bounded interval $[a, b]$ and $P = \{x_0, \ldots, x_k\}$ be a partition of $[a, b]$. Define the variation of f with respect to P by

$$V(f, P) = \sum_{i=1}^{k} |f(x_i) - f(x_{i-1})|,$$

and the *total variation* of f on $[a, b]$ by

$$TV(f) = \sup\{V(f, P) : P \text{ a partition of } [a, b]\}.$$

For a subinterval $[c, d]$ of $[a, b]$, $TV(f_{[c,d]})$ denotes the total variation of the restriction of f to $[c, d]$.

DEFINITION 7.20. *A real-valued function f on the closed, bounded interval $[a, b]$ is said to be of bounded variation on $[a, b]$ provided $TV(f) < \infty$.*

EXAMPLE 7.21. *Let f be an increasing function on $[a, b]$. Then, f is of bounded variation on $[a, b]$ and*

$$TV(f) = f(b) - f(a).$$

Indeed, for any partition $P = \{x_0, \ldots, x_k\}$ of $[a, b]$,

$$V(f, P) = \sum_{i=1}^{k} |f(x_i) - f(x_{i-1})| = \sum_{i=1}^{k} [f(x_i) - f(x_{i-1})] = f(b) - f(a).$$

EXAMPLE 7.22. *Let f be a Lipschitz function on $[a, b]$. Then, f is of bounded variation of $[a, b]$ and $TV(f) \leq c(b - a)$, where*

$$|f(u) - f(v)| \leq c|u - v| \text{ for all } u, v \text{ in } [a, b].$$

Indeed, for a partition $P = \{x_0, \ldots, x_k\}$ of $[a, b]$,

$$V(f, P) = \sum_{i=1}^{k} |f(x_i) - f(x_{i-1})| \leq c \sum_{i=1}^{k} (x_i - x_{i-1}) = c(b - a).$$

Thus, $c(b - a)$ is an upper bound of the set of all variations of f with respect to a partition of $[a, b]$ and hence $TV(f) \leq c(b - a)$.

EXAMPLE 7.23. *Define the function f on $[0, 1]$ by*

$$f(x) = \begin{cases} x \cos \frac{\pi}{2x}, & \text{if } 0 < x \leq 1, \\ 0, & \text{if } x = 0. \end{cases}$$

Then, f is continuous on $[0,1]$. But f is not of bounded variation on $[0,1]$. Indeed, for a natural number n, consider the partition $P_n = \{0, \frac{1}{2n}, \frac{1}{2n-1}, \ldots, \frac{1}{3}, \frac{1}{2}, 1\}$ of $[0,1]$. Then,

$$V(f, P_n) = 1 + \frac{1}{2} + \cdots + \frac{1}{n}.$$

Hence, f is not of bounded variation on $[0,1]$, since the harmonic series diverges.

Observe that if c belongs to (a,b), P is a partition of $[a,b]$, and P' is the refinement of P obtained by adjoining c to P, then, by the triangle inequality, $V(f,P) \leq V(f,P')$. Thus, in the definition of the total variation of a function on $[a,b]$, the supremum can be taken over partitions of $[a,b]$ that contain the point c. Now, a partition P of $[a,b]$ that contains the point c induces, and is induced by, partitions P_1 and P_2 of $[a,c]$ and $[c,b]$ respectively, and for such partitions

$$V(f_{[a,b]}, P) = V(f_{[a,c]}, P_1) + V(f_{[c,b]}, P_2).$$

Take the supremum among such partitions to conclude that

$$TV(f_{[a,b]}) = TV(f_{[a,c]}) + TV(f_{[c,b]}).$$

We infer from this that if f is of bounded variation on $[a,b]$, then

$$TV(f_{[a,v]}) - TV(f_{[a,u]}) = TV(f_{[u,v]}) \geq 0 \text{ for all } a \leq u < v \leq b.$$

Therefore, the function $x \mapsto TV(f_{[a,x]})$, which we call the *total variation function* for f, is a real-valued increasing function on $[a,b]$. Moreover, for $a \leq u < v \leq b$, if we take the crudest partition $P = \{u, v\}$ of $[u,v]$, we have

$$f(u) - f(v) \leq |f(v) - f(u)| = V(f_{[u,v]}, P) \leq TV(f_{[u,v]}) = TV(f_{[a,v]}) - TV(f_{[a,u]}).$$

Thus,

$$f(v) + TV(f_{[a,v]}) \geq f(u) + TV(f_{[a,u]}) \text{ for all } a \leq u < v \leq b.$$

We have established the following lemma.

LEMMA 7.24. *Let the function f be of bounded variation on the closed, bounded interval $[a,b]$. Then, f has the following explicit expression as the difference of two increasing functions on $[a,b]$:*

$$f(x) = [f(x) + TV(f_{[a,x]})] - TV(f_{[a,x]}) \text{ for all } x \in [a,b].$$

THEOREM 7.25 (Jordan's theorem). *A function f is of bounded variation on the closed, bounded interval $[a,b]$ if and only if it is the difference of two increasing functions on $[a,b]$.*

PROOF. Let f be of bounded variation on $[a,b]$. The preceding lemma provides an explicit representation of f as the difference of increasing functions. To prove the converse, let $f = g - h$ on $[a,b]$, where g and h are increasing functions on $[a,b]$. For any partition

$P = \{x_0, \ldots, x_k\}$ of $[a, b]$,

$$\begin{aligned} V(f, P) &= \sum_{i=1}^{k} |f(x_i) - f(x_{i-1})| = \sum_{i=1}^{k} |[g(x_i) - g(x_{i-1})] + [h(x_{i-1}) - h(x_i)]| \\ &\leq \sum_{i=1}^{k} |g(x_i) - g(x_{i-1})| + \sum_{i=1}^{k} |h(x_{i-1}) - h(x_i)| \\ &= \sum_{i=1}^{k} [g(x_i) - g(x_{i-1})] + \sum_{i=1}^{k} [h(x_i) - h(x_{i-1})] \\ &= [g(b) - g(a)] + [h(b) - h(a)]. \end{aligned}$$

Thus, the set of variations of f with respect to partitions of $[a, b]$ is bounded above by $[g(b) - g(a)] + [h(b) - h(a)]$ and therefore f is of bounded variation of $[a, b]$. □

We call the expression of a function of bounded variation f as the difference of increasing functions a *Jordan decomposition* of f.

COROLLARY 7.26. *If the function f is of bounded variation on the closed, bounded interval $[a, b]$, then it is differentiable a.e. on the open interval (a, b) and f' is integrable over $[a, b]$.*

PROOF. According to Jordan's theorem, f is the difference of two increasing functions on $[a, b]$. Thus, Lebesgue's theorem tells us that f is the difference of two functions which are differentiable a.e. on (a, b). Therefore, f is differentiable a.e. on (a, b). The integrability of f' follows from Corollary 7.19. □

7.3. Finishing remarks

Section 7.1 is prepared mainly based on Section 7.2 of [**5**], and note that the proof of Radon-Nikodym theorem follows from the treatment in [**3**]. On the other hand, Section 7.2 is prepared based on Sections 6.2 and 6.3 of [**13**].

APPENDIX A

Mathematical background

A.1. Sets and functions

In our operations with sets, we shall always deal with collection of subsets of some universal set Ω; the nature of this set will be clear from the context. The notation $\{x \in A : P(x)\}$ is used to denote the set of elements of A with property P. The set of all subsets of A (its *power set*) is denoted by $P(A)$.

The *complement* A^c of A consists of the elements of Ω which are not members of A, i.e. $A^c = \Omega \setminus A$. More generally, we have the *difference* $B \setminus A = \{x \in B : x \notin A\} = B \cap A^c$ and the *symmetric difference* $A \triangle B = (A \setminus B) \cup (B \setminus A)$. Note that $A \triangle B = \emptyset$ if and only if $A = B$.

For an index set Λ, we have

$$\bigcap_{\alpha\in\Lambda} A_\alpha = \{x : x \in A_\alpha \text{ for all } \alpha \in \Lambda\} = \{x : \forall\, \alpha \in \Lambda,\ x \in A_\alpha\},$$

$$\bigcup_{\alpha\in\Lambda} A_\alpha = \{x : x \in A_\alpha \text{ for some } \alpha \in \Lambda\} = \{x : \exists\, \alpha \in \Lambda,\ x \in A_\alpha\}.$$

These are linked by *de Morgan's laws*:

$$\left(\bigcup_{\alpha\in\Lambda} A_\alpha\right)^c = \bigcap_{\alpha\in\Lambda} A_\alpha^c, \quad \left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)^c = \bigcup_{\alpha\in\Lambda} A_\alpha^c.$$

If $A \cap B = \emptyset$, then A and B are *disjoint*. A family of sets $(A_\alpha)_{\alpha\in\Lambda}$ is *pairwise disjoint* if $A_\alpha \cap A_\beta = \emptyset$ whenever $\alpha \neq \beta$ $(\alpha, \beta \in \Lambda)$, and in this case, we usually write $\uplus_{\alpha\in\Lambda} A_\alpha$ instead of $\cup_{\alpha\in\Lambda} A_\alpha$ to emphasize it is a disjoint union.

The *Cartesian product* $A \times B$ of sets A and B is the set of ordered pairs $A \times B = \{(a, b) : a \in A, b \in B\}$. $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ denotes the plane, more generally, $\mathbb{R}^n$ is the $n-$fold Cartesian product of $\mathbb{R}$ with itself, i.e. the set of all $n-$tuples $(x_1, x_2, \ldots, x_n)$ composed of real numbers.

Given two sets A and B, a *function* from A to B is a rule or mapping that takes each element $x \in A$ and associates with it a single element of B. In this case, we write $f : A \to B$. The set $\mathcal{D}_f = A$ is called the *domain* of f. The *range* of f, denoted by $\mathcal{R}_f$, is not necessarily equal to B but refers to the subset of B given by $\{y \in B : y = f(x) \text{ for some } x \in A\}$. The function g *extends* f if $\mathcal{D}_f \subseteq \mathcal{D}_g$ and $g = f$ on $\mathcal{D}_f$; alternatively we say that f *restricts* g to $\mathcal{D}_f$. A function $f : A \to B$ is called

injective (or one-one)	$\iff f(x) = f(x')$ implies $x = x'$,
surjective (or onto)	$\iff \mathcal{R}_f = B$,
bijective (or one-to-one correspondence)	$\iff f$ is injective and surjective.

Set operations and direct images under a map f are not necessarily compatible: indeed, we have, in general,

$$f(E_1 \cup E_2) = f(E_1) \cup f(E_2),$$
$$f(E_1 \cap E_2) \neq f(E_1) \cap f(E_2),$$
$$f(E_1 \setminus E_2) \neq f(E_1) \setminus f(E_2).$$

Inverse images and set operations are always compatible. The inverse mapping f^{-1} maps subsets $C \subseteq B$ into subsets of A and it is defined as

$$f^{-1}(C) := \{x \in A : f(x) \in C\} \subseteq A \quad \text{for all} \quad C \subseteq B;$$

it is, in general, multi-valued, the notation $f^{-1}(y)$ is used only if $f^{-1}(\{y\})$ has at most one element, i.e. if f is injective. For $C, C_i, D \subseteq B$, one has

$$f^{-1}\left(\bigcup_{i\in I} C_i\right) = \bigcup_{i\in I} f^{-1}(C_i),$$
$$f^{-1}\left(\bigcap_{i\in I} C_i\right) = \bigcap_{i\in I} f^{-1}(C_i),$$
$$f^{-1}(C \setminus D) = f^{-1}(C) \setminus f^{-1}(D).$$

The *indicator* function $\mathbf{1}_A$ of the set A is the function

$$\mathbf{1}_A(x) = \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{if } x \notin A. \end{cases}$$

Note that $\mathbf{1}_{A\cap B} = \mathbf{1}_A \cdot \mathbf{1}_B$, $\mathbf{1}_{A\cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A\mathbf{1}_B$, and $\mathbf{1}_{A^c} = 1 - \mathbf{1}_A$.

For any set E, an *equivalence relation* on E is a relation (i.e. a subset R of $E \times E$, where we write $x \sim y$ to indicate that $(x, y) \in R$) with the following properties:

1. *reflexive*: for all $x \in E$, $x \sim x$,
2. *symmetric*: $x \sim y$ implies $y \sim x$,
3. *transitive*: $x \sim y$ and $y \sim z$ implies $x \sim z$.

An equivalence relation $\sim$ on E partitions E into disjoint *equivalence classes*: given $x \in E$, write $[x] = \{z : z \sim x\}$ for the equivalence class of x, i.e. the set of all elements of E that are equivalent to x. Thus $x \in [x]$, hence $E = \cup_{x\in E}[x]$. If $[x] \cap [y] \neq \emptyset$, then there is $z \in E$ with $x \sim z$ and $z \sim y$, hence $x \sim y$, so that $[x] = [y]$. We shall denote the set of all equivalence classes by $E/\sim$.

A.2. Countable and uncountable sets in $\mathbb{R}$

We say that a set A is *countable* if there is a one-to-one correspondence between A and a subset of $\mathbb{N}$, i.e. a function $f : A \to \mathbb{N}$ that takes distinct points to distinct points. Informally, A is *finite* if this correspondence can be set up using only an initial segment $\{1, 2, \ldots, N\}$ of $\mathbb{N}$ (for some $N \in \mathbb{N}$), while we call A *countably infinite* or *denumerable* if all of $\mathbb{N}$ is used. It is not difficult to see that countable unions of countable sets are countable; in particular, the set $\mathbb{Q}$ of rational numbers is countable. Let us illustrate these by the following example and lemma.

EXAMPLE A.1. *(i) Finite sets are countable: $a, b, ..., z \to 1, 2, ..., 26$, where $a \leftrightarrow 1, \ldots, z \leftrightarrow 26$, is bijective.*

(ii) The even numbers are countable. This follows from the fact that the map

$$f : \{2, 4, 6, \dots, 2i, \dots\} \to \mathbb{N}, \quad i \mapsto \frac{i}{2},$$

is an injection and even a bijection. This means that there are 'as many' even numbers as there are natural numbers.

(iii) The set of integers $\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \dots\}$ is countable. A possible counting scheme is shown on Figure A.1 below or, more formally,

$$g : i \in \mathbb{Z} \mapsto \begin{cases} 2i, & \text{if } i > 0 \\ 2|i| + 1, & \text{if } i \leq 0. \end{cases}$$

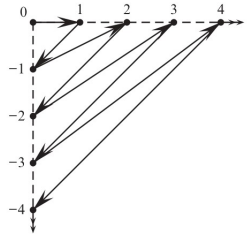


FIGURE A.1. Illustration for Example A.1 (iii).

(iv) The Cartesian product $\mathbb{N} \times \mathbb{N} := \{(i, k) : i, k \in \mathbb{N}\}$ is countable. To see this, arrange the pairs (i, k) in an array and count along the diagonals: The map for

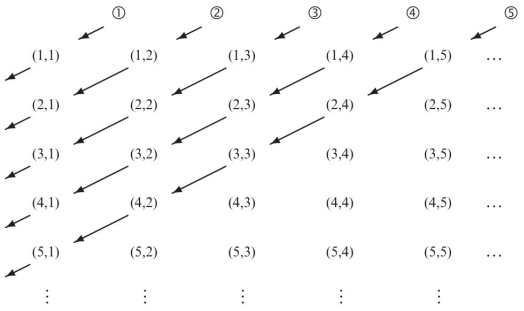


FIGURE A.2. Illustration for Example A.2 (iv).

the above counting scheme is given by

$$h : (i, k) \to \frac{(i + k)(i + k - 1)}{2} - k + 1 \in \mathbb{N}, \quad (i, k) \in \mathbb{N} \times \mathbb{N}.$$

(v) The rational numbers $\mathbb{Q}$ are countable: set $Q_\pm := \{q \in \mathbb{Q} : \pm q > 0\}$. Every element $\frac{m}{n} \in Q_+$ can be identified with at least one pair $(m, n) \in \mathbb{N} \times \mathbb{N}$, so that

$$Q_+ \subseteq \{ \underbrace{\tfrac{1}{1}}_{①}, \underbrace{\tfrac{1}{2}, \tfrac{2}{1}}_{②}, \underbrace{\tfrac{1}{3}, \tfrac{2}{2}, \tfrac{3}{1}}_{③}, \underbrace{\tfrac{1}{4}, \tfrac{2}{3}, \tfrac{3}{2}, \tfrac{4}{1}}_{④}, \dots \};$$

in the set ... on the right we distinguish between cancelled and uncancelled forms of a rational, e.g. $\frac{1}{3}, \frac{2}{6}, \frac{3}{9}, \dots$, *which are counted whenever they appear. The numbers* ① *refer to the corresponding diagonals in the counting scheme in part (iv). This shows that we can find injections* $Q_+ \xrightarrow{i} \{\dots\} \xrightarrow{i} \mathbb{N} \times \mathbb{N}$*; the set* $\mathbb{N} \times \mathbb{N}$ *is countable, thus* $Q_\pm$ *are countable. Finally,*

$$\mathbb{Q} = Q_- \uplus \{0\} \uplus Q_+ = \{r_1, r_2, r_3, \dots\} \uplus \{0\} \uplus \{q_1, q_2, q_3, \dots\}$$

and $p_1 := 0, p_{2i} := q_i, p_{2i+1} := r_i$ *gives an enumeration* $(p_1, p_2, p_3, \dots)$ *of* $\mathbb{Q}$*.*

LEMMA A.2. *Let* $A_1, A_2, A_3, \dots$ *be countably many countable sets. Then their union* $A = \bigcup_{i \in \mathbb{N}} A_i$ *is countable, i.e. countable unions of countable sets are countable.*

PROOF. Since each A_i is countable, we can find an enumeration

$$A_i = (a_{i,1}, a_{i,2}, \dots, a_{i,k}, \dots)$$

(if A_i is a finite set, we repeat the last element of the list infinitely often), so that

$$A = \bigcup_{i \in \mathbb{N}} A_i = (a_{i,k} : (i,k) \in \mathbb{N} \times \mathbb{N}).$$

Using Example A.1(iv) we can relabel $\mathbb{N} \times \mathbb{N}$ by $\mathbb{N}$ and (after deleting all duplicates) we have found an enumeration. □

Cantor showed that the set $\mathbb{R}$ *cannot* be placed in one-to-one correspondence with (a subset of) $\mathbb{N}$; thus it is an example of an *uncountable* set. Cantor's proof assumes that we can write each real number uniquely as a decimal (always choosing the non-terminating version). We can also restrict ourselves to showing that the interval $[0, 1]$ is uncountable.

If this set were countable, then we could write its elements as a sequence $(x_n)_{n \geq 1}$, and since each x_n has a unique decimal expansion of the form

$$x_n = 0.a_{n1}a_{n2}a_{n3} \dots a_{nn} \dots$$

for digits a_{ij} chosen from the set $\{0, 1, 2, \dots, 9\}$, we could therefore write down the array

$$\begin{aligned} x_1 &= 0.a_{11}a_{12}a_{13} \dots \\ x_2 &= 0.a_{21}a_{22}a_{23} \dots \\ x_3 &= 0.a_{31}a_{32}a_{33} \dots \\ &\dots \end{aligned}$$

Now write down $y = 0.b_1b_2b_3 \dots$, where the digits b_n are chosen to differ from a_{nn}. Such a decimal expansion defines a number $y \in [0, 1]$ that differs from each of the x_n (since its expansion differs from that of x_n in the nth place). Hence our sequence does not exhaust $[0, 1]$, and the contradiction shows that $[0, 1]$ cannot be countable.

Since the union of two countable sets must be countable, and $\mathbb{Q}$ is countable, it follows that $\mathbb{R} \setminus \mathbb{Q}$ is uncountable, i.e. there are far 'more' irrational numbers than the rational ones! Moreover, countable union of countable sets is still countable.

A.3. Topological properties of sets in $\mathbb{R}^n$

Given $a \in \mathbb{R}^n$ and $\varepsilon > 0$, recall that the $\varepsilon-$neighborhood of a is the set

$$V_\varepsilon(a) = \{x \in \mathbb{R}^n : |x - a| < \varepsilon\}$$

where $|\cdot|$ denotes the usual Euclidean *norm* on $\mathbb{R}^n$, i.e.

$$|x| = \sqrt{x_1^2 + \dots + x_n^2}.$$

DEFINITION A.3. *A set $O \subseteq \mathbb{R}^n$ is open if for all points $a \in O$ there exists an $\varepsilon-$neighborhood $V_\varepsilon(a) \subseteq O$. A set is closed if its complement is open.*

The open set in $\mathbb{R}$, i.e. $n = 1$, has a nice structure which is the disjoint union of a countable collection of open intervals[1]. However, for $n \geq 2$, we do not have such results.

If O_α is open for all $\alpha \in \Lambda$, then $\cup_{\alpha\in\Lambda} O_\alpha$ is open. It is easy to see that a finite intersection of open sets is open, but a countable intersection of open sets need not be open. For example, let $O_n = (-\frac{1}{n}, 1)$ for $n \in \mathbb{N}$, then $E = \cap_{n=1}^{\infty} O_n = [0, 1)$ is not open.

Note that $\mathbb{R}$, unlike $\mathbb{R}^n$ or more general spaces, has a *linear order*, i.e. given $x, y \in \mathbb{R}$, we can decide whether $x \leq y$ or $y \leq x$. Thus u is an *upper bound* for a set $A \subseteq \mathbb{R}$ if $a \leq u$ for all $a \in A$, and a *lower bound* is defined similarly. The *supremum* (or least upper bound) is then the minimum of all upper bounds and written $\sup A$. The *infimum* (or greatest lower bound) $\inf A$ is defined as the maximum of all lower bounds. The *completeness property* of $\mathbb{R}$ can be expressed by the statement that every set which is bounded from above has supremum.

DEFINITION A.4. *A set $A \subseteq \mathbb{R}^n$ is called* compact *if every open cover of A admits a finite subcover.*

In other words: whenever

$$A \subseteq \bigcup_{i\in I} G_i$$

for some collection of open sets $\{G_i\}_{i\in I}$, there exist finitely many indices $i_1, \ldots, i_N \in I$ such that

$$A \subseteq \bigcup_{k=1}^{N} G_{i_k}.$$

We present one important result concerning the compact sets in $\mathbb{R}^n$.

THEOREM A.5 (Heine-Borel theorem). *A set in $\mathbb{R}^n$ is compact if and only if it is closed and bounded.*

PROOF. See Pages 11-13 of [**10**]. □

A function $f : D \to \mathbb{R}^n$ is said to be *continuous* if $f^{-1}(O)$ is open for each open set $O \subseteq \mathbb{R}^n$. (The equivalence between this definition and the one induced from the $\varepsilon - \delta$ definition is shown in Theorem 4.8 of [**14**].)

We now state an important theorem concerning the maximum and minimum values of a continuous real-valued function.

THEOREM A.6 (Weierstrass's theorem). *Let E be a compact (or closed and bounded) set in $\mathbb{R}^N$, and suppose the function $f : E \to \mathbb{R}$ is continuous. Then, f attains its maximum and minimum values in E, i.e., there exist $a, b \in E$ such that*

$$f(a) \geq f(x) \quad \textit{and} \quad f(b) \leq f(x)$$

for all $x \in E$.

PROOF. See Theorem 1.51 in [**12**]. □

Specializing to real sequences $(x_n)_{n=1}^{\infty}$, we can further define the *upper limit* $\limsup_{n\to\infty} x_n$ as

$$\inf_{n\in\mathbb{N}} \sup_{m\geq n} x_m$$

[1] See Proposition 9 in Page 16 of [**13**] for the proof.

and the *lower limit* $\liminf_{n\to\infty} x_n$ as

$$\sup_{n\in\mathbb{N}} \inf_{m\geq n} x_m.$$

The sequence (x_n) has a limit if and only if $\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n$ (their common value is then its limit)[2]. We say (x_n) converges if its limit is a real number. A series $\sum_{n=1}^{\infty} a_n$ has a limit if the sequence $x_m = \sum_{n=1}^{m} a_n$ of its partial sums has a limit, and its limit is then the *sum* $\sum_{n=1}^{\infty} a_n$ of the series.

A.4. Finishing remarks

I have prepared this chapter mainly based on Chapter 1.1 of [**5**]. Other books like [**1, 10, 14, 15**] are also consulted.

[2]If you are not familiar with the upper and lower limits, then you may have a look at Section 1.5 of [**9**].

Bibliography

[1] S. Abbott. *Understanding analysis*. Undergraduate Texts in Mathematics. Springer, New York, second edition, 2015.

[2] S. Axler. *Measure, integration & real analysis*, volume 282 of *Graduate Texts in Mathematics*. Springer, Cham, 2020.

[3] R. C. Bradley. An elementary treatment of the Radon-Nikodým derivative. *Amer. Math. Monthly*, 96(5):437–440, 1989.

[4] H. Brezis. *Functional analysis, Sobolev spaces and partial differential equations*. Universitext. Springer, New York, 2011.

[5] M. Capiński and E. Kopp. *Measure, integral and probability*. Springer Undergraduate Mathematics Series. Springer-Verlag, London, second edition, 2004.

[6] J. B. Conway. *A course in functional analysis*, volume 96 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, second edition, 1990.

[7] O. Dovgoshey, O. Martio, V. Ryazanov, and M. Vuorinen. The Cantor function. *Expo. Math.*, 24(1):1–37, 2006.

[8] I. Fonseca and G. Leoni. *Modern methods in the calculus of variations: L^p spaces*. Springer Monographs in Mathematics. Springer, New York, 2007.

[9] O. Hijab. *Introduction to calculus and classical analysis*. Undergraduate Texts in Mathematics. Springer, New York, third edition, 2011.

[10] F. Jones. *Lebesgue integration on Euclidean space*. Jones and Bartlett Publishers, Boston, MA, 2001. Revised edition.

[11] M. Laczkovich and V. T. Sós. *Real analysis—foundations and functions of one variable*. Undergraduate Texts in Mathematics. Springer, New York, fifth edition, 2015. Translated from the third (2005) Hungarian edition by the authors.

[12] M. Laczkovich and V. T. Sós. *Real analysis—series, functions of several variables, and applications*. Undergraduate Texts in Mathematics. Springer, New York, 2017. Translated from the second (2013) Hungarian edition by Gergely Bálint.

[13] H. L. Royden and P. M. Fitzpatrick. *Real analysis*. Pearson, fifth edition, 2023.

[14] W. Rudin. *Principles of mathematical analysis*. International Series in Pure and Applied Mathematics. McGraw-Hill Book Co., New York-Auckland-Düsseldorf, third edition, 1976.

[15] R. L. Schilling. *Measures, integrals and martingales*. Cambridge University Press, Cambridge, second edition, 2017.

[16] T. Tao. *An introduction to measure theory*, volume 126 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 2011.

[17] K. Yosida. *Functional analysis*, volume 123 of *Grundlehren der Mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences]*. Springer-Verlag, Berlin-New York, sixth edition, 1980.